MIDDLE YEARS PROGRAMME
YEAR 1 · MATHEMATICS

IMPROVED FIRST EDITION
2026

MYP

MATHEMATICS

1

Student Coursebook

A concept-driven inquiry course in number, proportion, algebra, shape and data.

Number systems · Place value and powers · Factors and primes
Fractions and decimals · Percentages · Patterns and equations
Lines, angles and constructions · Sets · Perimeter, area and volume · Data and graphs

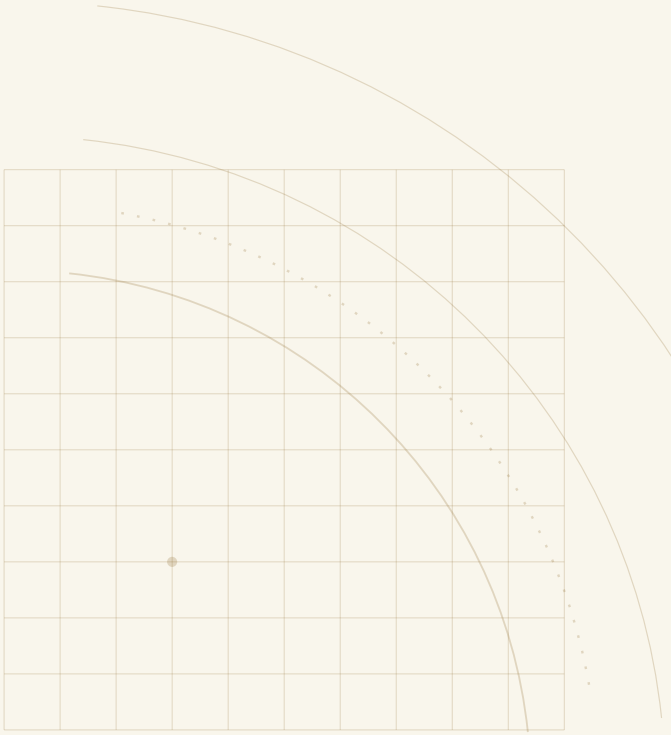
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AUTHOR

MYP MATHEMATICS 1

Mathematics 1

Student Coursebook

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SADIKOU MOUKODDIM

MYP YEAR 1

Mathematics 1

Student Coursebook

An inquiry-led introduction to number, place value, proportion, algebra, geometry, measurement, sets and data, built around the concepts, global contexts and assessment criteria of the Middle Years Programme.

IMPROVED FIRST EDITION

2026

SADIKOU MOUKODDIM

HEAD OF MATHEMATICS DEPARTMENT · MATHEMATICS TEACHER

MYP MATHEMATICS 1

Student Coursebook · Improved First Edition

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Edition

Improved First Edition, 2026. This edition revises the first edition of the same course: the mathematical scope, chapter sequence and learning outcomes are unchanged, while the explanations, investigations, worked examples, practice, illustration and page design have been rebuilt.

Independence

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Originality

All explanatory prose, worked examples, exercises, investigations, diagrams and illustrations in this book were written or drawn for this edition. The book is not a reproduction of any published coursebook. Where a concept is standard mathematics — place value, equivalent fractions, the angle sum of a triangle — the treatment and wording are the author's own.

Typefaces

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Answers

Answers and brief worked solutions to the practice sets, mixed practice and chapter reviews appear in the answer section at the back of the book. Attempt a question fully before checking: the value of an answer key lies in comparing reasoning, not in matching a number.

FRONT MATTER

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ORIENTATION

How to use this coursebook

This is a working book. Expect to write in the margins, test an idea, compare your method with a classmate's, and return to a question after your thinking has changed.

Year 1 is where many familiar ideas become precise. You already know how numbers are written, how shapes are named and how data can be displayed. This course asks you to look underneath those familiar forms: why a digit's position changes its value, why equivalent fractions name the same quantity, why an angle fact must hold for every triangle rather than merely one drawing.

The book is arranged in five parts. Part I establishes the structure of the number system and the rules that keep calculation reliable. Part II develops fractions, decimals and percentages as connected ways of describing parts of a whole. Part III introduces algebraic thinking through patterns and balance. Part IV brings together sets, shape and measurement. Part V examines data, graphs and the claims people make from them.

Every chapter follows one rhythm, and the rhythm is visible on the page: Engage, Explore, Explain, Worked Examples, Practice, Investigate and Apply, Reflect and Consolidate. The stage labels tell you where you are, so that a lesson can be planned around a stage rather than a page number.

Stage	What you do	What it is for
01 Engage	Meet a situation, photograph or puzzle.	Give the mathematics a reason to exist.
02 Explore	Try something before being told the rule.	Build a conjecture you can test.
03 Explain	Read definitions, notation and properties.	Turn the conjecture into a structure you can use.
04 Worked examples	Follow reasoning line by line, including checks.	See what a complete answer looks like.
05 Practice	Work from fluency to application and extension.	Make the method reliable and portable.
06 Investigate and apply	Use mathematics on a real task.	Meet the MYP criteria in a genuine setting.
07 Reflect	Summarise, check vocabulary, self-assess.	Know precisely what you can and cannot yet do.

The seven stages of every chapter.

NOTE

How the labels work

Definition boxes state precise meanings. Key idea boxes connect a method to a larger principle. Common error boxes show a mistake worth anticipating. ATL boxes name a learning skill and give a way to practise it. Criterion strips name the MYP criterion and strands developed by a section.

ATL SKILL

The exercise tags

F Fluency — short, deliberate practice to make a method accurate and efficient. R Reasoning — explain, compare, justify or diagnose an error. P Problem solving — multi-step and unfamiliar contexts. E Extension — deeper structure, generalisation or a more open task.

KEY IDEA

Three habits that make the book work

Estimate first. Predict the size of an answer before calculating it. Ask what a symbol means. Every letter, bracket and index is doing a job. Write the check. A calculation you cannot check is a calculation you do not yet trust.

Writing in this book

The wide outer margin is part of the page, not decoration. Use it for a question you could not finish, a note that a method reminded you of something, or the reason a step works. When you return to a chapter before an assessment, the margin is the fastest way back into your own thinking.

When you are stuck	What to do before asking
The question is unreadable	Underline the noun the question is asking about; the verb is usually the operation.
The method is forgotten	Read the most recent worked example in the same section and copy its first line.
The answer looks wrong	Estimate the size first, then check the units and the last step.
The question is ambiguous	Write down both readings and say which one you are answering.

FRAMEWORK

The MYP framework in this course

The Middle Years Programme asks students to learn mathematics as a connected, conceptual and applicable discipline. Concepts organise the content, contexts give it purpose, and criteria make the quality of thinking visible.

Key concepts

Key concept	What it asks in Year 1 mathematics
Form	What is this number, figure or display made of, and how is it arranged?
Logic	When must a statement be true, and what would count as a counterexample?
Relationships	How do quantities, variables and shapes depend on one another?

Related concepts used in this course

Quantity, Representation, Approximation, Equivalence, Pattern, Simplification, Generalisation, Space, Measurement, System, Model and Change. Each chapter names the two related concepts it develops most directly, so that the concept is a lens on the content rather than a label attached to it.

Global contexts

Global context	Chapters
Identities and relationships	4, 8
Orientation in space and time	1, 7
Personal and cultural expression	6
Scientific and technical innovation	2, 9
Fairness and development	3, 5
Globalisation and sustainability	10

Inquiry

Each chapter opens with a statement of inquiry, one factual, one conceptual and one debatable question, and a pathway that shows how the inquiry develops. A factual question has a settled answer. A conceptual question requires you to connect ideas. A debatable question has more than one defensible answer and must be argued.

ATL SKILL

ATL skills in this course

Approaches to learning are written in the form the MYP uses: “In order for a student to …, students must …”. Every chapter contains at least two ATL boxes: one for a thinking skill and one for a self-management, communication or research skill.

ASSESSMENT

Assessment criteria in practice

MYP mathematics uses four criteria. They describe different qualities of mathematical work, not four separate syllabuses: a single rich task can provide evidence for more than one criterion.

Criterion	What it looks for	Typical evidence in this book
A · Knowing and understanding	Selecting and applying the right mathematics, accurately.	Practice sets, chapter reviews, fluency items.
B · Investigating patterns	Recognising, describing and justifying a pattern, then generalising it.	The numbered investigations, extension items.
C · Communicating	Using notation, language and representations precisely, for an audience.	Explanations, diagrams, mini-tasks written for a named reader.
D · Applying mathematics	Using mathematics in a real context and judging whether the answer is reasonable.	Real-life applications and criterion-mapped mini-tasks.

Command words that shape the evidence

Command word	What a complete answer contains
State	The value, name or fact, with no working required.
Describe	What happens, in order, with the key features named.
Explain	A reason, connected to a property, definition or example.
Justify	Evidence: a calculation, a counterexample or a diagram.
Show that	A chain of working that ends at the stated result.
Investigate	Cases, a pattern, a general rule and a test of that rule.
Evaluate	A judgement with stated criteria and a limitation.

NOTE

Formative practice is not a formal assessment

The exercises in this book build the understanding that assessed tasks later require. Marks are not the point of a practice set; a question you cannot yet do is information, not a verdict.

PROGRESSION

Course map and progression

The ten chapters form a cumulative course. Each chapter names the knowledge it assumes, the new structure it builds, and the Year 2 territory it deliberately does not yet enter.

Ch	Chapter	Assumes	Builds
1	Number systems and place value	Counting, place value to thousands	Number families, decimals, rounding, estimation
2	Powers, expanded form and order of operations	Chapter 1	Index notation, squares and roots, order of operations
3	Factors, multiples and primes	Chapter 1–2	Divisibility, prime factorisation, HCF and LCM
4	Fractions and decimals	Chapters 1–3	Equivalence, four operations on fractions, decimal calculation
5	Percentages and applications	Chapter 4	Percentage of a quantity, increase and decrease, financial contexts
6	Patterns, expressions and equations	Chapters 1–5	Sequences, expressions, simplifying, solving linear equations
7	Lines, angles, triangles and constructions	Measurement language	Angle facts, triangle and polygon results, ruler and compass work
8	Sets and classification	Chapter 1	Set notation, Venn diagrams, two-set problems
9	Perimeter, area, surface area and volume	Chapter 7	Area formulae, circles, nets, volume and capacity
10	Data, tables and graphs	Chapters 4–5	Data types, displays, averages, misleading graphs

KEY IDEA

How the four criteria accumulate

Criterion A grows as the toolkit grows: by Chapter 10 you are selecting among many methods rather than applying one. Criterion B appears in every investigation, from counting strings of digits to the relationship between a cylinder's radius and its volume. Criterion C is practised in every written explanation and every diagram. Criterion D appears in each chapter's application, where the answer has to be judged as well as calculated.

CONVENTIONS

Notation, working and habits

Clear notation is part of the mathematics. It lets another person follow your reasoning, and it helps you notice when a claim is not yet precise.

Notation	Means	Example
$<$ $>$	is less than / is greater than	$3 < 8$, $-8 < -3$
$\le$ $\ge$	is at most / is at least	$x \le 5$
$\approx$	is approximately equal to	$48 \times 21 \approx 1\,000$
$\times$ $\div$	multiplied by, divided by	$24 \div 4 = 6$
a/b	a divided by b	¾ means $3 \div 4$
a^n	a raised to the power n	$10^3 = 1\,000$
$\sqrt{}$	square root	$\sqrt{49} = 7$
%	per hundred	$25\ \% = 0.25 = \frac{1}{4}$
$\in$ $\notin$	is an element of / is not an element of	$5 \in Z$ (the integers)
$\cup$ $\cap$	union / intersection	$A \cup B$
$\angle$	angle	$\angle ABC = 90°$
$\cong$	is congruent to	$\triangle ABC \cong \triangle DEF$

Five habits that protect accuracy

- Align equal signs. Write one equals sign beneath another so that each line states a true equality.
- Carry units. A number without its unit is an incomplete answer in any measurement question.
- Estimate first. Write the estimate before the calculation, then compare.
- Show the check. Substitute back, reverse the operation, or test with a simpler case.
- Name the rounding. Say “to the nearest hundred” rather than silently changing the value.

ATL SKILL

ATL · Self-management

In order for a student to keep a page of working readable, students must decide the layout before starting: number the question, align the equals signs, leave one line between parts, and box the final answer.

NOTE

Reading a mathematics question

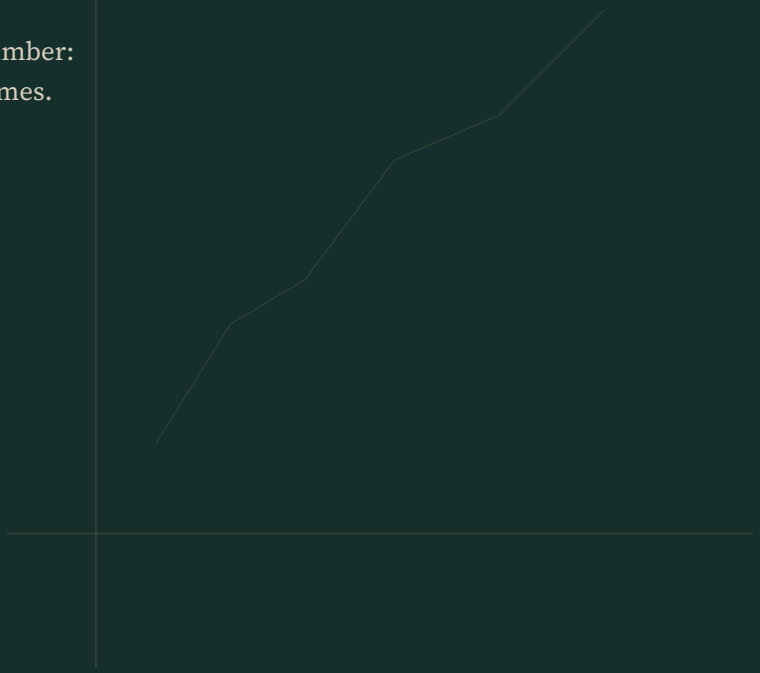
Read it twice. The first reading finds the quantities; the second finds the question. Underline the instruction — calculate, estimate, explain, justify — because the instruction decides what a complete answer contains.

PART I

Number, structure and proof

What a written number actually says, and the rules that keep calculation reliable.

The first three chapters establish the grammar of number: position, power, and the structure of factors and primes.





1

NUMBER, STRUCTURE AND PROOF

Number systems and place value

Ten digits are enough to write any number we can name. Their power comes from position: the same symbol can mean seven, seventy or seven thousandths.

KEY CONCEPT · FORM | RELATED · QUANTITY | RELATED · REPRESENTATION | CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How does a digit's position change its value, and how can we describe that structure so precisely that comparisons and calculations become reliable?

01 · MYP UNIT PLANNER

The whole chapter in one view: concepts, context and the evidence that will be looked for.

Statement of inquiry	A number's written form is an organised system of positions: a small set of digits can represent any magnitude when place and zero are used consistently, and the way we record a number decides how precisely we can compare, calculate and communicate it.

KEY IDEA

Key and related concepts

Key concept · Form. A number's written form is not one symbol but an organised arrangement of positions; changing a position changes the quantity it records.

Related · Quantity. Place value lets ten digits stand for any size, while rounding records an appropriate level of precision.

Related · Representation. The same value can appear as words, digits, an expanded sum, a point on a number line or a rounded claim.

NOTE

Global context · Orientation in space and time

Exploration: how different cultures have recorded quantity, and why the invention of a place-value system with zero changed what numbers could do. Every tally system works; only a positional system scales.

Inquiry questions

- Factual: What is the value of a digit in a given position, and how does multiplying or dividing by ten change a number's digits?
- Conceptual: Why is zero essential in a place-value system, and how can one number have several equivalent representations?
- Debatable: When a number is reported as “about 2 000”, is the result more useful than the exact count?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Read, write, compare, round and rewrite integers and decimals; choose sensible accuracy.
B · Investigating patterns	i, ii, iii	Investigate how many numbers a given number of digits can write; generalise the count.
C · Communicating	i, ii, iii, iv	Record working with correct notation; explain how place value works to a reader who has not met it.
D · Applying mathematics	i, ii, iii, iv	Design a community noticeboard that reports counts and costs at a sensible accuracy.

ATL SKILL

ATL · Critical thinking

In order for a student to judge whether a reported figure is precise enough, students must identify what each digit contributes and decide which digits still carry meaning in the context.

02 · LEARNING INTENTIONS AND SUCCESS CRITERIA

You are learning to	You have succeeded when you can
Understand how natural numbers, whole numbers and integers are related.	Classify a given number correctly and give an example that belongs to one set but not the other.
Know that the value of a digit depends on its place in a base-ten number.	State the value of any digit in a whole number or decimal, including zeros.
Write numbers in expanded form and return to standard form.	Move between 3 048, $3 \times 1000 + 0 \times 100 + 4 \times 10 + 8$ and words without error.
Compare and order integers and decimals using place value.	Place five mixed values in order and justify each decision with one sentence.
Round to a stated place and choose sensible accuracy in context.	Round to a named place and explain why a different place would be misleading.
Estimate before calculating and judge whether an answer is reasonable.	Produce an estimate within about 10 % and use it to catch an error.

Inquiry pathway for this chapter

Stage	What happens
1 · Context	A market stall and a community noticeboard report quantities in many different forms.
2 · Question	What does each digit in a displayed number actually contribute?
3 · Concept	Base-ten place value, zero as a placeholder, precision.
4 · Investigation	Count the numbers that different digit-lengths can write; generalise the count.
5 · Communication	Expanded form, number lines and justified rounding.
6 · Reflection	When does extra precision help, and when does it mislead?

NOTE

Prerequisite check · 10 minutes, no calculator

1. Write “four thousand and sixty” using digits.
2. Which is greater, 3.5 or 3.45? Explain in one sentence.
3. Calculate 7×8 and $56 \div 7$.
4. What is 100 more than 2 950?
5. Write ¾ as a decimal.
6. Put 0.6, 0.06 and 0.66 in order from least to greatest.

If more than two answers are incorrect, review place value to thousands and simple fractions of a whole before continuing. The answers, with the reasoning for each, are in the answer section.

03 · ENGAGE

A morning at the market, and a noticeboard that has to be read by everybody.

The trader at the next stall has three boxes of tomatoes: one labelled “120”, one labelled “1 200” and one labelled “12.00”. A customer reads all three as “twelve”. She is wrong about at least two of them, and the trader loses money either way. Nothing about the digits changed. Only their positions did.

That small scene is the whole chapter. Our number system does not invent a new symbol for every size. It reuses ten symbols and lets position do the work. Understand position, and you can read any number in the market, on a meter, on a bank statement or on a medicine bottle.



FIGURE 1

A market stall organises quantity before anyone writes a number: produce is stacked in rectangular arrays, weights are nested by size, and the counting frame records position. Place value formalises what the stall already does.

Original illustration created for this edition.

04 · EXPLORE

Before any definitions, find the pattern in your own writing.

Write the number 3 048.76 on a strip of paper. Now carry out these changes and write what happens to the digits each time: multiply by 10; multiply by 100; divide by 10; divide by 100. Do not use a calculator.

1. What happens to the digits themselves?
2. What happens to the position of every digit relative to the decimal point?
3. Which digit never changes its value, and why?
4. Predict what happens if you multiply by 1 000. Test your prediction.

KEY IDEA

What you should have noticed

The digits are unchanged in every case. Only their positions move. Multiplying by 10 moves every digit one place to the left; dividing by 10 moves every digit one place to the right. The decimal point stays where it is — it marks the boundary between whole values and parts of one.

1.1 Number systems and the number line

CRITERION A · i ◆ FORM ◆ REPRESENTATION

A number line turns “which is bigger?” into a question about position. Everything else in this chapter is a way of writing down what the line already shows.

Mathematicians keep several number families because each one answers a different question. The counting numbers tell you how many objects there are. Add zero and you can describe an empty shelf. Add the negatives and you can describe a debt, a temperature below freezing, or a depth below sea level.

DEFINITION

The number families

The natural numbers are the counting numbers 1, 2, 3, 4, … The whole numbers are the natural numbers together with zero: 0, 1, 2, 3, … The integers are the whole numbers and their negatives: …, −3, −2, −1, 0, 1, 2, 3, …

WHY IT MATTERS

A number line is not a picture of a number: it is a picture of order. Once two values sit on it, every comparison between them has an answer.

Every integer is also a rational number, because it can be written over 1: $7 = 7/1$ and $-3 = -3/1$. Terminating decimals such as 0.25 are rational too, since $0.25 = 25/100$. The families are nested, like boxes inside boxes: the natural numbers sit inside the whole numbers, which sit inside the integers, which sit inside the rationals.

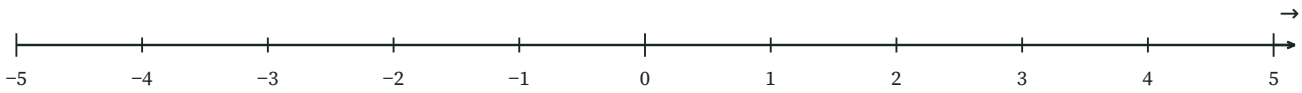


FIGURE 2 On a horizontal number line, values increase from left to right. Negative numbers lie to the left of zero; positive numbers lie to the right. Distance from zero gives magnitude, direction gives sign.

A number line is more than a display device. It gives a shared order to every integer, fraction and decimal. To compare two values, locate both: the one further to the right is greater. That rule never fails, which is why it is worth carrying a mental number line for the rest of your mathematical life.

KEY IDEA

Reading the symbols

The symbols $<$ and $>$ describe position, not size alone. "$3 < 8$" is correct because 3 lies to the left of 8. "$-8 < -3$" is also correct — even though 8 is greater than 3 — because -8 lies further left. Reversing the symbol reverses the claim.

COMMON ERROR

Reversing a sign when comparing negatives

A student writes $-8 > -3$ because $8 > 3$. On the number line -8 sits to the left of -3, so $-8 < -3$. Check every negative comparison against a sketch of the line, not against the size of the digits.

Working with the line

- Distance from zero is the magnitude. -8 and 8 are equally far from zero; they are different numbers because their directions differ.
- Between any two values there is another value. Between 3 and 4 lies 3.5; between 3.5 and 3.6 lies 3.55. This is why decimals can always be refined.
- Inequalities read left to right like a sentence: $-5 < -2 < 0 < 3$.

PRACTICE 1.1 SUPPORT 1–4 · CORE 3–7 · EXTENSION 7–9

1. Write down the value of (a) the number 4 more than −6, (b) the number 9 less than 2. F
2. Copy and complete: −7 … −3 using <, > or =. F
3. List the integers greater than −4 and less than 2. F
4. Draw a number line from −6 to 6 and mark −4.5, −1, 0.5 and 5. F
5. Which is greater: −2.5 or −2.05? Justify your answer using the line. R
6. A submarine is at −40 m and rises 15 m. A second submarine is at −22 m and dives 10 m. Which is deeper, and by how much? P
7. Explain why −3 is smaller than −2 even though 3 is larger than 2. R
8. Find two integers whose sum is −5 and whose difference is 7. E
9. The temperature was −3 °C at 06:00 and 5 °C at midday. Describe the change in two different ways, one using subtraction and one using a number line. E

1.2 Place value and the role of zero

CRITERION A · i, ii ◆ FORM ◆ QUANTITY

Our system is positional. The digit 3 means three, thirty or three hundred depending on where it stands, and each move to the left multiplies the value by ten. Zero is what makes this possible: it holds a place open so that every other digit keeps its position.

1000	100	10	1		0.1	0.01
3	0	4	8	•	7	6
thousands	hundreds	tens	ones		tenths	hundredths

FIGURE 3 Each position in 3 048.76 carries a different value; the point marks where values fall below one.

Read the table from the right. The ones column is worth 1; every column to its left is worth ten times as much. Moving right from the point, each column is worth one tenth as much. A table that works this way can be extended indefinitely in either direction, which is why no number is ever too large or too small to write.

DEFINITION

Place value

In a base-ten number, the value of a digit is the digit multiplied by ten raised to the power of its position from the decimal point. The number 3 048.76 therefore means $3 \times 1000 + 0 \times 100 + 4 \times 10 + 8 \times 1 + 7 \times 0.1 + 6 \times 0.01$.

Why zero is not “nothing”

Compare 305, 35 and 3.05. All three use the digits 3 and 5 only. The zero in 305 keeps the 3 in the hundreds column and the 5 in the ones column, forcing an empty tens column. In 3.05 the zero holds the tenths column open so that the 5 lands in the hundredths. Remove the zero and the number changes completely — the digits have not changed, but every value they carry has.

COMMON ERROR

Deleting an “unimportant” zero

Writing 4.5 for 4.05, or 4 500 for 4 005, is a place-value error, not a slip of the pen. Test any suspicious zero by asking: which column does it hold open? If deleting it moves another digit, the zero is doing work.

WORKED EXAMPLE 1.2.1 — ZERO AS A PLACEHOLDER

PROBLEM

Write the number “four thousand and sixty” in digits, and explain why a zero is needed.

1. The thousands digit is 4, so the number begins 4 _ _ _.
2. There is no “hundreds” in the name, so the hundreds digit is 0.
3. The tens digit is 6 and the ones digit is 0.
4. Writing 4 060 keeps every digit in its correct column: four thousands, no hundreds, six tens, no ones.

ANSWER 4 060. Without the zero, 46 would mean four tens and six ones.

$4 \times 1000 + 0 \times 100 + 6 \times 10 + 0 \times 1 = 4\,060$ ✓

WORKED EXAMPLE 1.2 — MULTIPLYING AND DIVIDING BY POWERS OF TEN

PROBLEM

Describe what happens to the digits of 3.48 when the number is multiplied by 10, multiplied by 100 and divided by 10.

1. × 10: every digit moves one place to the left, so 3.48 becomes 34.8.
2. × 100: every digit moves two places to the left, so 3.48 becomes 348.
3. ÷ 10: every digit moves one place to the right, so 3.48 becomes 0.348.
4. In each case the digits are unchanged; only their positions relative to the point move.

ANSWER 34.8, 348 and 0.348 respectively.

Estimate: 3.5 × 10 ≈ 35 and 34.8 is close ✓

PRACTICE 1.2 SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Write in digits: (a) two thousand and seven (b) nine hundred and forty (c) six thousand, two hundred and ten. F
2. State the value of the digit 7 in each number: (a) 2 714 (b) 507 (c) 7.35 (d) 12.07. F
3. Write 6 045 in expanded form in two different ways. F
4. Insert the missing digit: 5 _ 2 is between 5 300 and 5 400. R
5. A student says 0.7 is larger than 0.65 because 7 > 65. Explain the error and give the correct comparison. R
6. A museum ticket costs 12.50. Estimate the cost of 18 tickets, then calculate it exactly and compare the two results. P
7. A crate is labelled 2.4 tonnes. Write this in kilograms and state which position the decimal point moved to. P
8. Find two different three-digit numbers that use 4, 0 and 9 exactly once each. Which is larger, and why? E
9. Explain why “three thousandths” is written 0.003 rather than 0.300. E

1.3 Expanded form and equivalent representations

CRITERION A · i, ii ◆ REPRESENTATION ◆ FORM

Expanded form unpacks a number into the sum of its column values. It is not a different number, only a different way of writing the same

one, and it makes the structure visible: every zero in the original disappears into a multiplication by zero, and every digit shows exactly what it contributes.

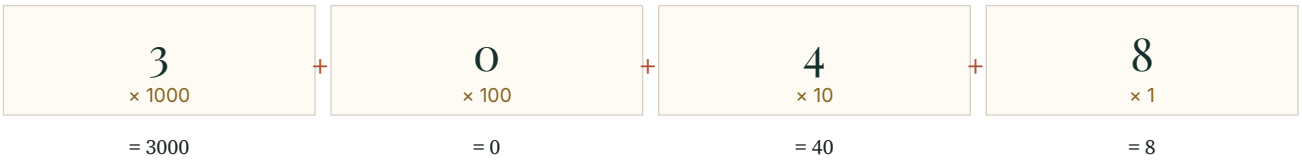


FIGURE 4 3 048 written as the sum of its column values. Each column contributes digit × value.

Three equivalent representations of the same quantity:

- Words: three thousand and forty-eight.
- Digits: 3 048.
- Expanded: $3 \times 1000 + 0 \times 100 + 4 \times 10 + 8 \times 1$, or $3\,000 + 40 + 8$.

DEFINITION

Equivalent representations

Two representations are equivalent when they name exactly the same value. 3 048, $3\,000 + 40 + 8$, 3 048.0 and "three thousand and forty-eight" are all equivalent; 3 480 is not, however similar it looks.

WORKED EXAMPLE 1.3.1 FROM EXPANDED FORM BACK TO DIGITS

PROBLEM

Write $5 \times 10\,000 + 0 \times 1000 + 2 \times 100 + 9 \times 1 + 3 \times 0.01$ as a single number.

1. List the columns the expression uses: ten-thousands, thousands, hundreds, ones, hundredths.
2. The thousands column contributes nothing, so it must be held open by a zero.
3. The tenths column is empty too, so it also needs a zero.
4. Assemble the digits in order: 5 0 2 0 9 . 0 3.

ANSWER 50 209.03

Read back: fifty thousand, two hundred and nine, and three hundredths ✓

WORKED EXAMPLE 1.3.2 SPOTTING A FALSE EQUIVALENCE

PROBLEM

A student claims $4 \times 100 + 7 \times 10 + 3 \times 0.1 = 473$. Is the claim correct?

1. Evaluate each term: 400, 70 and 0.3.
2. Add: $400 + 70 + 0.3 = 470.3$.
3. 473 would need 3×1, not 3×0.1.

ANSWER The claim is false. The correct value is 470.3.

Estimate: the answer must lie between 470 and 471 √

PRACTICE 1.3 SUPPORT 1–3 · CORE 3–7 · EXTENSION 6–8

1. Write in expanded form: (a) 8 502 (b) 7.04 (c) 60.5. F
2. Write as a single number: (a) $9\,000 + 300 + 7$ (b) $2 \times 10 + 5 \times 0.1$. F
3. Write 30 407 in words and in expanded form. F
4. Which is larger, $6 \times 100 + 4 \times 0.1$ or 604? Explain. R
5. Find the mistake: $2\,030 = 2 \times 1000 + 3 \times 10$. Correct it. R
6. A receipt reads $12 \times 100 + 5 \times 1 + 9 \times 0.01$. Write the amount in currency and explain why a zero is needed after the decimal point. P
7. Write a four-digit number and a four-decimal-place number that use the digits 1, 2, 3, 4 once each and have the smallest possible value. E
8. Explain why expanded form makes rounding easier, using 4 876 as an example. E

1.4 Comparing, ordering and rounding

CRITERION A · ii, iii ◆ QUANTITY ◆ FORM

Comparing decimals by digit length is the most common place-value error. 0.65 is not larger than 0.7 merely because it has more digits: 0.65 means six tenths and five hundredths; 0.7 means seven tenths. Line the decimal points up, fill empty columns with zeros, and compare like with like.

KEY IDEA

A reliable comparison routine

1. Write both numbers with the decimal points aligned, filling empty columns with zeros: 0.70 and 0.65. 2. Compare the columns from left to right. 3. Stop at the first difference — that column decides. Here 7 > 6 in the tenths column, so 0.70 > 0.65.

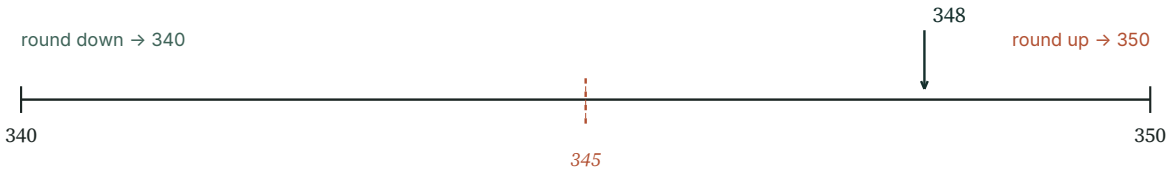


FIGURE 5 Rounding 348 to the nearest ten. The number sits between 340 and 350; the midpoint 345 decides which way it goes.

DEFINITION

Rounding

To round to a stated place, find the digit in that place and look at the digit immediately to its right. If that digit is 5 or more, increase the place digit by 1; otherwise leave it. Replace every digit to the right of the place with zeros (or drop them after the decimal point).

WORKED EXAMPLE 1.4.1 — ROUNDING TO A NAMED PLACE

PROBLEM

Round 3 847.265 to (a) the nearest thousand (b) the nearest ten (c) two decimal places.

1. (a) The thousands digit is 3; the next digit is 8, so round up: 4 000.
2. (b) The tens digit is 4; the next digit is 7, so round up: 3 850.
3. (c) Keep two decimal places; the third digit is 5, so round up: 3 847.27.

ANSWER (a) 4 000 (b) 3 850 (c) 3 847.27

Each answer is closer to the original than the neighbouring rounded value ✓

WORKED EXAMPLE 1.4.2 — ORDERING A MIXED SET

PROBLEM

Write 0.7, 0.07, 0.707 and 0.77 in order from least to greatest.

1. Give every number three decimal places: 0.700, 0.070, 0.707, 0.770.
2. Compare tenths: 0 is smallest, so 0.070 comes first.
3. The remaining three all have 7 tenths; compare hundredths: 0, 0 and 7, so 0.700 and 0.707 are next.
4. Compare thousandths of those two: 0 < 7, then 0.770 is largest.

ANSWER $0.07 < 0.7 < 0.707 < 0.77$

Four numbers, four positions, no repeats ✓

PRACTICE 1.4 — SUPPORT 1–5 · CORE 4–8 · EXTENSION 8–9

1. Insert < or >: (a) 0.6 … 0.58 (b) 2.07 … 2.7 (c) −3.5 … −3.05. F
2. Round to the nearest whole number: (a) 7.4 (b) 12.8 (c) 0.62. F
3. Round 45 678 to the nearest ten, hundred and thousand. F
4. Order from least to greatest: 3.4, 3.04, 3.44, 3.404. F
5. Round 9.996 to two decimal places. Explain why the whole-number part changes. R
6. A runner's time is recorded as 12.36 s. Round it to one decimal place and explain what information is lost. P
7. Give a number that rounds to 5.6 to one decimal place but not to 5.60 to two decimal places. R
8. The population of a town is 48 512. Write two different rounded versions and state a context in which each would be appropriate. P
9. Is it possible for a number to round to 100 and also to 0.1? Explain. E

1.5 Estimation and sensible accuracy

CRITERION A · iii · D · i ◆ QUANTITY ◆ APPROXIMATION

An estimate is not a guess. It is a deliberately simpler calculation whose answer is close enough to be useful and rough enough to be done in your head. Estimates do real work: they tell you whether a calculator answer is plausible before you trust it.

KEY IDEA

Round, calculate, adjust

Round each number to one significant figure, do the simplified calculation, then adjust for the direction of your rounding. $48 \times 21 \approx 50 \times 20 = 1\,000$. Because both numbers were rounded up, the exact answer must be a little below 1 000: 1 008.

WORKED EXAMPLE 1.5.1 ESTIMATING BEFORE CALCULATING

PROBLEM

A school orders 19 boxes of exercise books. Each box holds 48 books. Estimate the total, then calculate it exactly.

1. Round: $19 \approx 20$ and $48 \approx 50$.
2. Estimate: $20 \times 50 = 1\,000$ books.
3. Exact: $19 \times 48 = 19 \times (50 - 2) = 950 - 38 = 912$.
4. The exact answer is a little below the estimate, as predicted.

ANSWER Estimate 1 000; exact 912.

912 is within 10 % of 1 000 ✓

COMMON ERROR

Reporting more precision than you have

A student measures a desk as 1.2 m and reports its area as 1.44 m^2. The measurement is accurate to the nearest 0.1 m, so the area is only meaningful to about one decimal place: 1.4 m^2. Extra digits are not extra accuracy; they are a claim you cannot support.

Choosing a sensible level of accuracy

- Money is usually reported to the nearest 0.01, because that is the smallest unit in use.
- Populations and attendance are usually reported to the nearest ten, hundred or thousand.
- Measurements should be reported to the precision of the instrument, not beyond it.
- Intermediate calculations should keep more digits than the final answer, so that rounding errors do not accumulate.

PRACTICE 1.5 SUPPORT 1–3 · CORE 3–7 · EXTENSION 6–8

1. Estimate: (a) 29×41 (b) $612 \div 19$ (c) $98 + 203 + 397$. F
2. Estimate 4.9×20.2 and then calculate it exactly. F

PRACTICE 1.5 continued

SUPPORT 1–3 · CORE 3–7 · EXTENSION 6–8

1. A packet holds 24 pens. Estimate the number of packets needed for 500 pens. F
2. Explain why $3\,987 \times 0.51$ is roughly 2 000, without calculating it. R
3. A student calculates $5.2 \times 3.1 = 161.2$. Use an estimate to show the answer is wrong and give the correct value. R
4. A community noticeboard reports attendances of 247, 1 083 and 96. Round each to a sensible accuracy and explain your choices. P
5. A tank holds 1 250 litres. A pump delivers 18 litres a minute. Estimate the time to fill it, then calculate it to the nearest minute. P
6. Describe a situation in which rounding a number makes it less useful, and explain why. E
7. Estimate (a) 8.1×4.9 (b) $3\,950 \div 19$ (c) 0.48×61. F
8. A school orders 312 books at 4.95 each. Estimate the total cost, then say whether a calculator answer of 1 544.40 is plausible. R
9. Round 46 738 to the nearest (a) ten (b) hundred (c) thousand, and state which is most useful for a newspaper headline. R
10. A car travels 148 km in 2.4 hours. Estimate the average speed, then calculate it to one decimal place. P

05 · INVESTIGATE

Count the possibilities, then generalise.

INVESTIGATION 1

How many numbers can a given number of digits write?

FORM ◆ QUANTITY

Work in pairs. You will count possibilities systematically and look for a pattern rather than listing every case. A whole number may not begin with zero, because 007 is simply 7.

Digits used	No leading zero	Leading zero allowed	Difference
1	9	10	1
2	90	100	10
3	900	1 000	100
4	9 000	10 000	1 000

TABLE 4 Counts of whole numbers that can be written with exactly this many digits.

Your investigation

1. Check the first two rows by listing the numbers directly.
2. Describe in words how each row is obtained from the row above it.
3. Predict the seven-digit entries before you calculate them.
4. Write a rule for the count when n digits are allowed, both with and without a leading zero.
5. Explain why the two counts differ by exactly 10^{n-1}.
6. Test your rule for $n = 1$ and $n = 2$, then state whether it holds for every whole number n.

KEY IDEA

Conjecture to test

With zero allowed there are 10^n strings of n digits. Without a leading zero there are $9 \times 10^{n-1}$. The difference, 10^{n-1}, is exactly the number of strings that begin with zero — the same count for $n - 1$ digits.

ATL SKILL

ATL · Transfer

In order for a student to generalise a counting pattern, students must move from listing cases to describing the operation that produces each case, and then test the description against an unfamiliar case.

06 · APPLY

A task that leaves the page and goes into a community.

REAL-LIFE APPLICATION

Numbers for a community noticeboard

CRITERION D · i–iv ◆ QUANTITY ◆ REPRESENTATION

A neighbourhood association is putting up a noticeboard. It has to report four things: the number of households registered, the total collected for a repair fund, the cost of one repair and the percentage of the target reached. The board will be read from several metres away, by people in a hurry.

- Collected: 486 350 units of currency. Target: 600 000.
- Households registered: 1 284. Repairs completed: 37 of 45.
- One repair costs 12 500.

Your task

1. Decide the accuracy at which each figure should be shown, and justify each choice in one sentence.
2. Rewrite each figure at that accuracy without losing its meaning.
3. Compute the percentage of the target reached, to a sensible accuracy.
4. Explain why showing 486 350 next to a rounded target of 600 000 would be inconsistent.
5. Draft the four figures as they would appear on the board, then ask someone outside your class to read them back to you.

CRITERION

Criterion D · strands i–iv

- Identify the relevant elements of the real-life situation.
- Select and apply appropriate mathematical strategies.
- Justify the degree of accuracy chosen.
- Evaluate whether the board communicates what it intended.

07 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The noticeboard card

CRITERION C · ii, iii · D · ii

Context. The association wants a single card, no larger than A5, that explains place value to a reader who has never thought about it. It will be pinned beside the noticeboard.
Product. A card with no more than 120 words of explanation, one diagram and one worked example, readable from about one metre.

1. State what place value is in one sentence, using 3 048 as your example.
2. Show 3 048 in expanded form and explain what the zero is doing.
3. Include a diagram that makes the columns visible.
4. Work one example of rounding, naming the place each time.
5. Explain, in no more than two sentences, why 0.7 is larger than 0.65.
6. Ask a classmate to use the card to round 27 486 to the nearest hundred. Record whether they succeed, then revise one part of the card.
7. Evaluate the card: one strength, one limitation and one change for a younger reader.

CRITERION

Criteria and strands developed

- Criterion A · Knowing and understanding — ii, iii: accurate place-value statements and correct expanded form.
- Criterion C · Communicating — ii, iii: precise language, notation, diagram and audience awareness.
- Criterion D · Applying mathematics in real-life contexts — ii: applying place value to a real communication problem.

08 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Numbers belong to nested families; each extension of the family makes a new kind of question answerable.
- In base ten, a digit's value is the digit multiplied by a power of ten determined by its position.
- Zero holds a column open and is therefore essential, not decorative.
- Expanded form is an equivalent representation that exposes structure.
- Rounding replaces a value with a nearby value at a stated precision, and the choice of precision is a mathematical decision, not a presentation one.

Vocabulary

Term	Meaning
natural number	A counting number: 1, 2, 3, …
whole number	A natural number or zero.
integer	A whole number or its negative.
rational number	A number that can be written as a fraction of two integers, with a non-zero denominator.
place value	The value a digit carries because of its position.
expanded form	A number written as the sum of its column values.
rounding	Replacing a number with a nearby value at a stated precision.
estimate	A deliberately simplified calculation used to judge size or reasonableness.

Misconceptions to watch

COMMON ERROR

Three that survive into later years

Longer decimal, larger number. 0.65 is not larger than 0.7. Align columns and compare like with like.

Zero is nothing. A zero in the middle of a number does essential work; deleting it changes the value.

More digits, more accuracy. Digits beyond the precision of the measurement are a claim, not a fact.

Self-assessment

I can …	Not yet	Getting there	Confident
classify numbers as natural, whole or integer	☐	☐	☐
state the value of any digit in a decimal	☐	☐	☐
write and reverse expanded form	☐	☐	☐
order decimals correctly	☐	☐	☐
round to a named place and justify it	☐	☐	☐
estimate before calculating	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to know which habit still needs work, students must mark the table honestly and then attempt one question from the weakest row before reading the answer.

Looking ahead · Chapter 2 · Powers, expanded form and order of operations takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

09 · MIXED PRACTICE

PRACTICE Mixed practice 1 SUPPORT 1–5 · CORE 4–12 · EXTENSION 10–15

1. Write 8 007 040 in words. F
2. Write the value of the digit 6 in 3.062. F
3. Round 6.478 to one decimal place and to the nearest whole number. F
4. Insert < or >: (a) 0.08 ... 0.1 (b) −1.4 ... −1.35. F
5. Write $7 \times 1000 + 0 \times 100 + 9 \times 10 + 2 \times 1 + 5 \times 0.001$ as a single number. F
6. A student writes 2.5 for 2.05. Explain what has gone wrong and give the correct statement. R
7. Order from least to greatest: −0.5, 0.05, −0.05, 0.5, 0. R
8. Estimate 38×51 and then compute it. By what percentage does your estimate differ? P
9. A shop sells rice at 6.75 per kilogram. Estimate the cost of 14 kg and calculate the exact price. P
10. The distance to a village is quoted as “about 12 km”. What range of actual distances is consistent with this statement if the quote is to the nearest kilometre? R
11. Explain why —0.0” and “0” can be written differently but represent the same value. E
12. Write the largest and smallest numbers you can make with the digits 3, 0, 7, 5 and a decimal point, using each digit once. E
13. A pump delivers 0.75 litres per second. Estimate how long it takes to deliver 1 000 litres, then calculate it to a sensible accuracy. P
14. Show that 4 096 and 4 096.000 are equivalent, then explain what the extra digits might legitimately indicate. E
15. Design a five-question place-value diagnostic for a younger student. Include the answers and say what each question tests. E

10 · CHECK YOUR PROGRESS

Question	If you cannot do this ...
1–3	Revisit 1.2 and 1.4: read each digit's column aloud before writing.
4–6	Revisit 1.4: align the decimal points and fill empty columns.
7–9	Revisit 1.5: round first, calculate second, adjust third.
10–12	Revisit 1.3: expanded form exposes what rounding removes.
13–15	Revisit the chapter review; write your own version of each summary point.

2

NUMBER, STRUCTURE AND PROOF

Powers, expanded form and order of operations

A power is a compressed instruction: it records how many copies of a factor to multiply. A shared order of operations keeps that compression unambiguous.

KEY CONCEPT · FORM

RELATED · PATTERN

RELATED · SIMPLIFICATION

CONTEXT · SCIENTIFIC AND TECHNICAL INNOVATION

INQUIRY

How does a repeated multiplication become a single symbol, and what rules let everyone read that symbol the same way?

01 · MYP UNIT PLANNER

Concepts, context and the evidence the chapter will produce.

Statement of inquiry	Compressing repeated multiplication into a power, and agreeing on the order in which operations apply, allows complex calculations to be recorded compactly and reproduced reliably by anyone.

KEY IDEA

Key and related concepts

Key concept · Form. A power is a form that hides its own history: 2^5 does not show the five factors, but they are what the symbol means.

Related · Pattern. Powers of ten generate the pattern behind place value and behind scientific notation.

Related · Simplification. An agreed order of operations is what makes a compressed expression mean one thing rather than several.

NOTE

Global context · Scientific and technical innovation

Exploration: how measurement and computing depend on notation that compresses very large and very small values without losing them. A scientist writing 3×10^8 m/s is using the same structure as a student writing 10^3.

Inquiry questions

- Factual: What do the base and the exponent tell you, and how do you evaluate a power?
- Conceptual: Why do we need an agreed order of operations, and what would change if there were none?
- Debatable: Is “BIDMAS” a mathematical rule or a convention that could reasonably have been decided differently?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Evaluate powers, roots and multi-step expressions accurately.
B · Investigating patterns	i, ii, iii	Investigate the pattern in powers of ten and in the last digit of powers.
C · Communicating	i, ii, iii, iv	Record a calculation with one line per operation and name the rule used.
D · Applying mathematics	i, ii, iii, iv	Use powers to describe storage, growth and distance in real contexts.

ATL SKILL

ATL · Thinking skills

In order for a student to evaluate a complex expression without losing track, students must annotate each line with the rule being applied, so that the working can be audited by someone else.

02 · ENGAGE

A warehouse where the counting never stops.

A grain warehouse receives sacks in pallets. A pallet holds 10 rows of 10 sacks, so it holds 100. A block of pallets is 10 pallets long, 10 wide and 10 high: 1 000 pallets, or 100 000 sacks. The storekeeper does not write 100 000. She writes 10^5, and everyone in the warehouse reads the same number.



FIGURE 1 Stacking makes powers visible. A 10 × 10 layer is 10^2; ten such layers make 10^3; the warehouse simply keeps multiplying by ten, and the notation keeps up.

Original illustration created for this edition.

03 · EXPLORE

Build the pattern before naming it.

Copy and complete this table without a calculator. Write each product as a single number, then look for what changes between rows.

Power	Meaning	Value
10^1	10	10
10^2	10 × 10	100
10^3	10 × 10 × 10	1 000
10^4	…	…
10^5	…	…
10^6	…	…

1. How many zeros does 10^n have, in terms of n?
2. Predict 10^9 without calculating it.
3. Now complete the same table for powers of 2: 2^1 to 2^6. Does the same kind of pattern hold? What is different?
4. What would 10^0 have to be for the pattern to continue downward?

KEY IDEA

Two patterns, one structure

Powers of ten grow by adding one zero per step, so the exponent counts the zeros. Powers of two double at each step, so the exponent counts the doublings. In both cases the exponent counts how many copies of the base are multiplied — that is the definition, and every pattern follows from it.

2.1 From repeated multiplication to powers

CRITERION A · i ◆ FORM ◆ PATTERN

A power is not a new kind of number. It is a shorter way of writing a multiplication in which every factor is the same.

DEFINITION

Base, exponent, power

In a^n, the base a is the number being multiplied and the exponent n is the number of copies. So $a^n = a \times a \times \ldots \times a$, with n factors. The whole expression is called a power of a.

Read 3^4 as “three to the power four” or “three to the fourth”. It means $3 \times 3 \times 3 \times 3 = 81$. Read 3^4 carefully: it is not 3×4. The exponent counts factors, not additions.

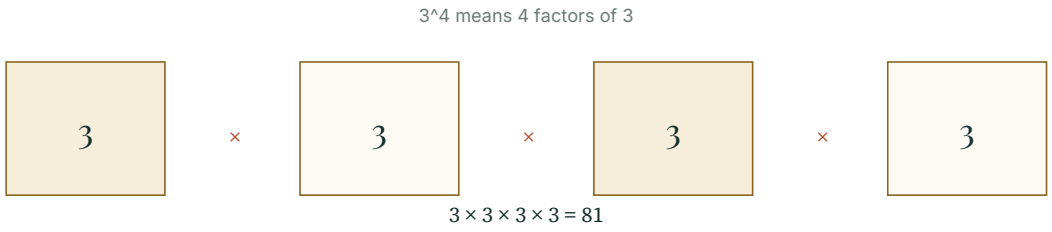


FIGURE 2 3^4 as four factors of 3. The exponent counts the factors, not their total.

COMMON ERROR

Multiplying the base by the exponent

$3^4 = 81$, not 12. A useful check: $3^2 = 9$ and $3^3 = 27$, so 3^4 must be larger again. If your answer is only 12, you have multiplied instead of repeated the multiplication.

The two special exponents

- Exponent 1 means one factor: $a^1 = a$. There is nothing to repeat.
- Exponent 0 means no factors at all, and the value is 1 for every non-zero base. This is not arbitrary: $10^3 \div 10 = 10^2$, $10^2 \div 10 = 10^1$, and continuing the pattern forces $10^1 \div 10 = 10^0 = 1$.
- Exponent 2 is read "squared" and exponent 3 is read "cubed", for reasons that become visible in Chapter 9.

WORKED EXAMPLE 2.1.1 — WRITING A PRODUCT AS A POWER

PROBLEM

Write each product as a single power and evaluate it: (a) $5 \times 5 \times 5$ (b) $2 \times 2 \times 2 \times 2 \times 2 \times 2$ (c) 7.

1. (a) Three factors of 5, so $5^3 = 125$.
2. (b) Six factors of 2, so $2^6 = 64$.
3. (c) One factor of 7, so $7^1 = 7$.

ANSWER (a) $5^3 = 125$ (b) $2^6 = 64$ (c) $7^1 = 7$

$2^5 = 32$, so $2^6 = 64$ ✓

WORKED EXAMPLE 2.1.2 — READING A POWER IN CONTEXT

PROBLEM

A sheet of paper is folded in half six times. How many layers are there, and how does the answer connect to a power?

1. Each fold doubles the number of layers.
2. After 1 fold: 2 layers = 2^1. After 2 folds: $4 = 2^2$. After 3 folds: $8 = 2^3$.
3. After 6 folds: $2^6 = 64$ layers.
4. The exponent equals the number of folds, because each fold is one more doubling.

ANSWER 64 layers, which is 2^6.

After 5 folds there are 32; $32 \times 2 = 64$ ✓

PRACTICE 2.1 — SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Write as a power: (a) $4 \times 4 \times 4$ (b) 9×9 (c) 6. F
2. Evaluate: (a) 2^3 (b) 5^2 (c) 10^4 (d) 3^0. F
3. Evaluate: (a) 7^2 (b) 2^7 (c) 4^3. F
4. Write 81 as a power of 3, and also as a power of 9. F
5. A student writes $4^3 = 12$. Identify the error and give the correct value. R
6. Explain why $10^0 = 1$ using the pattern of dividing by 10. R

PRACTICE 2.1 continued SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. A rumour is passed to 2 people, each of whom tells 2 more, and so on. How many people hear it at the fifth telling? Write your answer as a power and as a number. P
2. Which is larger, 2^{10} or 10^2? Explain without calculating both fully. E
3. Find all the powers of 2 between 30 and 100. E

2.2 Powers of ten and expanded form

CRITERION A · ii · B · i ◆ PATTERN ◆ REPRESENTATION

Place value is a system built on powers of ten. Every column is ten times the column on its right, so expanded form can be written with a power in every term, which is exactly how large numbers are handled in science.

DEFINITION

Expanded form with powers

A number in base ten can be written as a sum of digits multiplied by powers of ten: $4\,507 = 4 \times 10^3 + 5 \times 10^2 + 0 \times 10^1 + 7 \times 10^0$. To the right of the point the exponents are negative: $0.25 = 2 \times 10^{-1} + 5 \times 10^{-2}$.

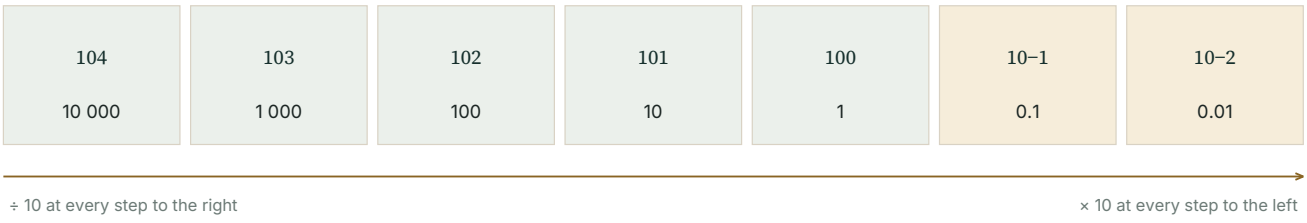


FIGURE 3 Moving left multiplies by ten; moving right divides by ten. The exponent records how many moves.

WORKED EXAMPLE 2.2.1 WRITING A LARGE NUMBER WITH POWERS

PROBLEM

Write 6 030 000 in expanded form using powers of ten, then as a single digit multiplied by a power of ten.

1. Count the columns: 6 in the millions, 0 in hundred-thousands, 3 in ten-thousands, then five zeros.
2. Expanded: $6 \times 10^6 + 0 \times 10^5 + 3 \times 10^4$.
3. Because everything after the 3 is zero, the number is 603×10^4.
4. The most compact form keeps one digit before the point: 6.03×10^6.

ANSWER $6 \times 10^6 + 3 \times 10^4$, which equals 6.03×10^6.

$6.03 \times 1\,000\,000 = 6\,030\,000$ ✓

WORKED EXAMPLE 2.2.2 COMPARING BY EXPONENT

PROBLEM

Which is larger, 4.1×10^5 or 3.9×10^6?

1. $4.1 \times 10^5 = 410\,000$.
2. $3.9 \times 10^6 = 3\,900\,000$.
3. The second is larger: a higher power of ten outweighs a smaller leading digit.

ANSWER 3.9×10^6 is larger, by a factor of nearly ten.

Compare exponents first; only compare leading digits when the exponents agree ✓

PRACTICE 2.2

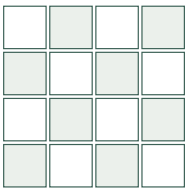
SUPPORT 1–3 · CORE 3–7 · EXTENSION 6–8

1. Write in expanded form using powers of ten: (a) 3 400 (b) 70 205 (c) 8 000 000. F
2. Write as a single number: (a) $5 \times 10^3 + 2 \times 10^2$ (b) 9×10^5. F
3. Write 45 000 as a single digit multiplied by a power of ten. F
4. Which is larger, 6.2×10^4 or 5.8×10^5? Explain your reasoning. R
5. A student writes $3\,000 = 3 \times 10^4$. Explain the error using a place-value argument. R
6. The distance from Earth to the Sun is about 1.5×10^8 km. Write this as an ordinary number and state how many zeros it has. P
7. Write the year 2026 in expanded form using powers of ten, then explain why the 10^0 term matters. E
8. Create a number whose expanded form contains exactly three non-zero terms and whose value is between 500 000 and 600 000. E

2.3 Squares, cubes and roots

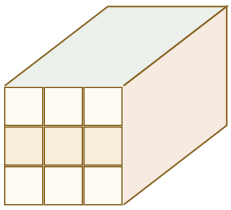
CRITERION A · i, ii ◆ FORM ◆ QUANTITY

A square number is a number multiplied by itself. The name is not decorative: 16 counters can be arranged in a 4 by 4 square, and no other square arrangement uses exactly 16. A cube number fills a cube in the same way.



4^{2} = 16

a square of 16 counters



3^{3} = 27

a cube of 27 small cubes

Square numbers count a flat square.

Cube numbers count a solid cube.

FIGURE 4 Square numbers are the counters in a square; cube numbers are the cubes in a cube. $5^2 = 25$ and $3^3 = 27$.

DEFINITION

Square and cube roots

The square root of n, written $\sqrt{n}$, is the positive number that multiplies by itself to give n. The cube root, written $\sqrt[3]{n}$, is the number that multiplies by itself twice to give n. So $\sqrt{49} = 7$ because $7 \times 7 = 49$, and $\sqrt[3]{27} = 3$ because $3 \times 3 \times 3 = 27$.

Facts worth knowing by heart

Square numbers	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144
Cube numbers	1, 8, 27, 64, 125, 216, 343, 512, 729, 1 000

KEY IDEA

Why the square root is always positive

Both 7×7 and $(-7) \times (-7)$ equal 49, but the root sign $\sqrt{}$ is defined to give the positive value only. In Year 1 you will meet $\sqrt{49} = 7$; the negative solution belongs to equation solving in later years.

WORKED EXAMPLE 2.3.1 SQUARES, CUBES AND ROOTS TOGETHER

PROBLEM

Evaluate (a) 6^2 (b) 6^3 (c) $\sqrt{81}$ (d) $\sqrt[3]{64}$.

1. (a) $6 \times 6 = 36$.
2. (b) $6 \times 6 \times 6 = 216$.
3. (c) $9 \times 9 = 81$, so $\sqrt{81} = 9$.
4. (d) $4 \times 4 \times 4 = 64$, so $\sqrt[3]{64} = 4$.

ANSWER 36, 216, 9 and 4.

$\sqrt{81} = 9$ and $9^2 = 81$; $\sqrt[3]{64} = 4$ and $4^3 = 64$ ✓

WORKED EXAMPLE 2.3.2 A SQUARE WITH A KNOWN AREA

PROBLEM

A square courtyard has an area of 144 m^2. Find the length of one side and the perimeter.

1. For a square, area = side^2, so side = $\sqrt{144}$.
2. $\sqrt{144} = 12$, so each side is 12 m.
3. Perimeter = $4 \times 12 = 48$ m.

ANSWER Side 12 m; perimeter 48 m.

$12^2 = 144$ ✓ and the perimeter has four equal sides ✓

PRACTICE 2.3

SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Evaluate: (a) 9^2 (b) 11^2 (c) 2^3 (d) 5^3. F
2. Evaluate: (a) $\sqrt{64}$ (b) $\sqrt{121}$ (c) $\sqrt[3]{125}$. F
3. Which of these are square numbers: 48, 64, 81, 90, 100? F
4. Find two consecutive square numbers whose difference is 15. R
5. Explain why 30 cannot be a square number, without using a calculator. R
6. A square tile has an area of 196 cm^2. Find its side length and its perimeter. P
7. A cubic tank holds 512 litres. Find its edge length in centimetres, given that 1 litre = 1 000 cm^3. P
8. Is the square of an even number always even? Test three cases and explain the pattern. E
9. Find the smallest whole number that must be added to 50 to make a square number, and the smallest that must be subtracted. E

2.4 Order of operations

CRITERION A · iii · C · i ◆ SYSTEM ◆ SIMPLIFICATION

Consider 2 + 3 × 4. Read left to right, it gives 20. Read with multiplication first, it gives 14. Both readings are possible in English; only one is mathematics. The convention exists because an expression written once has to mean one thing to everybody who reads it.

DEFINITION

The agreed order

1 Brackets and other grouping symbols. 2 Indices (powers and roots). 3 Multiplication and division, from left to right. 4 Addition and subtraction, from left to right. An expression with brackets inside brackets is worked from the innermost pair outward.

KEY IDEA

Multiplication and division have equal priority

24 ÷ 4 × 2 is 12, not 3, because division and multiplication rank equally and are worked left to right. The rule is not "division before multiplication"; it is "whichever comes first, working left to right".

WORKED EXAMPLE 2.4.1 — TWO OPERATIONS

PROBLEM

Evaluate $5 + 2 \times 3^2$.

1. Indices first: $3^2 = 9$, so the expression becomes $5 + 2 \times 9$.
2. Multiplication next: $2 \times 9 = 18$, giving $5 + 18$.
3. Addition last: $5 + 18 = 23$.

ANSWER 23

Estimate: 3^2 is 9 and $2 \times 9 = 18$, so the total is about 23 √

WORKED EXAMPLE 2.4.2 — BRACKETS CHANGE THE STRUCTURE

PROBLEM

Evaluate $(5 + 2) \times 3^2$ and explain why it differs from Worked example 2.4.1.

1. Brackets first: $5 + 2 = 7$.
2. Indices: $3^2 = 9$, so the expression is 7×9.
3. $7 \times 9 = 63$.
4. In 2.4.1 the 5 was added to the product; here the 5 is added to the 2 before multiplying, so the 5 is multiplied by 9 as well.

ANSWER 63, compared with 23 in 2.4.1.

$63 - 23 = 40 = 5 \times (9 - 1)$: the bracket multiplied the 5 by 9 rather than by 1 √

WORKED EXAMPLE 2.4.3 — A LONGER CHAIN

PROBLEM

Evaluate $36 \div 6 + 3 \times (8 - 2)^2$.

1. Innermost bracket: $8 - 2 = 6$.
2. Index: $6^2 = 36$.
3. Multiplication and division, left to right: $36 \div 6 = 6$ and $3 \times 36 = 108$.
4. Addition: $6 + 108 = 114$.

ANSWER 114

Reverse the order of the two middle steps and check that the answer is unchanged, because the two operations are independent √

COMMON ERROR

Working strictly left to right

A student evaluates $4 + 6 \div 2$ as 5 by adding first. The correct value is 7, because the division happens before the addition. If you want the addition first, you must write $(4 + 6) \div 2$, which is 5. The brackets are the instruction.

PRACTICE 2.4 — SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate $8 + 2 \times 5$. F
2. Evaluate $20 - 3 \times 4$. F
3. Evaluate: (a) $6 + 4^2$ (b) $(6 + 4)^2$ (c) $2^3 + 3^2$. F
4. Evaluate: (a) $24 \div 4 \times 2$ (b) $24 \div (4 \times 2)$ (c) $36 \div 6 + 3$. F
5. Evaluate $5 \times 2^3 - 3^2$, showing each stage of your working. F

PRACTICE 2.4 continued SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Insert brackets to make each statement true: (a) $3 + 5 \times 2 = 16$ (b) $12 - 4 \times 2 = 16$ (c) $18 \div 3 + 6 = 2$. R
2. A student calculates $12 + 6 \div 3$ as $18 \div 3 = 6$. Identify the mistake, give the correct value, and explain which operation should be performed first. R
3. Use the digits 2, 3 and 4 once each, with any operations and brackets, to make as many different values as you can. Record at least four expressions and their values; one expression must contain an index. E

2.5 Substitution and expressions with powers

CRITERION A · iii ◆ FORM ◆ MODEL

A letter in an expression stands for a number. Substitution means replacing the letter with a value and evaluating the resulting expression using the order of operations. The position of the letter inside a power matters.

WORKED EXAMPLE 2.5.1 SAME SUBSTITUTION, DIFFERENT STRUCTURE

PROBLEM

Evaluate $3x^2$ and $(3x)^2$ when $x = 4$, and explain why the results differ.

1. In $3x^2$ the exponent applies only to x: $3x^2 = 3 \times (4)^2 = 3 \times 16 = 48$.
2. In $(3x)^2$ the brackets make $3x$ the base: $(3 \times 4)^2 = 12^2 = 144$.
3. The bracket changes the base of the power. Without it, only x is squared. The second result is three times the first because $(3x)^2 = 3^2 \times x^2 = 9x^2$.

ANSWER $3x^2 = 48$ and $(3x)^2 = 144$.

$48 \times 3 = 144$ ✓

WORKED EXAMPLE 2.5.2 EVALUATING FORMULAS

PROBLEM

(a) The area of a square is $A = s^2$. Find A when $s = 7$ cm. (b) The volume of a cube is $V = s^3$. Find V when $s = 4$ cm. (c) A model for the distance fallen by an object is $d = 5t^2$, with d in metres and t in seconds. Find d when $t = 3$.

1. (a) Substitute $s = 7$: $A = 7^2 = 49$ cm^2.
2. (b) Substitute $s = 4$: $V = 4^3 = 64$ cm^3.
3. (c) Substitute $t = 3$: $d = 5 \times 3^2 = 5 \times 9 = 45$ m.

ANSWER (a) 49 cm^2 (b) 64 cm^3 (c) 45 m

In (c), the index is applied before the multiplication by 5 √

NOTE

Why substitution is a test of structure

Two expressions that look similar can behave differently once a value is inserted. Comparing $3x^2$ with $(3x)^2$, or x^2 with $2x$, shows exactly where the exponent applies — which is the same question you will ask of every formula in Chapters 6 and 9.

PRACTICE 2.5 SUPPORT 1–3 · CORE 1–6 · EXTENSION 7–8

1. Evaluate x^2 for (a) $x = 3$ (b) $x = 6$ (c) $x = 10$. F
2. Evaluate $2x^2$ for (a) $x = 3$ (b) $x = 6$ (c) $x = 10$. Compare each result with the corresponding result in question 1. F
3. For $a = 2$ and $b = 5$, evaluate $a^2 + b^2$ and $(a + b)^2$. Are they equal? F
4. A square has side length 9 cm. Use $A = s^2$ to find its area, with units. P
5. Use $V = s^3$ to find the volume of a cube with edge (a) 3 cm (b) 5 cm. F
6. The distance-fallen model is $d = 5t^2$. Find d when (a) $t = 2$ (b) $t = 4$. Explain why doubling t does not double d. R
7. Insert one pair of brackets to make each statement true: (a) $2 + 3^2 = 25$ (b) $4 \times 3^2 = 144$. Explain how the brackets change the base. R
8. A square has area 121 cm^2. Find its side length. If the side length is tripled, find the new area and explain how 3^2 appears in your answer. E

04 · INVESTIGATE

A pattern that hides in the last digit.

INVESTIGATION 2

The doubling sheet

PATTERN ◆ FORM

Fold a sheet of paper in half repeatedly. Each fold doubles the number of layers. Record the layer count and the thickness, then look for structure rather than arithmetic.

Folds	Layers	As a power of 2	Thickness if each sheet is 0.1 mm
0	1	2^0	0.1 mm
1	2	2^1	0.2 mm
2	4	2^2	0.4 mm
3	8	2^3	0.8 mm
4	16	2^4	1.6 mm
…	…	…	…

Your investigation

1. Continue the table to 10 folds.
2. Describe how the thickness changes from row to row, in one sentence.
3. Predict the thickness after 20 folds, then calculate it.
4. Write a rule for the thickness after n folds, starting from 0.1 mm.
5. A ream of 500 sheets is about 5 cm thick. After how many folds does the stack exceed the height of your classroom (about 3 m)?
6. Explain why the answer to the last question surprises most people.

KEY IDEA

What the investigation shows

Doubling is a power of 2, and powers grow far faster than most people expect. After 20 folds the stack is $2^{20} \times 0.1$ mm $\approx$ 105 m. The pattern is exponential: each step multiplies rather than adds, so the graph curves upward rather than rising in a straight line.

ATL SKILL

ATL · Critical thinking

In order for a student to trust a surprising result, students must check it against a smaller case where the answer can be counted directly, and then confirm that the same operation produces both.

05 · APPLY

Storage, growth and distance in a real setting.

REAL-LIFE APPLICATION

The school data store

CRITERION D · i–iv ◆ QUANTITY ◆ MODEL

A media club stores video projects on a server with three drives. The labels use powers of ten: one drive holds 2×10^{11} bytes, a second holds 5×10^{11} bytes and a third holds 1.2×10^{12} bytes. A single lesson video is about 4×10^{8} bytes.

Your task

1. Write the total capacity in bytes, using a single power of ten.
2. Estimate how many lesson videos will fit, showing your estimate as a power of ten.
3. Calculate the number more precisely, to two significant figures.
4. The club records 30 lessons. Does the storage suffice? Justify with a calculation.
5. Explain why the answer to question 2 is an estimate rather than a fact.

CRITERION

Criterion D · strands i–iv

- Recognise the quantities involved and their units.
- Select powers of ten as an appropriate strategy.
- Apply the strategy accurately, keeping track of exponents.
- Justify whether the conclusion about storage is reasonable.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

A power primer for the science lab

CRITERION A · i · C · ii, iii

Context. The science department is producing a one-page reference card for Year 1 students. It will be pinned beside the microscopes. The card must explain powers, squares, cubes, roots and the order of operations for a reader who has not yet studied this chapter.
Product. A card using no more than 180 words of explanation, plus diagrams and one fully worked example, readable from about one metre.

1. Define power, base and exponent, using 2^5 as your example.
2. Explain 10^3 and 10^0 with a pattern that makes clear why 10^0 is 1 rather than 0.
3. Show 4^2 as a square of 16 unit squares and explain why “square” is a suitable name.
4. Work through $4 + 3 \times 2^2$, one line per operation, labelling the rule used each time.
5. Explain the difference between $3x^2$ and $(3x)^2$ when $x = 2$. State both values.
6. Include a science example: a culture of 500 bacteria doubles every hour. Write an expression for the number after h hours and find the number after 5 hours.
7. Ask a classmate to use your card to evaluate $2 + 4 \times 3^2$. Record whether they succeed, then revise one part of the card.
8. Evaluate your card: one strength, one limitation and one change you would make for a younger audience.

CRITERION

Criteria and strands developed

- Criterion A · Knowing and understanding — i: accurate definitions and calculations throughout the card.
- Criterion C · Communicating — ii, iii: precise notation, a clear diagram and awareness of the reader.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- A power records repeated multiplication; the exponent counts factors, not additions.
- Powers of ten generate place value, so expanded form can be written with exponents.
- Square and cube numbers correspond to square and cube arrangements; roots reverse them.
- One agreed order of operations makes a written expression mean exactly one thing.
- Substitution tests the structure of an expression: where a symbol sits decides what it does.

Vocabulary

Term	Meaning
base	The number that is multiplied repeatedly in a power.
exponent / index	The number of copies of the base.
power	A product of equal factors, or the whole expression a^n.
square number	The product of a number with itself.
cube number	The product of three equal factors.
square root	The positive number whose square is the given number.
substitution	Replacing a letter by a number and evaluating.

Misconceptions to watch

COMMON ERROR

Three that persist

$3^4 = 12$. Powers are not multiplications of base by exponent.

Multiplication always before division. They rank equally; work left to right.

$-144 = 12$. Roots give positive values in Year 1; the negative solution arises from equations, not from the root sign.

Self-assessment

I can …	Not yet	Getting there	Confident
write a repeated product as a power	☐	☐	☐
evaluate powers, squares, cubes and roots	☐	☐	☐
write a number using powers of ten	☐	☐	☐
apply the order of operations	☐	☐	☐
substitute into an expression with a power	☐	☐	☐
explain the difference between $3x^2$ and $(3x)^2$	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to identify which rule still causes errors, students must redo one question from the weakest row without looking at the worked example, then compare their lines with the model.

Looking ahead · Chapter 3 · Factors, multiples and primes takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 2 — SUPPORT 1–5 · CORE 4–11 · EXTENSION 10–14

1. Evaluate: (a) 4^3 (b) 10^3 (c) $\sqrt{100}$ (d) $\sqrt[3]{8}$. F
2. Write $5 \times 5 \times 5 \times 5$ as a power and evaluate it. F
3. Evaluate $7 + 3 \times 2^2$. F
4. Write 90 300 in expanded form using powers of ten. F
5. Evaluate (a) 2^5 (b) $3^4 - 3^2$ (c) $100 - 4^2$. F
6. Insert brackets to make $2 + 4 \times 3 = 18$ true. R
7. Explain why 2^{10} is greater than 10^2 even though 10 is greater than 2. R
8. A square garden has an area of 225 m^2. Find the length of fencing needed to enclose it. P
9. Evaluate $3x^2 + 2x$ when $x = 4$. P
10. A sheet of paper is 0.08 mm thick. How thick is a stack of 2^{10} sheets? Give your answer in millimetres and in metres. P
11. Explain, using an example, why $10^0 = 1$ and not 0. R
12. Find a two-digit number that is both a square number and a cube number. Explain why it is the only one below 100. E
13. The number 64 can be written as a power of 2, a power of 4 and a power of 8. Write all three. E
14. Investigate the last digit of 2^1, 2^2, 2^3, … and describe the cycle. Predict the last digit of 2^{20}. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–3	Revisit 2.1 and 2.4: write the meaning of the power before evaluating.
4–6	Revisit 2.2: count the columns from the right and name each exponent.
7–9	Revisit 2.3 and 2.5: draw the square or substitute slowly.
10–12	Revisit the investigation: doubling is a power of 2.
13–14	Revisit 2.1 and 2.3: list the powers in order and look at the digits.



3

NUMBER, STRUCTURE AND PROOF

Factors, multiples and primes

Every whole number is built from smaller whole numbers. Learning to see that structure turns division, fair sharing and repeating schedules into questions with exact answers.

KEY CONCEPT · RELATIONSHIPS

RELATED · SYSTEM

RELATED · PATTERN

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How is a whole number built from smaller whole numbers, and how does that structure decide what can be shared fairly

01 · MYP UNIT PLANNER

Statement of inquiry	Whole numbers are built from factors in a structured way, and that structure makes it possible to divide quantities fairly, schedule repeating events and communicate results with exact precision.

KEY IDEA

Key and related concepts

Key concept · Relationships. Every number greater than 1 relates to the primes that build it, and that relationship does not change however the number is written.

Related · System. Divisibility rules form a system: each one tests a specific structure in the digits.

Related · Pattern. Multiples repeat on a regular grid, which is why the lowest common multiple can be read from the pattern.

NOTE

Global context · Fairness and development

Exploration: how the structure of numbers decides what can be shared equally, from splitting a harvest among households to scheduling two services that must not clash.

Inquiry questions

- Factual: What is a factor, what is a multiple, and how can you tell whether one number divides another?
- Conceptual: Why is prime factorisation called the “building block” representation of a number?
- Debatable: Is it fairer to divide a quantity using the highest common factor or the lowest common multiple?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Find factors, multiples, primes, HCF, LCM and prime factorisations.
B · Investigating patterns	i, ii, iii	Investigate the factor table and the pattern of primes.
C · Communicating	i, ii, iii, iv	Present a factor tree and a Venn diagram so another reader can follow it.
D · Applying mathematics	i, ii, iii, iv	Plan a fair relay schedule using common multiples.

ATL SKILL

ATL · Thinking skills

In order for a student to decide whether a large number is prime, students must test divisibility systematically and stop at the square root, rather than trying divisors at random.

02 · ENGAGE

The bakery trays.

A bakery lays loaves on rectangular trays. Twelve loaves fit as 1×12, 2×6 or 3×4. Thirteen loaves fit only as 1×13. The baker says thirteen is "a stubborn number". Mathematicians say it is prime — and the stubbornness is the same property that makes large primes valuable in protecting messages sent across the internet.

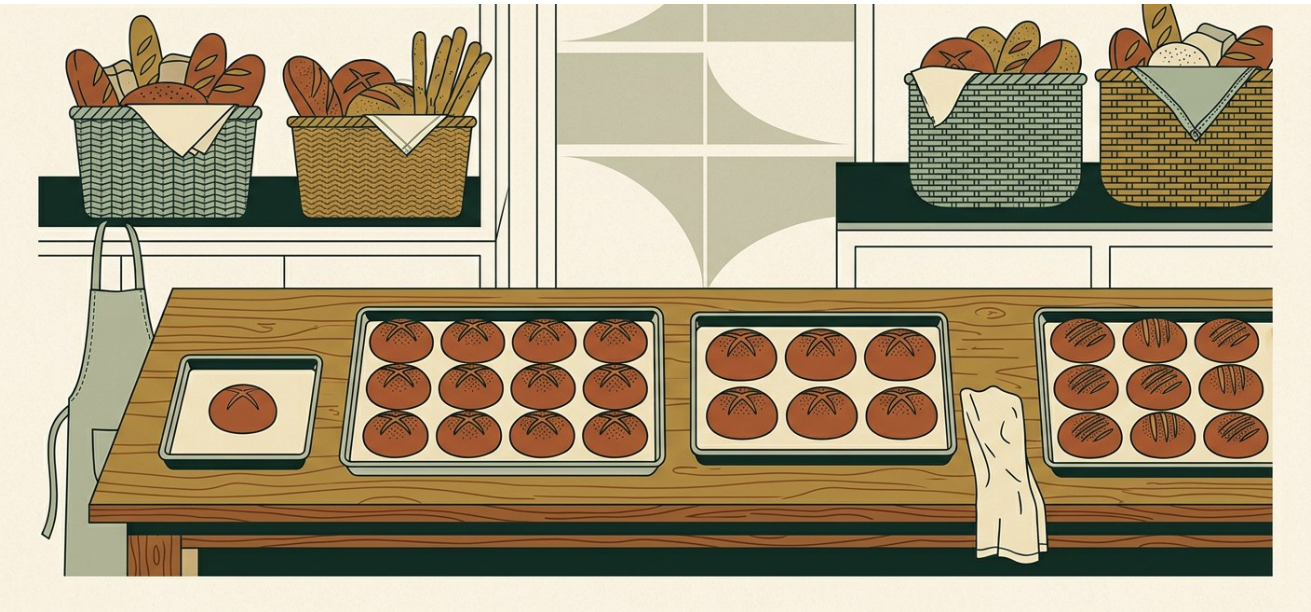


FIGURE 1 Arranging loaves in equal rows is a physical factorisation. Some quantities allow many rectangles; some allow only one.
Original illustration created for this edition.

03 · EXPLORE

Sort before defining.

Write the numbers 1 to 20 in a row. For each number, list every way it can be written as a product of two whole numbers. Then sort the twenty numbers into three groups: those with one product, those with two, and those with more.

1. Which numbers had only one product, and what do they have in common?
2. What is special about 1, and why is it usually excluded from the primes?
3. Which number has the most products below 21?
4. Predict where 24 and 25 will fall before checking.

KEY IDEA

The three families you found

Primes have exactly two different factors: 1 and the number itself. Composite numbers have more than two. 1 has only one factor, so it is neither, which keeps every statement about primes true without exception.

3.1 Factors, multiples and factor pairs

CRITERION A · i ◆ SYSTEM

DEFINITION

Factor and multiple

A factor of a whole number divides it exactly, leaving no remainder. A multiple is the result of multiplying that number by a whole number. So 6 is a factor of 24, and 24 is a multiple of 6. Factors and multiples are the same relationship seen from two directions.

Factors come in pairs. Since 24 = 3 × 8, both 3 and 8 are factors, and they belong together as a pair. The factor pairs of 24 are 1 × 24, 2 × 12, 3 × 8 and 4 × 6. The list stops when the two members of a pair meet or cross, which is why you never need to test beyond the square root.

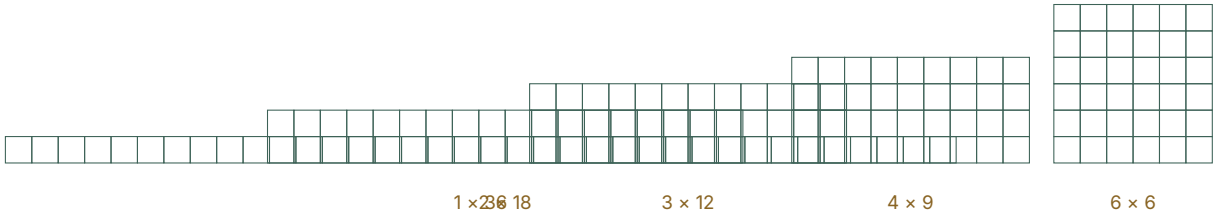


FIGURE 2 The factor pairs of 36 as rectangular arrays. Every pair is the same 36 counters arranged differently.

KEY IDEA

Why the square root is the stopping point

If $a \times b = n$ and $a < b$, then a must be less than $\sqrt{n}$. So every factor pair has one member below the square root. Searching up to $\sqrt{n}$ finds every pair, and searching further finds only repeats.

WORKED EXAMPLE 3.1.1 — LISTING THE FACTORS OF 48

PROBLEM

Find all the factors of 48 and write them as pairs.

1. Start from 1: 1 × 48 = 48, so 1 and 48 are a pair.
2. 2 × 24 = 48; 3 × 16 = 48; 4 × 12 = 48; 6 × 8 = 48.
3. 5 and 7 do not divide 48. The next number to test is $\sqrt{48} \approx 6.9$, and we have already passed it.
4. So the factors are 1, 2, 3, 4, 6, 8, 12, 16, 24 and 48.

ANSWER 1, 2, 3, 4, 6, 8, 12, 16, 24, 48 — ten factors.

Count them: ten. Pairs: 1×48, 2×24, 3×16, 4×12, 6×8 √

WORKED EXAMPLE 3.1.2 — MULTIPLES IN A REAL SCHEDULE

PROBLEM

One bus leaves every 6 minutes and another every 8 minutes. They leave together at 08:00. When will they next leave together?

1. List multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48.
2. List multiples of 8: 8, 16, 24, 32, 40, 48.
3. The first common multiple is 24, and every common multiple is a multiple of 24.
4. 08:00 + 24 minutes = 08:24.

ANSWER At 08:24, and again every 24 minutes.

24 is a multiple of 6 and of 8, and no smaller number is √

PRACTICE 3.1 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 7–9

1. List the factor pairs of (a) 18 (b) 28 (c) 45. F
2. Write the first five multiples of (a) 7 (b) 11. F
3. How many factors does 36 have? List them. F
4. Find the numbers between 20 and 40 with exactly four factors. F
5. Explain why 1 is a factor of every whole number. R
6. A red light flashes every 4 seconds and a blue light every 6 seconds. They flash together at the start. When do they next flash together? P
7. Is a multiple of a number always larger than the number? Explain using a counter-example or a justification. R
8. Find the smallest number with exactly six factors. E
9. Two numbers have a highest common factor of 6 and a lowest common multiple of 36. Suggest a possible pair and explain how you found it. E

3.2 Divisibility tests

CRITERION A · ii · B · i ◆ SYSTEM ◆ PATTERN

A divisibility test is a shortcut that reads the digits instead of performing the division. Each test works because of the structure of base ten, and each one can be explained rather than merely memorised.

2	last digit is even
3	digit sum is a multiple of 3
4	last two digits form a multiple of 4
5	last digit is 0 or 5
6	divisible by 2 and by 3
9	digit sum is a multiple of 9
10	last digit is 0

FIGURE 3 Seven divisibility rules. Each is a statement about the digits, not about the size of the number.

KEY IDEA

Why the digit-sum test works for 3 and 9

Any power of 10 is one more than a multiple of 9: $10 = 9 + 1$, $100 = 99 + 1$, $1\,000 = 999 + 1$. When a number is broken into its columns, every column contributes a multiple of 9 plus the digit itself. So the remainders depend only on the sum of the digits.

WORKED EXAMPLE 3.2.1 — TESTING 3 528 FOR DIVISIBILITY

PROBLEM

Test 3 528 for divisibility by 3, 4, 6 and 9.

1. Digit sum $3 + 5 + 2 + 8 = 18$, which is a multiple of 3, so 3 528 is divisible by 3.
2. 18 is not a multiple of 9? It is: $18 = 2 \times 9$, so 3 528 is also divisible by 9.
3. Last two digits are 28, and $28 \div 4 = 7$, so divisible by 4.
4. Divisible by 2 (it is even) and by 3, so divisible by 6.

ANSWER Divisible by 3, 4, 6 and 9.

$3\,528 \div 9 = 392$, and 392 is even ✓

COMMON ERROR

Using a digit rule on the wrong digit

The test for 4 uses the last two digits, not the last digit: 1 514 ends in 4, but 14 is not a multiple of 4, so 1 514 is not divisible by 4. Read each rule precisely before applying it.

PRACTICE 3.2 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Which of these are divisible by 3: 114, 205, 321, 400, 891? F
2. Which are divisible by 9: 234, 567, 808, 1 008? F
3. Test 2 340 for divisibility by 2, 3, 4, 5, 6 and 10. F
4. Is 7 254 divisible by 6? Show the two tests you used. F
5. Explain why a number ending in 0 is always divisible by both 2 and 5. R
6. A number is divisible by 3 but not by 9. What can you say about its digit sum? R

PRACTICE 3.2 continued SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the smallest three-digit number divisible by 2, 3 and 5. P
2. Use the structure of base ten to explain the divisibility test for 9. E

3.3 Prime and composite numbers

CRITERION A · ii · B · ii ◆ PATTERN ◆ SYSTEM

DEFINITION

Prime and composite

A prime number is a whole number greater than 1 with exactly two factors: 1 and itself. A composite number has more than two factors. 1 is neither prime nor composite, because it has only one factor.

Primes are the atoms of multiplication. Every composite number can be broken into primes, and no composite number is itself a building block. This is why primes appear in unexpected places: in the lengths of repeating patterns, in the design of calendars, and in the encryption that protects a bank transfer.

1	**2**	**3**	4	**5**	6	**7**	8	9	10
11	12	**13**	14	15	16	**17**	18	**19**	20
21	22	**23**	24	25	26	27	28	**29**	30
31	32	33	34	35	36	**37**	38	39	40
41	42	**43**	44	45	46	**47**	48	49	50
51	52	**53**	54	55	56	57	58	**59**	60

FIGURE 4 The primes below 60. Notice that after 2 and 5, every prime ends in 1, 3, 7 or 9 — but not every number ending in those digits is prime (21 and 51 are not).

Testing whether a number is prime

- Test divisibility by 2, 3, 5 and 7 first; most composites fail one of these.
- If nothing divides it, test the remaining primes up to its square root.
- Stop at the square root: if no factor has been found by then, none exists.
- A prime can be large; a test does not need to try every number below it.

WORKED EXAMPLE 3.3.1 — IS 97 PRIME?

PROBLEM

Decide whether 97 is prime, showing your reasoning.

1. $\sqrt{97} \approx 9.8$, so only 2, 3, 5 and 7 need be tested.
2. 97 is odd, so not divisible by 2. Digit sum 16, not a multiple of 3. It does not end in 0 or 5.
3. $97 \div 7 = 13$ remainder 6, so not divisible by 7.
4. No prime up to the square root divides 97.

ANSWER 97 is prime.

The next primes are 101 and 103, both larger than 97 √

NOTE

Why primes matter beyond the classroom

Modern encryption relies on the fact that multiplying two large primes is easy but recovering the primes from the product is not. A 200-digit number built from two 100-digit primes cannot realistically be factorised by testing, which is what protects the message.

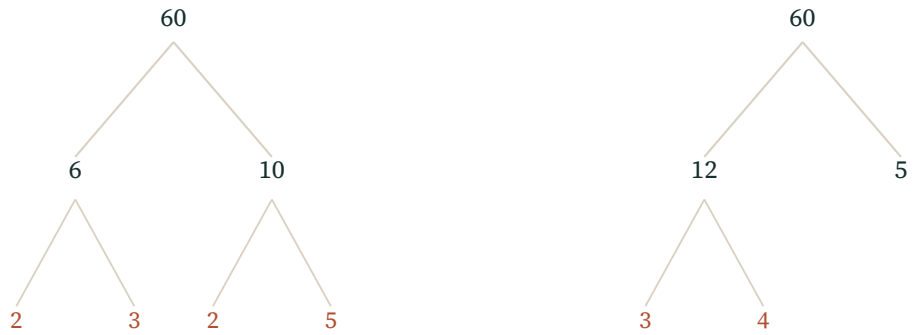
PRACTICE 3.3 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. List the primes between 20 and 50. F
2. Which of these are prime: 51, 53, 57, 59, 61? F
3. Explain why 2 is the only even prime. R
4. Is 91 prime? Show your test. F
5. Find two primes whose sum is 30. How many such pairs are there? P
6. A student says 87 is prime because it is not divisible by 2, 3 or 5. Identify the error. R
7. Find the smallest prime greater than 100, explaining each test. P
8. Investigate: is the sum of two primes always even? Test five cases and explain the exception. E

3.4 Prime factorisation

CRITERION A · iii · C · iii ◆ REPRESENTATION ◆ SYSTEM

Prime factorisation writes a number as a product of primes only. The result is unique: 60 is $2 \times 2 \times 3 \times 5$ however you find it, and no other product of primes gives 60. That uniqueness is called the fundamental theorem of arithmetic, and it is the reason factor trees always agree.



two routes, one result: $60 = 2 \times 2 \times 3 \times 5$

FIGURE 5 Two different factor trees for 60, arriving at the same prime factorisation: $2^2 \times 3 \times 5$.

DEFINITION

Prime factorisation

The prime factorisation of a number is its expression as a product of prime factors, usually written with indices and with the primes in order: $60 = 2^2 \times 3 \times 5$.

WORKED EXAMPLE 3.4.1 — FACTORISING 84 BY DIVISION

PROBLEM

Write 84 as a product of prime factors.

1. Divide by the smallest prime that works: $84 \div 2 = 42$.
2. $42 \div 2 = 21$. 21 is odd, so move to 3: $21 \div 3 = 7$.
3. 7 is prime, so stop.
4. Collect the divisors: $2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$.

ANSWER $84 = 2^2 \times 3 \times 7$

$2^2 \times 3 \times 7 = 4 \times 21 = 84$ ✓

WORKED EXAMPLE 3.4.2 — USING A FACTORISATION TO TEST DIVISIBILITY

PROBLEM

Use the prime factorisation of 360 to decide whether 360 is divisible by 24.

1. $360 = 2^3 \times 3^2 \times 5$.
2. $24 = 2^3 \times 3$.
3. Every prime power in 24 appears in 360, with at least the same exponent.
4. So $360 \div 24 = 15$ exactly.

ANSWER Yes. $360 \div 24 = 15$.

$24 \times 15 = 360$ ✓

PRACTICE 3.4 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Write as a product of prime factors: (a) 36 (b) 45 (c) 100. F
2. Write 72 in prime factorisation form, using indices. F
3. Write 128 as a power of 2. F
4. Two students factorise 90 differently but reach the same answer. Show two routes. R
5. Use prime factorisation to decide whether 210 is divisible by 14. P
6. Explain why the prime factorisation of a prime number is just the number itself. R
7. Find the smallest number whose prime factorisation contains exactly three different primes. P
8. A number has prime factorisation $2^a \times 3^b$. It has exactly six factors. Find all possible values of the number. E

3.5 Highest common factor and lowest common multiple

CRITERION A · iii · D · i ◆ RELATIONSHIPS ◆ SYSTEM

Two numbers always share factors and share multiples. The largest shared factor (HCF) answers "how can this be split into equal groups?" The smallest shared multiple (LCM) answers "when do these cycles meet again?" Both can be read directly from the prime factorisations.

KEY IDEA

Reading HCF and LCM from the factorisations

For HCF, take the primes common to both numbers and give each the lower exponent. For LCM, take every prime that appears and give each the higher exponent. With $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$: HCF $= 2 \times 3 = 6$ and LCM $= 2^2 \times 3^2 = 36$.



FIGURE 6 12 and 18 arranged as sets of factors. The shared factors sit in the overlap; the highest common factor is the largest of them.

WORKED EXAMPLE 3.5.1 — SPLITTING INTO EQUAL GROUPS

PROBLEM

A harvest of 84 mangoes and 60 oranges is to be packed into identical bags with no fruit left over, each bag holding only one kind of fruit. What is the largest number of bags possible?

1. The number of bags must divide both 84 and 60, so find the HCF.
2. $84 = 2^2 \times 3 \times 7$ and $60 = 2^2 \times 3 \times 5$.
3. Common primes with lower exponents: $2^2 \times 3 = 12$.
4. So 12 bags: 7 mangoes in each of 7 bags and 5 oranges in each of 5 bags.

ANSWER 12 bags.

$12 \times 7 = 84$ and $12 \times 5 = 60$ ✓

WORKED EXAMPLE 3.5.2 — WHEN DO THE CYCLES MEET?

PROBLEM

One water pump is checked every 12 days and another every 18 days. Both were checked today. In how many days will both be checked on the same day again?

1. Find the LCM of 12 and 18.
2. $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$.
3. LCM $= 2^2 \times 3^2 = 36$.
4. So both are checked together again after 36 days.

ANSWER 36 days.

$36 \div 12 = 3$ and $36 \div 18 = 2$, both whole numbers ✓

PRACTICE 3.5

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the HCF of (a) 12 and 20 (b) 18 and 27. F
2. Find the LCM of (a) 4 and 6 (b) 8 and 12. F
3. Find the HCF and LCM of 24 and 36. F
4. Why is the HCF of two numbers never larger than the smaller number? R
5. Two bells ring every 15 and every 20 minutes. If they ring together at 09:00, when do they next ring together? P
6. A teacher has 48 pencils and 36 rulers and wants identical packs with no remainder. What is the greatest number of packs? P
7. Explain why the product of the HCF and LCM of two numbers equals the product of the numbers, using 12 and 18. E
8. Find two numbers whose HCF is 4 and whose LCM is 60. E

04 · INVESTIGATE

INVESTIGATION 3

The factor table

PATTERN ◆ SYSTEM

Draw a table with the numbers 1 to 20 across the top and down the side. Shade the cell where the row number is a factor of the column number. Work systematically; do not shade cell by cell.

Number	1	2	3	4	5	6	8	10	12
Factors	all	even	digit sum	last two	ends 0/5	2 and 3	three 2s	2 and 5	3 and 4

Your investigation

1. Which column is fully shaded, and why?
2. Which rows have the fewest shaded cells, and what kind of number are they?
3. Describe the shading pattern for row 2, row 3 and row 4 in words.
4. Predict the shading for row 7 without drawing it.
5. Explain why a number with exactly two shaded cells in the table is prime.
6. Look down the diagonal. What do the shaded cells there represent?

KEY IDEA

The structure the table reveals

Multiples form a regular grid: row n is shaded every n columns. Primes are the rows with exactly two shaded cells. The diagonal is shaded because every number is a factor of itself — which is precisely what makes the definition of a prime depend on having exactly one other factor.

ATL SKILL

ATL · Transfer

In order for a student to generalise from a table, students must describe the rule in words before writing any symbol, then test the description on a row they have not shaded.

05 · APPLY

REAL-LIFE APPLICATION

The fairness relay

CRITERION D · i–iv ◆ RELATIONSHIPS ◆ SYSTEM

A school sports day runs three relay races around the same track. The junior race takes 4 minutes per lap, the middle race 6 minutes and the senior race 9 minutes. All three start together at 09:00. The organisers need to know when the track is free for the award ceremony.

Your task

1. Find the LCM of 4, 6 and 9, showing your prime factorisations.
2. State the first three times all three races finish together.
3. The ceremony needs 30 clear minutes. Recommend a start time and justify it.
4. Each race has 24 runners, 18 runners and 12 runners, and the organisers want equal teams within each race with no one left out. Find the largest possible team size shared by all three races.
5. Explain which of your two answers came from the LCM and which from the HCF, and why the context decided it.

CRITERION

Criterion D · strands i–iv

- Identify the quantities and the constraint (simultaneous, or no remainder).
- Choose HCF or LCM from the structure of the problem.
- Apply prime factorisation accurately.
- Justify that the recommended schedule is realistic.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The fairness poster

CRITERION C · i–iii · D · ii

Context. A community association wants a poster that explains, without using the words “highest common factor” or “lowest common multiple”, how to share a harvest equally and how to schedule two repeating events.

1. Choose two quantities (one shared, one scheduled) and state them.
2. Show the factorisation of each quantity using a diagram.
3. Explain, in no more than 60 words, how the diagram answers each question.
4. Include one worked example with a check.
5. Ask a classmate to use your poster on a new pair of numbers. Record whether they succeed, then revise one part.
6. Evaluate your poster: one strength, one limitation, one improvement.

CRITERION

Criteria and strands developed

- Criterion A · Knowing and understanding — ii: accurate factorisation and correct HCF and LCM.
- Criterion C · Communicating — i, ii, iii: precise language and a diagram a reader can follow.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Factors and multiples describe the same relationship from opposite directions.
- Divisibility tests read the structure of the digits, and each can be explained.
- Primes are the multiplicative building blocks; 1 is neither prime nor composite.
- Prime factorisation is unique, so different routes give the same answer.
- HCF answers “equal groups”; LCM answers “when do cycles meet”.

Vocabulary

Term	Meaning
factor	A whole number that divides another exactly.
multiple	The product of a number and a whole number.
prime	A whole number greater than 1 with exactly two factors.
composite	A whole number with more than two factors.
prime factorisation	A number written as a product of primes.
HCF	The largest number dividing two or more numbers exactly.
LCM	The smallest number that is a multiple of two or more numbers.

Misconceptions to watch

COMMON ERROR

Three that persist

1 is prime. It has only one factor, so it fails the definition.

The LCM is the product. Only when the numbers share no factor; LCM(4, 6) = 12, not 24.

A number ending in 3 is prime. 33, 63 and 93 all end in 3 and none is prime.

Self-assessment

I can …	Not yet	Getting there	Confident
list factors and multiples systematically	☐	☐	☐
apply divisibility tests	☐	☐	☐
decide whether a number is prime	☐	☐	☐
write a prime factorisation	☐	☐	☐
find the HCF and LCM	☐	☐	☐
choose HCF or LCM from a context	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to recognise which of HCF and LCM a word problem needs, students must state in one sentence what the answer represents — equal groups, or a repeat — before calculating anything.

Looking ahead · Chapter 4 · Fractions and decimals takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 3

SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. List all the factors of 60. F
2. Find the HCF of 30 and 45. F
3. Find the LCM of 6, 8 and 12. F
4. Write 180 as a product of prime factors. F
5. Which of 97, 111, 121 and 131 are prime? F
6. Explain why every even number greater than 2 is composite. R
7. A number has exactly three factors. What kind of number is it? Give two examples. R
8. Two runners complete a lap in 45 and 60 seconds. Starting together, when do they next pass the start line together? P
9. Pencils come in packs of 12 and pens in packs of 18. What is the smallest number of each that gives equal numbers of pencils and pens? P
10. Use divisibility tests to decide whether 4 356 is divisible by 12. R
11. Find the largest two-digit prime and justify that it is prime. E
12. Show that the product of two consecutive numbers is always even. E
13. Find two numbers whose LCM is 48 and whose HCF is 4. E
14. A rectangular courtyard is 84 tiles long and 60 tiles wide. Square tiles of the largest possible size are to cover it exactly. Find the tile size and the number of tiles. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–2	Revisit 3.1: write factor pairs and stop at the square root.
3–4	Revisit 3.4 and 3.5: build the prime factorisations first.
5–7	Revisit 3.3: test primes up to the square root only.
8–10	Revisit 3.5: decide whether the question asks for equal groups or for a repeat.
11–14	Revisit the investigation: patterns in factors explain all of these.

PART II

Fractions, decimals and percentages

Three ways of writing the same part of a whole, and the decisions each one makes easy.

Fractions, decimals and percentages are not three topics; they are three notations for one idea.



SADIKOU MOUKODDIM

4

FRACTIONS, DECIMALS AND PERCENTAGES

Fractions and decimals

A fraction is not a piece of a cake; it is a number with a position on the line. Once you can see that, the four operations stop being separate rules and become consequences of one idea.

KEY CONCEPT · RELATIONSHIPS

RELATED · EQUIVALENCE

RELATED · QUANTITY

CONTEXT · IDENTITIES AND RELATIONSHIPS

INQUIRY

How can two different-looking fractions name the same number, and what does that equivalence let us do that we could not do

01 · MYP UNIT PLANNER

Statement of inquiry	A fraction, a decimal and a percentage are three notations for the same relationship between a part and a whole, and equivalence is the idea that lets us choose whichever notation makes the current question easiest.

KEY IDEA

Key and related concepts

Key concept · Relationships. A fraction records a relationship between two quantities, which is why the same relationship has many names.

Related · Equivalence. Multiplying numerator and denominator by the same number changes the writing without changing the number.

Related · Quantity. A fraction of a quantity is found by dividing, then multiplying — in that order.

NOTE

Global context · Identities and relationships

Exploration: how shared resources, recipes and measurements are divided, and why a fair share needs an exact number rather than an impression.

Inquiry questions

- Factual: How do you convert between a fraction, a decimal and a percentage?
- Conceptual: Why does multiplying the numerator and denominator by the same number not change the value?
- Debatable: Is it more honest to report a share as 1/3 or as 33 %?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Convert, compare, order and calculate with fractions and decimals.
B · Investigating patterns	i, ii, iii	Investigate which fractions give terminating decimals and why.
C · Communicating	i, ii, iii, iv	Show working with equivalent fractions so the method is auditable.
D · Applying mathematics	i, ii, iii, iv	Scale a recipe and cost a kitchen for a larger number of people.

ATL SKILL

ATL · Thinking skills

In order for a student to add fractions with different denominators, students must first decide on a common unit and rewrite both quantities in it, just as they would convert metres and centimetres before adding.

02 · ENGAGE

The tailor's strip.

A tailor has a strip of cloth 1 metre long and needs to cut lengths of ¾ m. She can cut one, and has ¼ m left over. Someone else needs ⅔ m from a second strip. How much cloth do the two customers need together? The answer — one whole metre and five twelfths — only appears once both fractions are measured in the same unit.

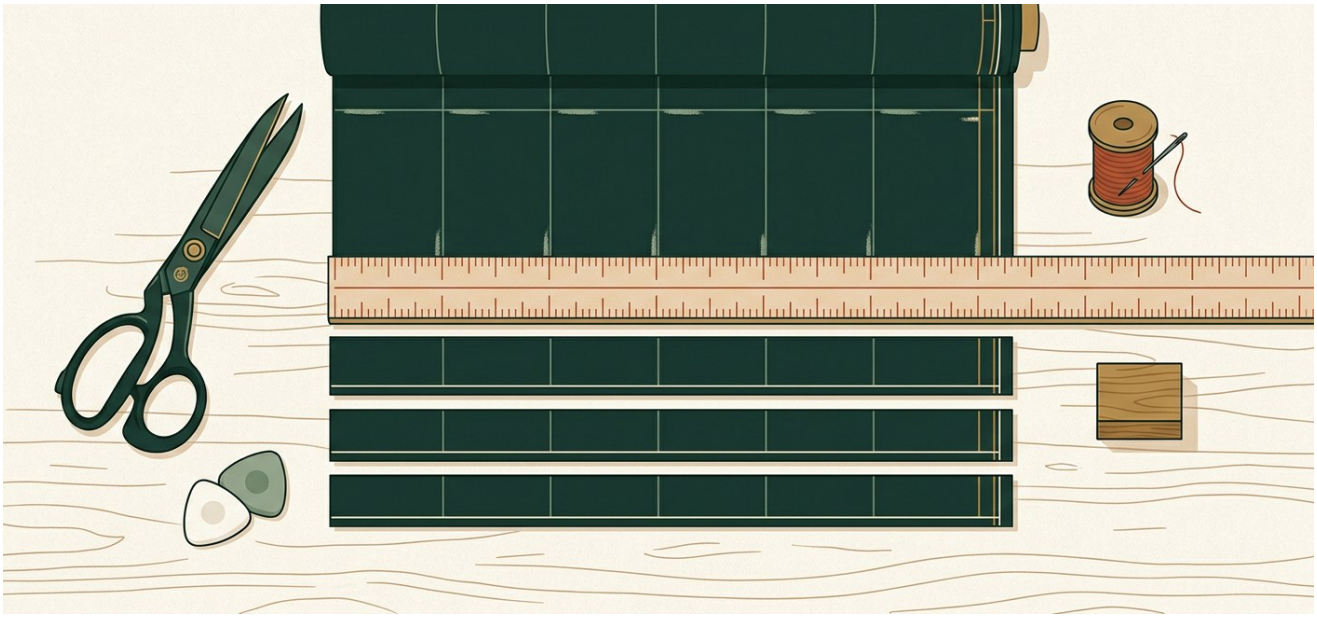


FIGURE 1 Cutting a strip into equal parts is the physical origin of a fraction: the denominator is the number of equal parts, the numerator the number taken.

Original illustration created for this edition.

03 · EXPLORE

Find the equivalence before being taught it.

Fold a strip of paper into halves, then fold again to make quarters, then again to make eighths. Write the fraction of the strip that is shaded at each stage, and look at what happens to the numbers.

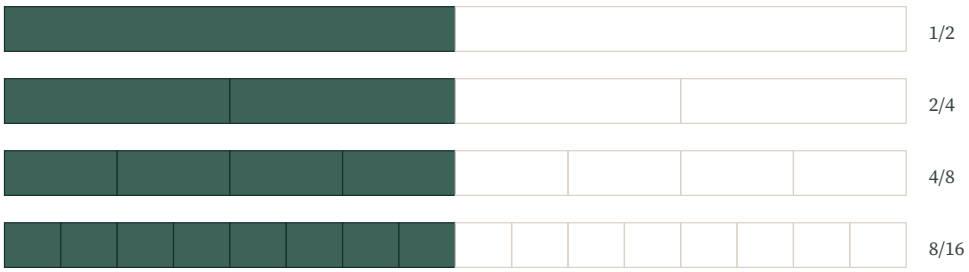


FIGURE 2 The same shaded length written three ways. The number of parts increases at every row; the amount of strip does not change.

1. What happens to the numerator and denominator when you double the number of parts?
2. Write the pattern as a rule using words, not symbols.
3. Predict the next two fractions in the sequence ½, 2/4, 4/8, …
4. Why can you not simply add the numerators and denominators to find an equivalent fraction?

KEY IDEA

The rule you found

Multiplying the numerator and the denominator by the same number produces an equivalent fraction, because the number of parts and the number taken both increase in the same proportion.

4.1 Equivalent fractions and simplest form

CRITERION A · i ◆ EQUIVALENCE ◆ SIMPLIFICATION

DEFINITION

Equivalent fractions

Two fractions are equivalent when they name the same point on the number line. Multiplying or dividing both the numerator and denominator by the same non-zero number produces an equivalent fraction: $3/4 = 6/8 = 9/12$.

Simplest form (also called lowest terms) is the equivalent fraction in which the numerator and denominator share no factor other than 1. It is not the "best" fraction in every context — 6/8 is easier to add to 1/8 — but it is the standard form for an answer.

WORKED EXAMPLE 4.1.1 — REDUCING TO SIMPLEST FORM

PROBLEM

Write 36/48 in simplest form.

1. Find the highest common factor of 36 and 48.
2. $36 = 2^2 \times 3^2$ and $48 = 2^4 \times 3$, so the HCF is $2^2 \times 3 = 12$.
3. Divide both parts by 12: $36 \div 12 = 3$ and $48 \div 12 = 4$.
4. So $36/48 = 3/4$.

ANSWER 3/4

$3 \times 12 = 36$ and $4 \times 12 = 48$, so the fractions are equivalent ✓

WORKED EXAMPLE 4.1.2 BUILDING AN EQUIVALENT FRACTION

PROBLEM

Write 2/5 with a denominator of 35.

1. Ask what 5 must be multiplied by to reach 35: $5 \times 7 = 35$.
2. Multiply the numerator by the same number: $2 \times 7 = 14$.
3. So $2/5 = 14/35$.

ANSWER 14/35

$14 \div 35 = 0.4$ and $2 \div 5 = 0.4$ ✓

COMMON ERROR

Adding the same number instead of multiplying

A student writes $2/5 = 3/6$ by adding 1 to each part. That changes the value: $2/5 = 0.4$ but $3/6 = 0.5$. Equivalence is produced by multiplying or dividing, never by adding or subtracting.

PRACTICE 4.1 SUPPORT 1–4 · CORE 3–7 · EXTENSION 7–9

1. Write two fractions equivalent to (a) 1/2 (b) 3/5 (c) 7/10. F
2. Write in simplest form: (a) 8/12 (b) 15/25 (c) 18/24. F
3. Write 4/7 with a denominator of 42. F
4. Which of these are equivalent to 2/3: 8/12, 10/16, 6/9, 12/20? F
5. Explain why multiplying numerator and denominator by the same number keeps the value unchanged. R
6. A recipe uses 3/8 of a bag of rice. Write this with a denominator of 24 and explain what the new numbers mean in the recipe. P
7. Is 6/8 always written as 3/4? Give a context in which 6/8 is the more useful form. R
8. Find a fraction equivalent to 2/3 with a numerator of 14. E
9. How many fractions with a denominator less than 30 are equal to 1/3? List them. E

4.2 Comparing and ordering fractions

CRITERION A · ii · C · iii ◆ EQUIVALENCE ◆ QUANTITY

Fractions can be compared in three ways, and the choice depends on the numbers. Use the same denominator when the parts are easy to find; compare against a half or a whole when the fractions are close to

an obvious landmark; and convert to decimals when precision matters.

KEY IDEA

Three strategies

Common denominator. 3/5 and 5/8 become 24/40 and 25/40, so 5/8 is larger.

Landmark. 5/9 is more than a half (since 4.5/9 is a half), and 3/7 is less than a half, so 5/9 > 3/7.

Decimal. 3/5 = 0.6 and 5/8 = 0.625.

WORKED EXAMPLE 4.2.1 ORDERING WITH A COMMON DENOMINATOR

PROBLEM

Write 2/3, 3/4 and 5/8 in order from smallest to largest.

1. The denominators 3, 4 and 8 have LCM 24.
2. 2/3 = 16/24; 3/4 = 18/24; 5/8 = 15/24.
3. Compare the numerators: 15 < 16 < 18.
4. So 5/8 < 2/3 < 3/4.

ANSWER 5/8 < 2/3 < 3/4

As decimals: 0.625 < 0.667 < 0.75 √

WORKED EXAMPLE 4.2.2 USING A LANDMARK TO DECIDE QUICKLY

PROBLEM

Which is larger, 7/12 or 9/16?

1. Both are more than a half, so the landmark does not separate them.
2. Use a common denominator: 48 is the LCM of 12 and 16.
3. 7/12 = 28/48 and 9/16 = 27/48.
4. So 7/12 is larger by 1/48.

ANSWER 7/12 is larger.

7/12 ≈ 0.583 and 9/16 ≈ 0.563 √

PRACTICE 4.2

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Insert < or >: (a) 3/5 … 2/3 (b) 5/8 … 7/12. F
2. Order from smallest: 1/2, 2/5, 3/8, 7/16. F
3. Which is larger, 4/7 or 5/9? Use a common denominator. F
4. Write a fraction between 1/2 and 2/3 with a denominator of 12. F

PRACTICE 4.2 continued SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Explain how you can tell that 11/20 is larger than 1/2 without calculating. R
2. A student says 3/8 > 2/5 because 3 > 2 and 8 > 5. Explain why this reasoning fails. R
3. Three students run 3/8, 0.4 and 2/5 of a kilometre. Who runs furthest? Justify. P
4. Find the fraction exactly halfway between 1/3 and 1/2. E

4.3 Adding and subtracting fractions

CRITERION A · iii ◆ EQUIVALENCE

You cannot add quantities measured in different units, and a fraction is a measurement in units of $1/n$. Adding 1/2 and 1/3 requires rewriting both in units of 1/6, exactly as adding 1 m and 30 cm requires converting to the same unit first.

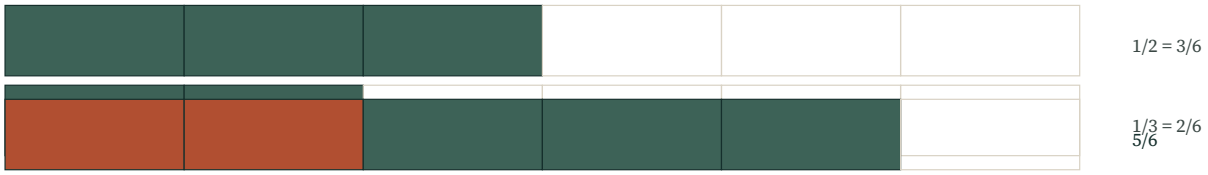


FIGURE 3 1/2 + 1/3 becomes 3/6 + 2/6. The common denominator is the common unit.

DEFINITION

The method

To add or subtract fractions: 1 find a common denominator (the LCM of the denominators is efficient but any common multiple works); 2 rewrite each fraction with that denominator; 3 add or subtract the numerators; 4 simplify if possible.

WORKED EXAMPLE 4.3.1 ADDING WITH DIFFERENT DENOMINATORS

PROBLEM

Calculate 3/4 + 5/6, giving the answer in simplest form.

1. LCM of 4 and 6 is 12.
2. 3/4 = 9/12 and 5/6 = 10/12.
3. 9/12 + 10/12 = 19/12.
4. 19/12 is already in simplest form; as a mixed number it is 1 and 7/12.

ANSWER 19/12 = 1 7/12

Estimate: 0.75 + 0.83 = 1.58, and 19/12 ≈ 1.58 √

WORKED EXAMPLE 4.3.2 SUBTRACTING FROM A WHOLE

PROBLEM

Calculate 3 − 5/8.

1. Write 3 as 24/8.
2. 24/8 − 5/8 = 19/8.
3. As a mixed number: 2 and 3/8.

ANSWER 19/8 = 2 3/8

2.375 + 0.625 = 3 √

COMMON ERROR

Adding the denominators

1/2 + 1/3 is not 2/5. Two-fifths is smaller than a half, but the sum of two positive fractions must be larger than either. An estimate catches the error immediately: 0.5 + 0.33 = 0.83, not 0.4.

PRACTICE 4.3 SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Calculate: (a) 1/5 + 2/5 (b) 3/8 + 1/8 (c) 7/10 − 3/10. F
2. Calculate: (a) 1/2 + 1/3 (b) 2/3 + 1/4 (c) 3/4 − 1/2. F
3. Calculate 5/6 + 3/8, in simplest form. F
4. Calculate: (a) 2 − 3/5 (b) 4 1/4 + 2 1/2. F
5. Explain why a common denominator is needed before adding. R
6. A tank is 3/8 full in the morning and 1/2 full after rain. What fraction of the tank is now full, and how much more can it hold? P
7. Find two fractions whose sum is 1 and whose difference is 1/3. E
8. Calculate 1/2 + 1/4 + 1/8 and describe the pattern in the answer. E

PRACTICE 4.3 continued SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. A student calculates 5/6 − 1/3 = 4/3. Diagnose the error and give the correct answer. R

4.4 Multiplying and dividing fractions

CRITERION A · iii ◆ QUANTITY

Multiplication of fractions needs no common denominator, because multiplying does not combine unlike units — it takes a part of a part. A third of a half is a sixth: 1/3 × 1/2 = 1/6, and the picture makes the reason obvious.

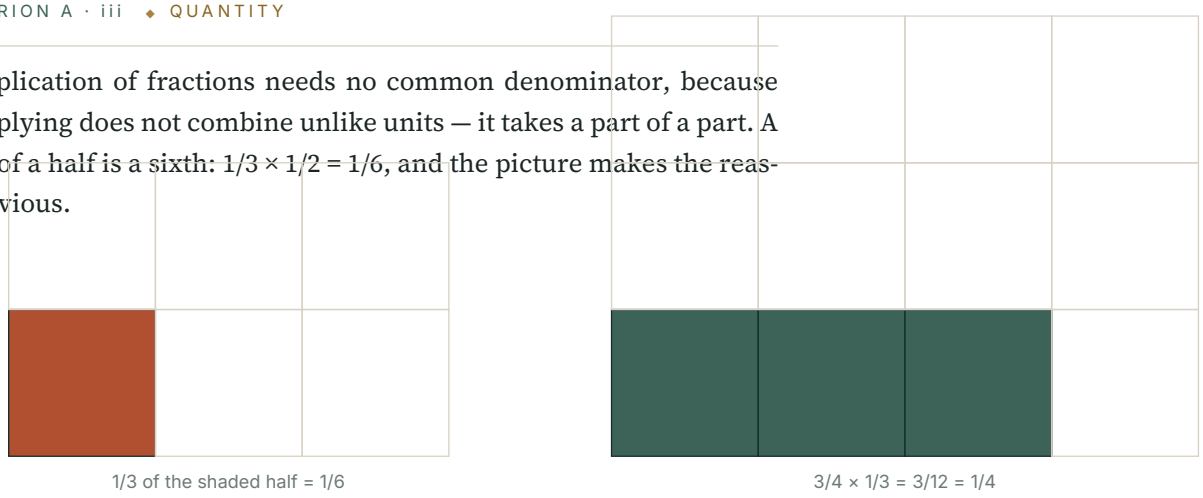


FIGURE 4 A third of a half is one sixth. Multiplying numerators and denominators corresponds to multiplying the numbers of rows and columns in the grid.

DEFINITION

The rules

Multiply: multiply numerators together and denominators together. Simplify before multiplying if you can. Divide: multiply by the reciprocal of the second fraction — 3/4 ÷ 2/5 = 3/4 × 5/2. The reciprocal of a fraction is the fraction turned upside down.

KEY IDEA

Why division becomes multiplication

Dividing by 2/5 asks “how many two-fifths fit into the quantity?” One whole contains 5/2 halves of two-fifths, so dividing by 2/5 is the same as multiplying by 5/2. Division by a fraction counts how many of that fraction fit.

WORKED EXAMPLE 4.4.1 — MULTIPLYING AND SIMPLIFYING EARLY

PROBLEM

Calculate 5/8 × 4/15, in simplest form.

1. Look for common factors before multiplying: 4 and 8 share 4; 5 and 15 share 5.
2. Cancel: 5/8 × 4/15 = (1 × 1)/(2 × 3).
3. Multiply: 1/6.

ANSWER 1/6

5/8 = 0.625 and 4/15 ≈ 0.267, and 0.625 × 0.267 ≈ 0.167 = 1/6 √

WORKED EXAMPLE 4.4.2 — DIVIDING BY A FRACTION

PROBLEM

Calculate 3/5 ÷ 2/7.

1. The reciprocal of 2/7 is 7/2.
2. 3/5 × 7/2 = 21/10.
3. As a mixed number, 2 and 1/10.

ANSWER 21/10 = 2 1/10

2/7 of 21/10 = 42/70 = 3/5 √

PRACTICE 4.4

SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Calculate: (a) 1/2 × 2/3 (b) 3/4 × 8/9 (c) 5/6 × 3/10. F
2. Write the reciprocal of (a) 3/4 (b) 5 (c) 1/7. F
3. Calculate: (a) 1/2 ÷ 1/4 (b) 3/4 ÷ 3/8 (c) 2 ÷ 1/3. F
4. Calculate 2/3 of 45. F
5. Explain why 1/2 ÷ 1/4 is 2 rather than 1/8, using words or a diagram. R
6. A recipe for 4 people uses 3/4 cup of oil. How much is needed for 6 people? P
7. A ribbon 5/6 m long is cut into pieces of 1/12 m. How many pieces are there? P
8. Without calculating, decide whether 3/4 × 5/6 is greater or less than 3/4. Explain. E
9. Find a fraction that gives 1 when multiplied by itself, other than 1 and −1. E

4.5

Decimals: place value, ordering and calculation

CRITERION A · ii, iii ◆ FORM ◆ QUANTITY

A decimal is a fraction written in place value. 0.37 means 37/100, and this single fact connects every decimal rule to something already known about fractions. Adding decimals works column by column because adding fractions with the same denominator works numerator by numerator.

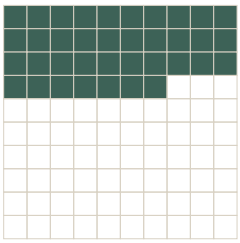


FIGURE 5 A hundred square with 37 cells shaded: 0.37 = 37/100. The grid makes the connection between place value and fraction visible.

Calculating with decimals

- Adding and subtracting: align the decimal points, fill empty columns with zeros, then calculate as for whole numbers.
- Multiplying: multiply as whole numbers, then count the total number of decimal places in both factors and place the point accordingly. 0.3 × 0.4 = 0.12, not 1.2.
- Dividing: make the divisor a whole number by multiplying both numbers by the same power of ten, then divide.

WORKED EXAMPLE 4.5.1 MULTIPLYING DECIMALS

PROBLEM

Calculate 2.4 × 0.35.

1. Ignore the points: 24 × 35 = 840.
2. Count decimal places: 1 + 2 = 3.
3. Place the point three places from the right: 0.840 = 0.84.

ANSWER 0.84

Estimate: 2.5 × 0.35 ≈ 0.875, and 0.84 is close ✓

WORKED EXAMPLE 4.5.2 — DIVIDING BY A DECIMAL

PROBLEM

Calculate 4.5 ÷ 0.15.

1. Multiply both numbers by 100 to make the divisor whole: 450 ÷ 15.
2. 450 ÷ 15 = 30.
3. Check by multiplying back: 30 × 0.15 = 4.5.

ANSWER 30

30 × 0.15 = 4.5 ✓

COMMON ERROR

Counting decimal places wrongly

0.02 × 0.3 has three decimal places altogether, so the answer is 0.006, not 0.06. Confirm with an estimate: 0.02 × 0.3 is about a fiftieth, and 0.06 would be three times too large.

PRACTICE 4.5 — SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Calculate: (a) 3.6 + 1.45 (b) 8 − 2.7 (c) 0.4 + 0.06. F
2. Calculate: (a) 0.6 × 0.5 (b) 1.2 × 0.3 (c) 0.25 × 4. F
3. Calculate: (a) 4.8 ÷ 0.6 (b) 1.2 ÷ 0.04. F
4. Order from least to greatest: 0.4, 0.04, 0.404, 0.44. F
5. Explain why 0.5 × 0.5 = 0.25 rather than 2.5 or 0.025. R
6. A metre of cloth costs 3.75. Find the cost of 2.4 metres. P
7. A bottle holds 0.75 litres. How many full cups of 0.15 litres can be poured? P
8. Without calculating, decide whether 0.9 × 0.9 is greater or less than 0.9. Explain. E
9. Insert a decimal point in 1250 so that the number lies between 12 and 13. E

4.6 Fractions, decimals and the same number

CRITERION A · i · B · ii ◆ REPRESENTATION ◆ PATTERN

DEFINITION

Converting

Fraction to decimal: divide the numerator by the denominator. Decimal to fraction: write over the appropriate power of ten and simplify — 0.36 = 36/100 = 9/25. Fraction to percentage: convert to a decimal and multiply by 100.

Because the conversion is division, every fraction must do one of two things: terminate, or repeat. Which happens depends not on the numerator but on the denominator once the fraction is in simplest form.

KEY IDEA

Which fractions terminate?

A fraction in simplest form gives a terminating decimal exactly when its denominator has no prime factors other than 2 and 5. So 3/8 terminates ($8 = 2^3$) and 7/20 terminates ($20 = 2^2 \times 5$), but 1/3 and 5/6 recur, because 3 and 6 contain the prime 3.

WORKED EXAMPLE 4.6.1 — CONVERTING AND DECIDING

PROBLEM

Write (a) 5/8 and (b) 2/3 as decimals, and state whether each terminates.

1. (a) $5 \div 8 = 0.625$; $8 = 2^3$, so the decimal terminates.
2. (b) $2 \div 3 = 0.666...$; the remainder 2 repeats at every step, so the decimal recurs: 0.6 with a dot over the 6.
3. A recurring decimal is usually written to three decimal places with a note, or with a dot or bar to show the repeating block.

ANSWER (a) 0.625, terminating (b) 0.666..., recurring

$0.625 \times 8 = 5$ ✓

WORKED EXAMPLE 4.6.2 — FROM DECIMAL TO FRACTION

PROBLEM

Write 0.45 as a fraction in simplest form.

1. 0.45 = 45/100.
2. HCF of 45 and 100 is 5.
3. $45 \div 5 = 9$ and $100 \div 5 = 20$.
4. So 0.45 = 9/20.

ANSWER 9/20

$9 \div 20 = 0.45$ ✓

PRACTICE 4.6

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Write as decimals: (a) 3/4 (b) 7/10 (c) 5/8. F
2. Write as fractions in simplest form: (a) 0.6 (b) 0.25 (c) 0.05. F
3. Write 4/5 as a percentage. F
4. Which of 1/4, 1/6, 3/8, 2/9 give terminating decimals? R
5. Explain why 1/3 cannot be written exactly as a terminating decimal. R
6. A share is reported as 0.33 of a total. Is it exactly one third? Explain the difference. P
7. Order 2/5, 0.41 and 3/7 from smallest to largest, showing your method. P
8. Find a fraction that gives a recurring decimal with a repeating block of two digits. E
9. Convert 7/8 to a decimal by division and state whether it terminates. F
10. Write 0.375 as a fraction in simplest form. F
11. Explain why 0.9 recurring is equal to 1, using the fraction method. R
12. A calculator shows 0.333333333. Explain what the display has done and what the exact value is. P

04 · INVESTIGATE

INVESTIGATION 4

Which fractions stop?

PATTERN ◆ FORM

Write the fractions 1/2 to 1/12 in simplest form and convert each to a decimal by division. Record the denominator's prime factors and whether the decimal terminates or recurs.

Fraction	Simplest form	Denominator factors	Decimal	Type
1/6	1/6	2×3	0.1666...	recurs
1/8	1/8	2^3	0.125	terminates
1/9	1/9	3^2	0.1111...	recurs
1/10	1/10	2×5	0.1	terminates
1/12	1/12	$2^2 \times 3$	0.08333...	recurs

Your investigation

1. Complete the table for every fraction from 1/2 to 1/12.
2. Sort the fractions into two groups by whether the decimal terminates.
3. What do the denominators in the terminating group have in common?
4. Test your rule on 1/16, 1/25 and 1/40.
5. Find a denominator with only 2s and 5s that makes a decimal with four places.
6. Explain why the prime 3 forces a repeating pattern.

KEY IDEA

The rule the investigation produces

A fraction in simplest form terminates exactly when the denominator is built from 2s and 5s only. Any other prime in the denominator forces a repeating block, because the remainders must eventually cycle.

ATL SKILL

ATL · Transfer

In order for a student to state a rule that holds beyond the cases examined, students must test the rule on fractions outside the table, including at least one counterexample attempt.

05 · APPLY

REAL-LIFE APPLICATION

The scaling kitchen

CRITERION D · i–iv ◆ QUANTITY ◆ EQUIVALENCE

A school kitchen has a recipe for 12 portions and needs 30. Scaling a recipe means multiplying every quantity by 30/12 = 5/2, and then deciding how to measure the result with the equipment actually available.

Ingredient	For 12 portions	For 30 portions	Practical measure
Rice	3/4 kg	…	…
Oil	1/3 litre	…	…
Tomatoes	2 1/2 kg	…	…
Stock	0.6 litre	…	…

Your task

1. Calculate the scaled quantity for each ingredient, in simplest form.
2. Decide a practical measure for each: the kitchen has a 100 ml jug and scales accurate to 10 g.
3. Calculate the percentage increase in each ingredient and explain why every one is the same.
4. The kitchen has 2 kg of rice. Is that enough for 30 portions? Justify with a calculation.
5. Explain why the scaled quantities are exact even when the practical measure is rounded.

CRITERION

Criterion D · strands i–iv

- Identify the quantities and the scale factor.
- Select multiplication by a fraction as the appropriate strategy.
- Apply it accurately and convert between units.
- Justify the practical measures and evaluate suitability.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The fraction explainer

CRITERION C · i–iv · A · i, ii

Context. A younger class is convinced that 1/2 + 1/3 = 2/5 because "you add the tops and add the bottoms". Your task is a one-page explanation that corrects this without simply giving the rule.

1. Draw a diagram that shows why 2/5 is too small.
2. Show the correct calculation with a common denominator.
3. Explain, using a unit analogy, why a common denominator is needed.
4. Give one example where adding denominators accidentally gives a correct answer, and explain why it is a coincidence.
5. Test your explanation on a classmate: ask them to evaluate 1/4 + 1/6 and see whether they use your reasoning.
6. Evaluate your page: what worked, what did not, what would you change?

CRITERION

Criteria and strands developed

- Criterion A · Knowing and understanding — i, ii: accurate equivalence and calculation.
- Criterion C · Communicating — i–iv: notation, diagram and explanation for a named audience.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Equivalence comes from multiplying or dividing both parts by the same number.
- Comparing fractions needs a common unit, a landmark or a decimal.
- Adding and subtracting require a common denominator; multiplying and dividing do not.
- A decimal is a fraction in place-value notation, so decimal rules are fraction rules.
- A fraction terminates exactly when its denominator has no prime factors other than 2 and 5.

Vocabulary

Term	Meaning
numerator	The number of equal parts taken.
denominator	The number of equal parts in the whole.
equivalent fractions	Fractions naming the same value.
simplest form	The equivalent fraction with no common factor.
reciprocal	The fraction turned upside down; the number that multiplies to 1.
terminating decimal	A decimal that ends.
recurring decimal	A decimal with a repeating block of digits.

Misconceptions to watch

COMMON ERROR

Three that persist

Add the tops and the bottoms. A fifth is not a common unit for halves and thirds.

Multiplying makes numbers bigger. Multiplying by a fraction less than 1 makes the result smaller.

$0.5 \times 0.5 = 2.5$. Two decimal places in the factors give two places in the answer.

Self-assessment

I can …	Not yet	Getting there	Confident
write equivalent fractions and simplify	☐	☐	☐
compare and order fractions	☐	☐	☐
add and subtract fractions	☐	☐	☐
multiply and divide fractions	☐	☐	☐
calculate with decimals	☐	☐	☐
convert between fractions, decimals and percentages	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to choose the most efficient notation, students must notice which form the question already uses and which form the answer must take.

Looking ahead · Chapter 5 · Percentages and applications takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 4 — SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. Simplify: (a) 12/18 (b) 20/35 (c) 42/56. F
2. Calculate 2/3 + 1/6. F
3. Calculate 5/8 × 4/15. F
4. Calculate 3/4 ÷ 9/16. F
5. Write 0.35 as a fraction in simplest form and 7/20 as a decimal. F
6. Order from least to greatest: 3/5, 0.58, 11/20. R
7. Explain why 1/2 + 1/3 = 5/6 rather than 2/5. R
8. A recipe for 8 uses 2/3 cup of milk. Find the amount for 20, in simplest form. P
9. A rope 7/8 m long is cut into four equal pieces. Find the length of each. P
10. Calculate 1.25 × 0.8 and explain the position of the decimal point. F
11. Find a fraction between 2/5 and 1/2 with a denominator of 20. E
12. Show that 0.999… is equal to 1, using fractions. E
13. Explain why multiplying by 3/4 makes a number smaller but dividing by 3/4 makes it larger. R
14. A tank is 2/5 full. After adding 30 litres it is 3/4 full. Find the capacity. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–2	Revisit 4.1: multiply or divide both parts by the same number.
3–4	Revisit 4.3 and 4.4: common denominator for add, reciprocal for divide.
5–6	Revisit 4.5 and 4.6: a decimal is a fraction over a power of ten.
7–9	Revisit 4.2: draw the two fractions on one strip.
10–14	Revisit the investigation and the key idea on terminating decimals.

5

FRACTIONS, DECIMALS AND PERCENTAGES

Percentages and applications

Per cent means "per hundred", and that is the whole idea. Everything else — a discount, a tax, a budget, a price rise — is a fraction of a hundred wearing a different coat.

KEY CONCEPT · RELATIONSHIPS

RELATED · QUANTITY

RELATED · EQUIVALENCE

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How does a percentage make two different quantities comparable, and when does comparing them mislead?

01 · MYP UNIT PLANNER

Statement of inquiry	A percentage expresses a part in units of one hundred, which makes quantities with different wholes comparable; that same act of standardising can clarify a decision or conceal what produced it.

KEY IDEA

Key and related concepts

Key concept · Relationships. A percentage relates a part to its whole, so the whole must always be named.

Related · Quantity. A percentage of a quantity is found by dividing by 100 and multiplying by the percentage.

Related · Equivalence. 25 %, 0.25 and 1/4 are three names for one number, and each makes a different calculation easy.

NOTE

Global context · Fairness and development

Exploration: how discounts, taxes, interest and aid targets are communicated, and how a percentage can be accurate and still misleading when its base is hidden.

Inquiry questions

- Factual: How do you find a percentage of a quantity, and how do you express one quantity as a percentage of another?
- Conceptual: Why does a 20 % discount followed by a 20 % increase not restore the original price?
- Debatable: Is a “50 % off” label informative if the original price is unknown?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Convert between forms; find percentages of quantities; increase and decrease.
B · Investigating patterns	i, ii, iii	Investigate whether two successive discounts equal their sum.
C · Communicating	i, ii, iii, iv	Present a price comparison so that the base of each percentage is visible.
D · Applying mathematics	i, ii, iii, iv	Cost a market stall, set prices and evaluate a budget.

ATL SKILL

ATL · Critical thinking

In order for a student to judge a percentage claim, students must identify the base it refers to before accepting the figure, because the same number can describe very different quantities.

02 · ENGAGE

Two shops, two discounts.

Shop A offers 30 % off a shirt marked 40. Shop B offers 20 % off the same shirt marked 45. Both signs look generous. One shirt is cheaper, and the difference is only visible after both percentages have been applied to their own base — which is the whole point of a percentage, and the whole danger of one.



FIGURE 1 A shop window standardises everything to hundreds. The standardising is what makes comparison possible; the base is what makes it honest.

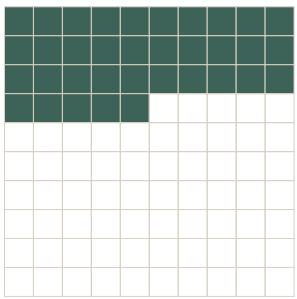
Original illustration created for this edition.

03 · EXPLORE

Build a hundred square.

Draw a 10 by 10 square of 100 cells. Shade 35 of them. Now answer without counting the shading again:

1. What fraction of the square is shaded?
2. What decimal is shaded?
3. What percentage is shaded?
4. Shade a further 15 cells. What percentage is shaded now, and how did you get the answer?
5. What percentage would you need to shade to reach exactly three quarters?



35 cells shaded out of 100

fraction	35/100
decimal	0.35
percentage	35 %

one region, three names

FIGURE 2 A hundred square makes 35 % visible as 35/100 = 0.35. The three notations describe one region.

5.1 Per cent: a proportion of 100

CRITERION A · i ◆ EQUIVALENCE

DEFINITION

Percentage

A percentage is a fraction with denominator 100. So 47 % = 47/100 = 0.47. The symbol % is shorthand for “divided by 100”.

Converting between the three forms

From	To	Method	Example
Fraction	Decimal	Divide numerator by denominator	3/8 = 0.375
Decimal	Percentage	Multiply by 100	0.375 = 37.5 %
Percentage	Decimal	Divide by 100	45 % = 0.45
Percentage	Fraction	Write over 100 and simplify	45 % = 9/20
Fraction	Percentage	Decimal first, then multiply by 100	3/8 = 37.5 %

WORKED EXAMPLE 5.1.1 THREE NAMES FOR ONE NUMBER

PROBLEM

Express 5/8 as a decimal and as a percentage.

1. 5 ÷ 8 = 0.625.
2. 0.625 × 100 = 62.5.
3. So 5/8 = 0.625 = 62.5 %.

ANSWER 0.625 and 62.5 %

62.5 % of 8 = 5, so the conversion is consistent √

COMMON ERROR

Forgetting that the base must be 100

A student writes 0.4 % for 0.4. Percentages are already hundredths, so 0.4 = 40 %. Reversing the direction of the conversion is the most common percentage error; check by asking whether the answer is bigger or smaller than the original number.

PRACTICE 5.1 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Write as percentages: (a) 0.6 (b) 0.08 (c) 1.25. F
2. Write as decimals: (a) 45 % (b) 7 % (c) 120 %. F
3. Write as fractions in simplest form: (a) 25 % (b) 60 % (c) 12.5 %. F
4. Write 7/20 as a percentage. F
5. Explain why 1 % of a quantity is not always small. R
6. A survey of 40 students finds 12 prefer football. Write this as a percentage. P
7. Which is larger, 35 % of 80 or 40 % of 70? Show your reasoning. R
8. Find a percentage that is equal to 2/3 when rounded to one decimal place, and state whether it is exact. E

5.2 Finding a percentage of a quantity

CRITERION A · ii ◆ QUANTITY

DEFINITION

The method

To find p % of a quantity: convert the percentage to a decimal and multiply, or multiply by p and divide by 100. So 15 % of 240 = 0.15 × 240 = 36.

The decimal method is faster with a calculator; the fraction method is often faster in the head. 25 % of 84 is a quarter of 84, which is 21 — no calculator needed. Choosing the notation that fits the numbers is a mathematical decision, not a shortcut.



FIGURE 3 10 % and 5 % land-markers give 15 % of 240 without a calculator: 24 + 12 = 36.

WORKED EXAMPLE 5.2.1 — TWO ROUTES TO THE SAME ANSWER

PROBLEM

Find 35 % of 260.

1. Decimal route: 35 % = 0.35, and 0.35 × 260 = 91.
2. Fraction route: 35/100 × 260 = 9 100/100 = 91.
3. Landmark route: 10 % is 26, so 30 % is 78, and 5 % is 13; 78 + 13 = 91.

ANSWER 91

35 % is a bit more than a third, and a third of 260 is about 87 √

WORKED EXAMPLE 5.2.2 — ONE QUANTITY AS A PERCENTAGE OF ANOTHER

PROBLEM

A class of 32 students has 24 who walk to school. What percentage walk?

1. Divide the part by the whole: 24 ÷ 32 = 0.75.
2. Multiply by 100: 75 %.
3. Check: 75 % of 32 = 24.

ANSWER 75 %

0.75 × 32 = 24 √

PRACTICE 5.2

SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Find (a) 10 % of 90 (b) 25 % of 84 (c) 50 % of 7. F
2. Find (a) 15 % of 240 (b) 35 % of 260 (c) 60 % of 45. F
3. Find 12.5 % of 96. F
4. A team plays 40 matches and wins 26. What percentage does it win? F
5. Explain how to find 15 % of a quantity using only 10 % and half of that. R
6. A shop has 180 items, of which 45 are sold. What percentage remains? P
7. In a school of 750 students, 54 % are girls. How many boys are there? P
8. A jacket costs 64. Its price is reduced by 12.5 %. Find the new price using fractions. E
9. A student says 20 % of 50 is 10 and 50 % of 20 is 10, so the two statements are the same. Are they? Explain. E

5.3

Increasing and decreasing by a percentage

CRITERION A · iii ◆ QUANTITY ◆ CHANGE

An increase of 20 % can be calculated in two ways: find 20 % and add it, or multiply by 1.20 directly. The second method is shorter and, more importantly, it generalises: a multiplier of 0.85 decreases by 15 % in one step.

KEY IDEA

The multiplier

An increase of p % multiplies by $1 + p/100$. A decrease of p % multiplies by $1 - p/100$. So +20 % is × 1.20, −15 % is × 0.85, and +100 % is × 2.

WORKED EXAMPLE 5.3.1 — A PRICE RISE

PROBLEM

A bus fare of 2.40 rises by 15 %. Find the new fare.

1. The multiplier is $1 + 0.15 = 1.15$.
2. $2.40 \times 1.15 = 2.76$.
3. Check: the increase is 0.36, which is 15 % of 2.40.

ANSWER 2.76

2.40 + 0.36 = 2.76 ✓

WORKED EXAMPLE 5.3.2 — COMPARING TWO CHANGES

PROBLEM

A quantity rises by 20 % and then falls by 20 %. Is it back to where it started?

1. Rising by 20 % multiplies by 1.20; falling by 20 % multiplies by 0.80.
2. Overall multiplier: $1.20 \times 0.80 = 0.96$.
3. So the quantity ends at 96 % of its original value: a 4 % decrease.
4. The two percentages applied to different bases, so they do not cancel.

ANSWER No — it finishes 4 % below the starting value.

On 100: 120, then 96 ✓

COMMON ERROR

Adding percentages that apply to different bases

A 20 % rise followed by a 20 % fall is not 0 %. The second percentage is taken from the new, larger value, so it removes more than the first added. Multiply the multipliers; never add the percentages.

PRACTICE 5.3 SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Increase 80 by 10 %. F
2. Decrease 250 by 20 %. F
3. Find the multiplier for (a) +35 % (b) −12 % (c) +150 %. F
4. A salary of 1 800 rises by 4 %. Find the new salary. F
5. Explain why a 10 % rise followed by a 10 % fall does not return to the original value. R
6. A phone costs 320 and is reduced by 15 % in a sale, then by a further 10 % at the till. Find the final price. P
7. After a 25 % increase a price is 75. Find the original price. P
8. A population grows by 5 % in year 1 and 8 % in year 2. Find the two-year percentage change. E
9. Find a single multiplier equivalent to +20 % followed by −25 %. E

5.4 Discounts, taxes and budgets

CRITERION D · i · A · iii ◆ QUANTITY ◆ MODEL

Financial contexts are where percentages meet decisions. A discount reduces a price; a tax adds to it; a budget allocates a whole into parts. Each requires the same three questions: what is the base, what is the percentage, and is the result sensible?

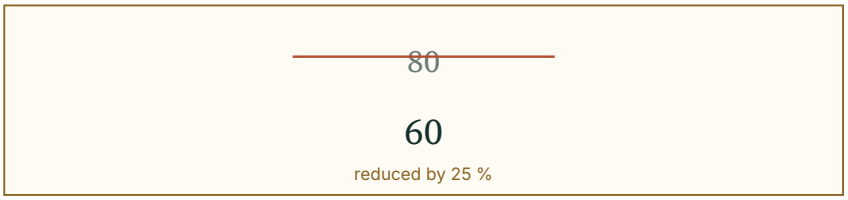


FIGURE 4 A reduction of 25 % applies to the marked price, not to the reduced price. Naming the base is part of the calculation.

WORKED EXAMPLE 5.4.1 — DISCOUNT THEN TAX

PROBLEM

An item marked 80 is reduced by 25 %, then a 10 % tax is added to the reduced price. Find the final amount.

1. Reduction: $80 \times 0.75 = 60$.
2. Tax on the reduced price: $60 \times 1.10 = 66$.
3. Combined multiplier: $0.75 \times 1.10 = 0.825$, so the final amount is 82.5 % of the marked price.

ANSWER 66

$80 \times 0.825 = 66$ ✓

WORKED EXAMPLE 5.4.2 — ALLOCATING A BUDGET

PROBLEM

A household income of 2 400 is allocated as 35 % housing, 25 % food, 15 % transport, 10 % school costs and the rest to savings. Find each amount.

1. Housing: $0.35 \times 2\,400 = 840$.
2. Food: $0.25 \times 2\,400 = 600$. Transport: $0.15 \times 2\,400 = 360$.
3. School: $0.10 \times 2\,400 = 240$.
4. Percentages used: $35 + 25 + 15 + 10 = 85$, so savings take 15 %: 360.

ANSWER 840, 600, 360, 240 and 360 to savings.

$840 + 600 + 360 + 240 + 360 = 2\,400$ ✓

PRACTICE 5.4

SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. A 60 item is reduced by 15 %. Find the sale price. F
2. A bill of 45 attracts 10 % tax. Find the total. F
3. A budget of 900 allocates 40 % to rent. How much is left? F
4. Find the total cost of a 120 item with 20 % discount and 5 % tax. F
5. Explain why a shop can honestly advertise “25 % off” and still make a profit. R
6. A stall sells 150 items at 4.50 each, with 8 % of revenue paid as a market fee. Find the amount the trader keeps. P
7. A family spends 30 % of income on rent and 45 % on food and transport. It saves 500. Find the income. P
8. Two shops discount the same 200 item: shop A by 30 %, shop B by 20 % then a further 10 % at the till. Which is cheaper, and by how much? E
9. Design a budget for 3 000 in which savings are exactly 12 % and no category exceeds 40 %. E

04 · INVESTIGATE

INVESTIGATION 5

Do two discounts add up?

PATTERN ◆ QUANTITY

A shop advertises “20 % off, then a further 20 % off at the till”. Most customers assume the total discount is 40 %. Test that assumption on a range of starting prices before deciding.

Starting price	After 20 %	After a further 20 %	Total discount %
50	…	…	…
100	…	…	…
250	…	…	…
400	…	…	…

Your investigation

1. Complete the table.
2. Is the total discount the same for every starting price? Explain why.
3. Write the total discount as a single multiplier.
4. Predict the total discount for two successive discounts of p % each, and test your rule with $p = 10$ and $p = 50$.
5. Find the single discount that is equivalent to 20 % followed by 20 %.
6. Explain, in one sentence, why the two discounts do not add.

KEY IDEA

The general result

Two successive discounts of p % and q % multiply: the combined multiplier is $(1 - p/100)(1 - q/100)$. For 20 % and 20 % this is $0.8 \times 0.8 = 0.64$, a total discount of 36 %, always — the difference from 40 % is the discount on the discount.

ATL SKILL

ATL · Transfer

In order for a student to test a claim that “feels obvious”, students must compute at least three cases with different starting values, because a single case can agree with a false rule by coincidence.

05 · APPLY

REAL-LIFE APPLICATION

The community market stall

CRITERION D · i–iv ◆ QUANTITY ◆ MODEL

A trader buys 200 kg of tomatoes at 1.20 per kilogram. Transport costs 45. About 8 % of the stock spoils before sale. The trader wants a 35 % profit on total cost and sells the rest in 2 kg bags.

Your task

1. Find the total cost, including transport.
2. Find the mass of tomatoes that can be sold.
3. Find the target revenue for a 35 % profit on cost.
4. Find the selling price per 2 kg bag, to the nearest 0.10.
5. State one reason the trader might choose a lower price, and calculate the profit at that price.
6. Explain how a small error in the spoilage percentage affects the price.

CRITERION

Criterion D · strands i–iv

- Identify costs, losses and the target.
- Select percentage increase and unit pricing strategies.
- Apply them accurately across several steps.
- Justify the price and evaluate the sensitivity to the spoilage rate.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The fair-price card

CRITERION C · i–iii · D · ii, iii

Context. Customers at the stall often misread percentage labels. Produce a card, no larger than A5, that explains how a discount and a tax combine, using one worked example and one diagram.

1. State what a percentage is in one sentence.
2. Show 80 with a 25 % discount and then 10 % tax, line by line.
3. Include a diagram showing that the two percentages apply to different bases.
4. Explain why “20 % then 20 %” is not 40 %.
5. Add one sentence telling a customer how to check a price quickly.
6. Test the card on someone outside the class and record one misunderstanding it did not prevent.
7. Revise one part of the card and evaluate the revision.

CRITERION

Criteria and strands developed

- Criterion C · Communicating — i–iii: precise language, a clear diagram and an identified audience.
- Criterion D · Applying mathematics in real-life contexts — ii, iii: an accurate worked example and a justified check.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- A percentage is a fraction over 100, so all three notations are interchangeable.
- A percentage always has a base, and the base must be named to interpret it.
- A percentage of a quantity is found by multiplying by a decimal or a fraction.
- Successive percentage changes multiply; they never add.
- Financial decisions require the calculation and a judgement about reasonableness.

Vocabulary

Term	Meaning
percentage	A fraction expressed in hundredths.
base	The whole to which a percentage refers.
multiplier	The single number by which a quantity is multiplied for a change.
discount	A reduction from a marked price.
tax	A percentage added to a price by an authority.
budget	An allocation of a whole amount into named parts.

Misconceptions to watch

COMMON ERROR

Three that persist

0.4 = 0.4 %. 0.4 is 40 %; percentages are already hundredths.

+20 % then −20 % is no change. The multipliers 1.2 and 0.8 give 0.96.

A bigger percentage means a bigger amount. 40 % of 70 (28) is less than 35 % of 80 (28 is equal, in fact) — always compare the calculated amounts.

Self-assessment

I can …	Not yet	Getting there	Confident
convert between fractions, decimals and percentages	☐	☐	☐
find a percentage of a quantity	☐	☐	☐
express one quantity as a percentage of another	☐	☐	☐
increase and decrease by a percentage	☐	☐	☐
use a multiplier for successive changes	☐	☐	☐
judge whether a percentage claim is informative	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to read a percentage claim critically, students must ask what the base is, whether the percentages were added or multiplied, and what the absolute amounts are.

Looking ahead · Chapter 6 · Patterns, expressions and equations takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 5

SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. Write 0.85 as a percentage and 6 % as a decimal. F
2. Find 15 % of 320. F
3. Increase 180 by 12 %. F
4. Decrease 90 by 30 %. F
5. Write 18 out of 24 as a percentage. F
6. A price of 48 rises by 25 % and then falls by 20 %. Find the final price and compare it with the original. R
7. Explain why a 50 % increase followed by a 50 % decrease is not a return to the original value. R
8. A jacket is reduced from 120 to 102. Find the percentage reduction. P
9. A school has 480 students; 45 % are in the lower school. How many are in the upper school? P
10. A meal costs 32 and a 15 % service charge is added. Find the total and the amount of the charge. P
11. A trader's cost is 80 and the selling price is 100. Find the profit as a percentage of cost and as a percentage of selling price. Explain why the two differ. E
12. Find the original price if 64 is the price after a 20 % discount. E
13. Investigate: after what single percentage increase does a 25 % decrease return the value to its start? E
14. A population of 2 500 grows by 4 % a year for two years. Find the population, and compare with a single 8 % increase. E

09 · CHECK YOUR PROGRESS

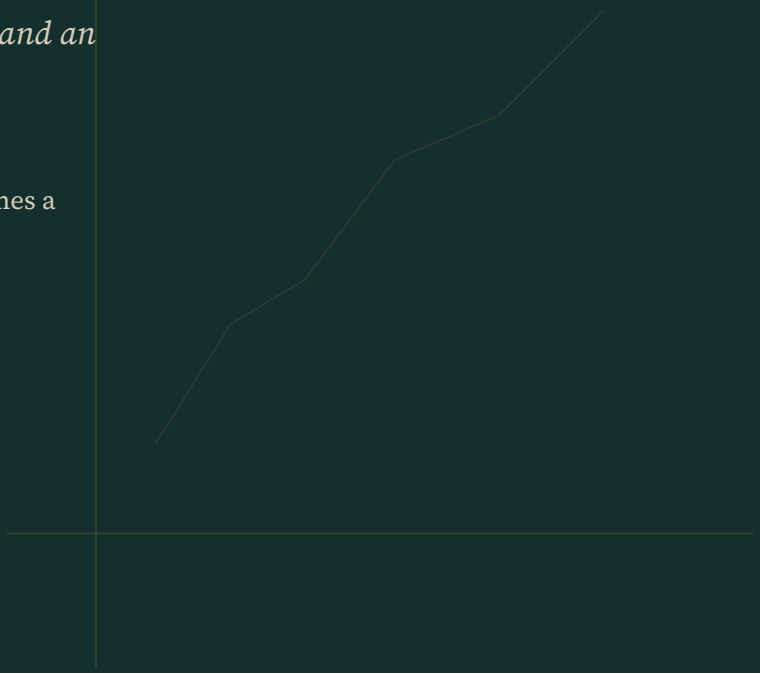
Question	If you cannot do this …
1–2	Revisit 5.1 and 5.2: write the percentage as a decimal first.
3–5	Revisit 5.3: use a multiplier, not two steps.
6–8	Revisit the investigation: successive changes multiply.
9–11	Revisit 5.4: name the base before calculating.
12–14	Revisit 5.3 and work backwards by dividing by the multiplier.

PART III

Algebra as compressed reasoning

A letter is a number you have not chosen yet, and an equation is a question written as a balance.

Algebra begins as shorthand for a pattern and becomes a method for finding what is not yet known.



6

ALGEBRA AS COMPRESSED REASONING

Patterns, expressions and equations

A letter is a number you have not chosen yet. An equation is a question written as a balance. Algebra is not a new arithmetic; it is arithmetic that keeps its options open.

KEY CONCEPT · RELATIONSHIPS

RELATED · PATTERN

RELATED · GENERALISATION

CONTEXT · PERSONAL AND CULTURAL EXPRESSION

INQUIRY

How does a pattern become a rule, and how does a rule let us answer a question without doing every step by hand?

01 · MYP UNIT PLANNER

Statement of inquiry	A pattern becomes mathematics when it is described by a general rule, and a rule becomes useful when it can be rearranged to answer a question about a value that is not yet known.

KEY IDEA

Key and related concepts

Key concept · Relationships. An expression records how one quantity depends on another, which is why substituting a value produces a predictable result.

Related · Pattern. A linear pattern changes by a constant difference, and that constant is the coefficient in the rule.

Related · Generalisation. A rule stated with a letter holds for every case, not only the cases examined.

NOTE

Global context · Personal and cultural expression

Exploration: how weaving, tiling and ornament use repeating structures, and how the same structure can be described in words, in a table, in a graph and in a rule.

Inquiry questions

- Factual: How do you find the next term of a sequence, and how do you substitute into an expression?
- Conceptual: Why does solving an equation preserve the equality when you do the same thing to both sides?
- Debatable: Is a rule more useful in words, in a table, in a graph or in symbols?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Substitute, simplify, expand and solve one- and two-step equations.
B · Investigating patterns	i, ii, iii	Investigate the matchstick chain and generalise the rule.
C · Communicating	i, ii, iii, iv	Explain a pattern in words, symbols and a table consistently.
D · Applying mathematics	i, ii, iii, iv	Model a courtyard paving project and judge the model.

ATL SKILL

ATL · Transfer

In order for a student to generalise a pattern, students must first record several cases in a table, then describe in words what changes by a constant amount, and only then introduce a letter.

02 · ENGAGE

The loom.

A narrow-strip weaver repeats one motif along a long band. Each repeat adds the same length of cloth. If she knows the length of one motif and the length of the starting border, she can predict the length of a band with any number of motifs without measuring it — provided she has found the rule.



FIGURE 1 A repeating motif is a linear pattern: the length grows by a constant amount per repeat. Algebra is the shortest way to write that sentence.

Original illustration created for this edition.

03 · EXPLORE

Find the rule from the table.

A band has a border of 6 cm, then motifs of 9 cm each. Complete the table, then describe how the length grows.

Motifs	1	2	3	4	10	n
Length (cm)	15	24	33	…	…	…

1. By how much does the length grow for each extra motif?
2. What would the length be with 0 motifs, and what does that number mean?
3. Write the rule in words.
4. Write the rule with a letter n for the number of motifs.
5. Use the rule to find the length for 25 motifs.

KEY IDEA

The structure you found

Length = $9n + 6$. The 9 is the amount added per motif; the 6 is what is there before any motif is added. Every linear pattern has those two parts: a constant difference and a starting value.

6.1 Patterns and rules

CRITERION B · i, ii · A · i ◆ PATTERN ◆ GENERALISATION

A sequence is an ordered list of numbers. If consecutive terms differ by the same amount, the sequence is linear, and the rule has the form term = difference × position + starting value.

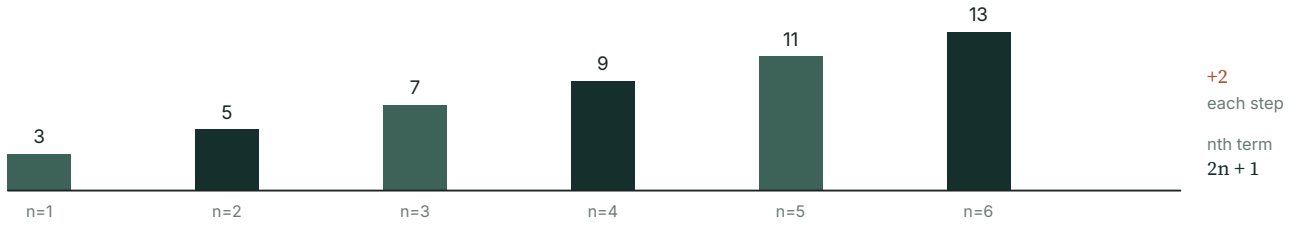


FIGURE 2 A linear sequence as steps of equal height: each step adds the same difference.

DEFINITION

nth term

The nth term of a linear sequence is an expression in n that gives the value at position n. If the first term is a and the common difference is d, then the nth term is $a + (n - 1)d$.

WORKED EXAMPLE 6.1.1 FINDING THE NTH TERM

PROBLEM

Find the nth term of the sequence 5, 8, 11, 14, ...

1. The difference between consecutive terms is 3, so the rule contains $3n$.
2. Compare $3n$ with the terms: for $n = 1$, $3n = 3$, but the term is 5.
3. The rule must add 2: $3n + 2$.
4. Check with $n = 3$: $3 \times 3 + 2 = 11$, which matches the third term.

ANSWER nth term = $3n + 2$

$n = 4$ gives 14 ✓ and $n = 1$ gives 5 ✓

WORKED EXAMPLE 6.1.2 USING THE RULE TO ANSWER A QUESTION

PROBLEM

A sequence has nth term $4n - 1$. Which position in the sequence holds 39?

1. Set the rule equal to the value: $4n - 1 = 39$.
2. Add 1 to both sides: $4n = 40$.
3. Divide both sides by 4: $n = 10$.
4. So 39 is the 10th term.

ANSWER The 10th term.

$4 \times 10 - 1 = 39$ ✓

PRACTICE 6.1 SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Write the next three terms: (a) 3, 7, 11, … (b) 20, 17, 14, … F
2. Find the nth term: (a) 2, 5, 8, … (b) 10, 14, 18, … F
3. Find the nth term of 1, 6, 11, 16, … F
4. A sequence has nth term $5n + 3$. Find the 10th term. F
5. Explain how you can tell from a table whether a sequence is linear. R
6. A tiling pattern uses 12 tiles for the first row and 5 more for each row after. Write a rule for the number of tiles in row n. P
7. Which term of the sequence 7, 12, 17, … is 62? P
8. Find a sequence whose nth term is $2n + 1$ and another whose nth term is $3n - 2$. Write the first four terms of each and compare. E
9. Explain why the nth term of an increasing linear sequence must contain n with a positive coefficient. E

6.2 Expressions and substitution

CRITERION A · i, ii ◆ FORM

DEFINITION

Expression, term, coefficient

An expression combines numbers, letters and operations without an equals sign: $3x + 5$. A term is one part of it separated by + or −. A coefficient is the number multiplying a letter: in $3x$, the coefficient is 3.

Substituting a value into an expression is a test of whether you can read its structure. In $3x + 5$, the multiplication happens before the addition, so $x = 2$ gives 11, not 16.

WORKED EXAMPLE 6.2.1 SUBSTITUTING INTO EXPRESSIONS

PROBLEM

Evaluate $4a - 2b$ when $a = 3$ and $b = 5$.

1. Replace each letter: $4 \times 3 - 2 \times 5$.
2. Multiply first: $12 - 10$.
3. Subtract: 2.

ANSWER 2

Pick easier numbers: $a = 1$, $b = 1$ gives $4 - 2 = 2$, the same value coincidentally; test $a = 2$, $b = 1$ to confirm the method gives 6 ✓

WORKED EXAMPLE 6.2.2 WRITING AN EXPRESSION FROM A SITUATION

PROBLEM

A taxi charges a fixed 3 plus 2 per kilometre. Write an expression for the cost of k kilometres and find the cost of a 7 km journey.

1. The fixed charge does not change with distance: 3.
2. The distance charge is 2 per kilometre: $2k$.
3. Expression: $2k + 3$.
4. For $k = 7$: $2 \times 7 + 3 = 17$.

ANSWER $2k + 3$; a 7 km journey costs 17.

A 1 km journey should cost 5; $2(1) + 3 = 5$ ✓

PRACTICE 6.2 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Evaluate $2x + 7$ when $x = 4$. F
2. Evaluate $5y - 3$ when $y = 2$. F
3. Evaluate $3a + 2b$ when $a = 2$ and $b = 5$. F
4. Write an expression for: (a) 5 more than n (b) 3 times m (c) 8 less than twice p. F
5. Explain why $3x + 2$ with $x = 4$ is 14 and not 18. R
6. A phone plan costs 12 a month plus 0.05 per minute. Write an expression for the monthly cost of m minutes. P
7. A rectangle has length $x + 3$ and width x. Write an expression for its perimeter. P
8. Find a value of n for which $4n$ and n^2 are equal, other than $n = 0$. E

6.3 Simplifying and expanding

CRITERION A · ii ◆ SIMPLIFICATION

Like terms contain the same letter with the same power, so they can be combined: $3x + 5x = 8x$ because three lots of x plus five lots of x is eight lots of x. Unlike terms such as $3x$ and $3x^2$ cannot be combined, because they count different things.

KEY IDEA

Expanding a bracket

A bracket is a multiplication: $3(x + 4)$ means three copies of $(x + 4)$. Multiplying each term inside by the number outside gives $3x + 12$. This is the distributive law, and it is the reverse of factorising.

WORKED EXAMPLE 6.3.1 — COLLECTING LIKE TERMS

PROBLEM

Simplify $7a + 3b - 2a + 5b$.

1. Collect the a terms: $7a - 2a = 5a$.
2. Collect the b terms: $3b + 5b = 8b$.
3. The simplified expression is $5a + 8b$; the two terms are unlike, so they stay separate.

ANSWER $5a + 8b$

Substitute $a = 1$, $b = 1$: original gives $7 + 3 - 2 + 5 = 13$; simplified gives $5 + 8 = 13$ ✓

WORKED EXAMPLE 6.3.2 — EXPANDING AND SIMPLIFYING

PROBLEM

Expand and simplify $4(x + 3) + 2(x - 1)$.

1. Expand each bracket: $4x + 12$ and $2x - 2$.
2. Add the results: $4x + 12 + 2x - 2$.
3. Collect like terms: $6x + 10$.

ANSWER $6x + 10$

$x = 2$: original $4(5) + 2(1) = 22$; simplified $12 + 10 = 22$ ✓

COMMON ERROR

$3x + 4 = 7x$

Adding unlike terms is not allowed. $3x + 4$ stays as it is; only if the second number also multiplies x can the two be combined. Substitute a value to see the difference: with $x = 10$, $3x + 4 = 34$, not 70.

PRACTICE 6.3 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Simplify: (a) $4x + 3x$ (b) $9y - 2y$ (c) $6a + a$. F
2. Simplify: (a) $3x + 2 + 5x$ (b) $8m - 3 - 2m$. F
3. Expand: (a) $3(x + 2)$ (b) $5(2y - 1)$. F
4. Expand and simplify $2(x + 4) + 3(x + 1)$. F
5. Explain why $2x + 3x$ can be simplified but $2x + 3x^2$ cannot. R
6. A rectangle has sides $x + 2$ and 3. Write and simplify an expression for its perimeter. P
7. A student writes $5(x + 2) = 5x + 2$. Diagnose the error and correct it. R
8. Find two expressions that simplify to $4x + 6$, one of which uses a bracket. E

6.4 Equations and formulas

CRITERION A · iii · C · ii ◆ RELATIONSHIPS ◆ MODEL

DEFINITION

Equation and solution

An equation states that two expressions are equal: $3x + 2 = 11$. A solution is the value of the letter that makes the statement true. Solving means finding that value, not guessing it.

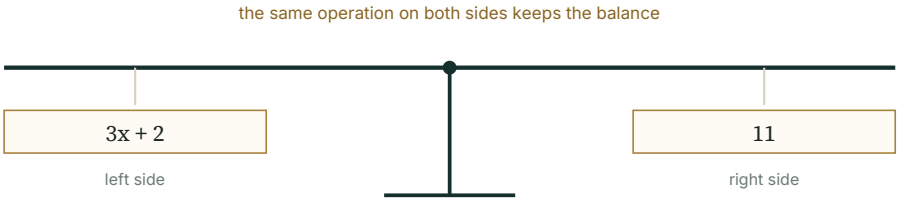


FIGURE 3 An equation is a balance. Whatever is done to one side must be done to the other, or the equality is destroyed.

The method

- Undo addition or subtraction first, using the inverse operation on both sides.
- Undo multiplication or division second.
- Check by substituting the solution back into the original equation.
- Keep the equals signs aligned so that each line states a true equality.

WORKED EXAMPLE 6.4.1 — A TWO-STEP EQUATION

PROBLEM

Solve $4x + 7 = 31$.

1. Subtract 7 from both sides: $4x = 24$.
2. Divide both sides by 4: $x = 6$.
3. Check: $4 \times 6 + 7 = 24 + 7 = 31$.

ANSWER $x = 6$

Substitution returns the original right-hand side ✓

WORKED EXAMPLE 6.4.2 — AN EQUATION WITH A BRACKET

PROBLEM

Solve $3(x - 2) = 18$.

1. Divide both sides by 3: $x - 2 = 6$.
2. Add 2 to both sides: $x = 8$.
3. Alternatively expand first: $3x - 6 = 18$, so $3x = 24$ and $x = 8$.

ANSWER $x = 8$

$3(8 - 2) = 18$ ✓

WORKED EXAMPLE 6.4.3 — REARRANGING A FORMULA

PROBLEM

The perimeter of a rectangle is $P = 2(l + w)$. Find l when $P = 26$ and $w = 4$.

1. Substitute: $26 = 2(l + 4)$.
2. Divide by 2: $13 = l + 4$.
3. Subtract 4: $l = 9$.

ANSWER $l = 9$

$2(9 + 4) = 26$ ✓

COMMON ERROR

Doing an operation to one side only

From $4x + 7 = 31$, subtracting 7 from the left alone gives $4x = 31$, which is false. The balance metaphor is precise: an equation is a statement of equality, and equality survives only if both sides change together.

PRACTICE 6.4

SUPPORT 1–4 · CORE 3–8 · EXTENSION 7–9

1. Solve: (a) $x + 5 = 12$ (b) $x - 3 = 9$ (c) $2x = 14$. F
2. Solve: (a) $3x + 1 = 16$ (b) $5x - 4 = 21$. F
3. Solve $2x + 9 = 3$. F
4. Solve $4(x + 1) = 20$. F
5. Explain why you subtract before dividing in $3x + 5 = 20$. R
6. A taxi charges 3 plus 2 per kilometre. A journey costs 23. How far was it? P
7. The perimeter of a square is 44 cm. Write an equation and solve it to find the side length. P
8. Solve $3x + 2 = x + 10$, explaining each step. E
9. Write an equation with solution $x = 5$ that contains a bracket and a constant. E
10. Solve $x/4 = 7$. F
11. Solve $5 - x = 2$. F
12. Explain what happens to the balance when you multiply both sides by 3. R
13. A rectangle has length $x + 4$ and width 3. Its perimeter is 30. Write and solve an equation. P

04 · INVESTIGATE

INVESTIGATION 6

The matchstick chain

PATTERN ◆ GENERALISATION

Squares are made from matchsticks in a chain, sharing one side each time. Build the first four shapes or draw them, then count the matchsticks carefully — the shared side is the whole difficulty.

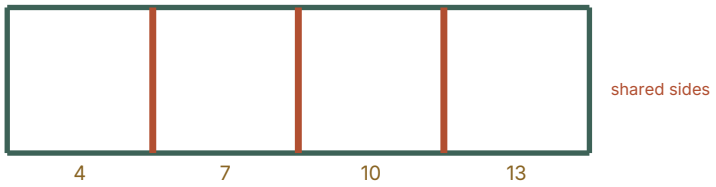


FIGURE 4 Three squares in a chain use 10 matchsticks, not 12, because each new square shares one side.

Squares	1	2	3	4	5	n
Matchsticks	4	7	10	…	…	…

Your investigation

1. Count the matchsticks for 1, 2, 3 and 4 squares.
2. Explain in words why the count does not grow by 4 each time.
3. Find the constant difference and write the rule.
4. Predict the number of matchsticks for 20 squares and check by calculation.
5. How many squares can be made with 100 matchsticks? Write the equation you solved.
6. Now build a different chain: an L-shape, or a row of triangles. Find its rule and compare the constant differences.

KEY IDEA

Why the rule has that shape

The first square needs 3 matchsticks plus the closing side: 4. Each extra square adds 3. So matchsticks = $3n + 1$. The constant 3 is the number of new matchsticks per square; the +1 is the side that closes the first square.

ATL SKILL

ATL · Critical thinking

In order for a student to trust a general rule, students must test it against the smallest case, where the count can be verified by hand.

05 · APPLY

REAL-LIFE APPLICATION

The courtyard pond

CRITERION D · i–iv ◆ RELATIONSHIPS ◆ MODEL

A courtyard is to be paved around a rectangular pond. The pond is 3 m by 2 m, and the paved border is x metres wide on every side. The builder needs the paved area to quote for tiles.

Your task

1. Draw the courtyard for $x = 1$ and label every dimension.
2. Write the overall length and width in terms of x.
3. Write an expression for the total area and a second expression for the pond area.
4. Hence write an expression for the paved area, simplified.
5. Find the paved area when $x = 1.5$ m.
6. Explain one limitation of the model for very large x.

CRITERION

Criterion D · strands i–iv

- Identify the variables and the fixed quantities.
- Select algebraic modelling as the strategy.
- Apply expansion correctly to obtain the paved area.
- Evaluate the model's limits in the real context.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The pattern explainer

CRITERION B · i, ii · C · i, ii

Context. Your class is producing a display of patterns for a parents' evening. Each student explains one pattern so that a visitor can predict the next term and the hundredth term without counting.

1. Choose a pattern that grows by a constant difference and draw its first four terms.
2. Present the pattern in three forms: a drawing, a table and a rule.
3. Explain the meaning of each number in your rule in words.
4. Predict the 100th term and show the substitution.
5. Ask a visitor or classmate to use your display to find the 50th term. Record the result.
6. Evaluate your display: what made the rule easy to see, and what obscured it?

CRITERION

Criteria and strands developed

- Criterion B · Investigating patterns — i, ii: correct generalisation from the cases examined.
- Criterion C · Communicating — i, ii: consistent presentation across three representations.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- A linear pattern has a constant difference, which appears as the coefficient.
- The nth term lets you answer questions about any position without listing.
- Substituting into an expression tests whether you can read its structure.
- Like terms can be combined; unlike terms cannot.
- Solving an equation means keeping a balance true at every step.

Vocabulary

Term	Meaning
sequence	An ordered list of numbers.
nth term	An expression giving the value at position n.
expression	A combination of numbers, letters and operations.
term	One part of an expression separated by + or −.
coefficient	The number multiplying a letter.
expand	Multiply out a bracket.
equation	A statement that two expressions are equal.
solution	The value that makes an equation true.

Misconceptions to watch

COMMON ERROR

Three that persist

$3x + 4 = 7x$. 4 is not a number of x.

$5(x + 2) = 5x + 2$. The 5 multiplies both terms.

Moving a term without changing its sign. Subtract from both sides rather than “moving”.

Self-assessment

I can …	Not yet	Getting there	Confident
find the next terms of a sequence	☐	☐	☐
find and use the nth term	☐	☐	☐
substitute into an expression	☐	☐	☐
simplify and expand	☐	☐	☐
solve one- and two-step equations	☐	☐	☐
form an equation from a situation	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to check an algebraic simplification, students must substitute one convenient value into both the original and the simplified expression and confirm the results agree.

Looking ahead · Chapter 7 · Lines, angles, triangles and constructions takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 6 — SUPPORT 1–5 · CORE 4–12 · EXTENSION 10–15

1. Write the next two terms: 6, 11, 16, ... F
2. Find the nth term of 4, 7, 10, ... F
3. Evaluate $3x - 4$ when $x = 5$. F
4. Simplify $5a + 2b - a + 3b$. F
5. Solve $2x + 7 = 19$. F
6. Expand $4(2x - 3)$. F
7. Solve $3(x - 4) = 15$. F
8. Explain why $6x$ and $6x^2$ cannot be added. R
9. A plumber charges 25 call-out plus 18 per hour. Write an expression for h hours and find the charge for 3 hours. P
10. Which term of 9, 13, 17, ... is 97? P
11. Solve $5x - 2 = 2x + 13$. E
12. The sum of three consecutive numbers is 72. Form an equation and solve it. E
13. A rectangle has length $x + 4$ and width $x - 1$. Write and simplify an expression for its perimeter, then find the perimeter when $x = 6$. P
14. Explain how the constant difference of a sequence appears in its graph. R
15. Write two different expressions that both give the perimeter of a rectangle of length l and width w. E

09 · CHECK YOUR PROGRESS

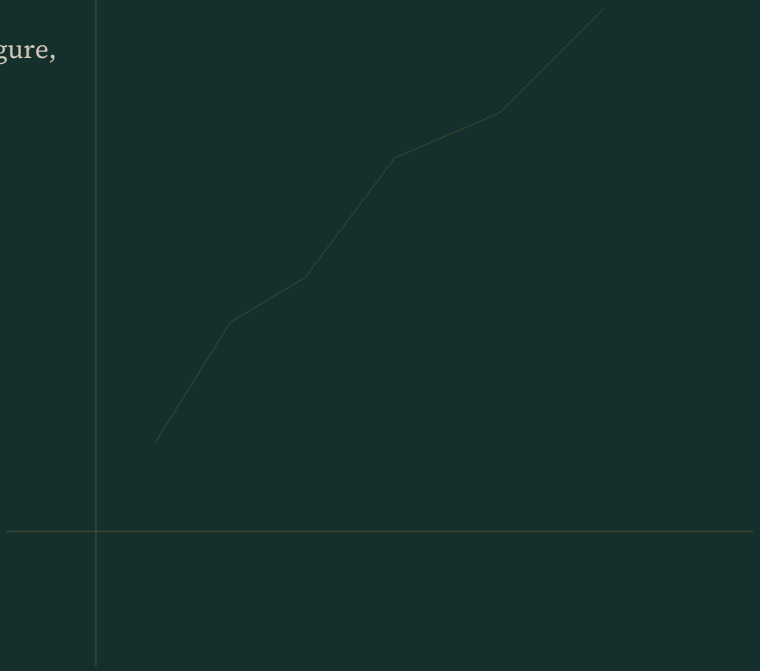
Question	If you cannot do this ...
1–2	Revisit 6.1: find the difference, then adjust the constant.
3–5	Revisit 6.2 and 6.3: multiplication before addition.
6–7	Revisit 6.4: undo the addition, then the multiplication.
8–10	Revisit 6.3 and the taxi example in 6.2.
11–15	Revisit the investigation: build the pattern in a table first.

PART IV

Describing space and belonging

Lines, angles, shapes, sets and measured space: the mathematics of arranging and classifying.

Geometry is the study of what must be true of any figure, and sets are the study of what must belong where.



SADIKOU MOUKODDIM

MYP MATHEMATICS 1

7

DESCRIBING SPACE AND BELONGING

Lines, angles, triangles and constructions

A single straight line contains 180°, and a full turn contains 360°. From those two facts, and one statement about parallel lines, a whole system of shape reasoning follows.

KEY CONCEPT · RELATIONSHIPS

RELATED · SPACE

RELATED · SYSTEM

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How do a few angle facts generate everything that can be known about a triangle or a polygon?

01 · MYP UNIT PLANNER

Statement of inquiry	Spatial relationships obey a small number of angle rules, and precise constructions let those rules be checked and reproduced in real structures.

KEY IDEA

Key and related concepts

Key concept · Relationships. Each angle fact relates two or more angles, which is why a single measurement can determine many others.

Related · Space. Angles describe direction and turn, so the same facts appear in roofs, roads and folded paper.

Related · System. Triangles and polygons are systems: a few properties force all the rest.

NOTE

Global context · Orientation in space and time

Exploration: how builders have established square corners and level lines without modern instruments, and why the 3–4–5 triangle was used for thousands of years.

Inquiry questions

- Factual: What are the angle facts for a straight line, a point, parallel lines and a triangle?
- Conceptual: Why must the angles of every triangle add to 180°?
- Debatable: Is a construction with ruler and compass more convincing than a measurement with a protractor?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Apply angle facts to find unknown angles and justify each step.
B · Investigating patterns	i, ii, iii	Investigate the triangle inequality and the angle sums of polygons.
C · Communicating	i, ii, iii, iv	Draw and label a construction so another student can repeat it.
D · Applying mathematics	i, ii, iii, iv	Design a roof truss and check that it can be built.

ATL SKILL

ATL · Communication skills

In order for a student to give a complete reason for an angle calculation, students must name the fact used at each step rather than only the arithmetic.

02 · ENGAGE

The carpenter's square.

A carpenter needs a right angle but owns no square. She marks 3 units along one edge, 4 along the other, and adjusts until the distance between the marks is exactly 5. The corner is now square. She is using a fact about triangles that was known in Egypt long before it was proved.

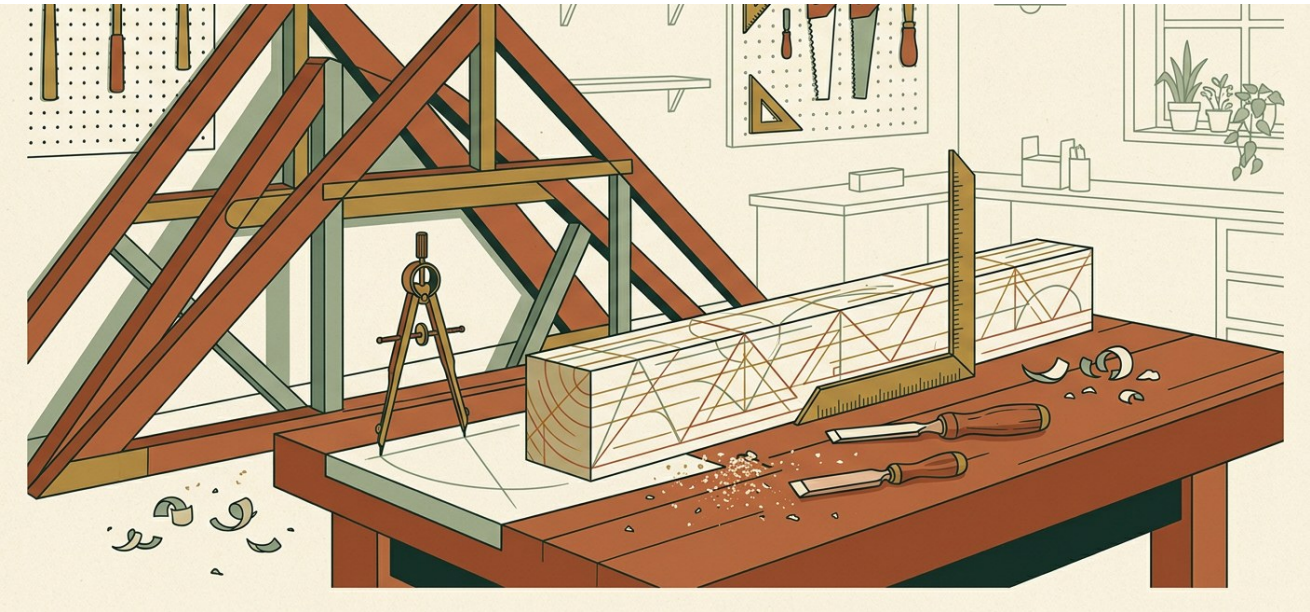


FIGURE 1 Roof trusses are triangles for a reason: a triangle cannot change shape without a side changing length, which is why it carries load so reliably.

Original illustration created for this edition.

03 · EXPLORE

Tear the corners off.

Draw three different triangles on paper: one acute, one with a right angle and one obtuse. Measure every angle with a protractor and record the three sums.

1. What do the three sums have in common?
2. Tear the corners off one triangle and place them together along a straight edge. What do you notice?
3. Repeat with a quadrilateral. What is the sum, and how does it relate to the triangle result?
4. Could a triangle have two obtuse angles? Explain using the sum.

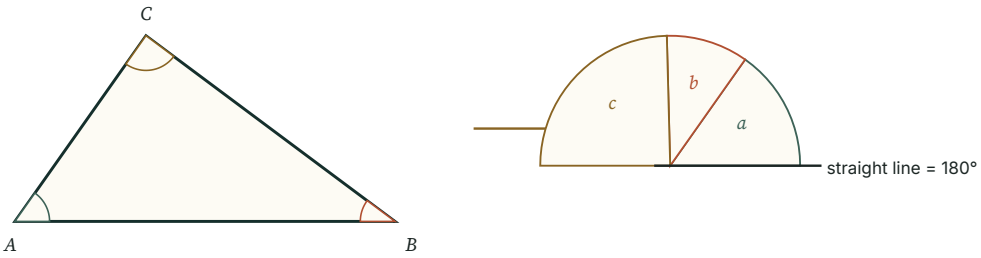


FIGURE 2 The three angles of a triangle rearrange to form a straight line, which is why their sum is 180°.

KEY IDEA

The first big result

The angles of any triangle add to 180°. Every other triangle fact in this chapter follows from it, and it follows in turn from the fact that angles on a straight line add to 180° together with the parallel-line facts.

7.1 Lines, angles and angle facts

CRITERION A · i · C · iii ◆ SPACE ◆ SYSTEM

DEFINITION

Angle vocabulary

An acute angle is less than 90°; a right angle is exactly 90°; an obtuse angle is between 90° and 180°; a straight angle is 180° and a reflex angle is between 180° and 360°. Two angles that add to 90° are complementary; two that add to 180° are supplementary.

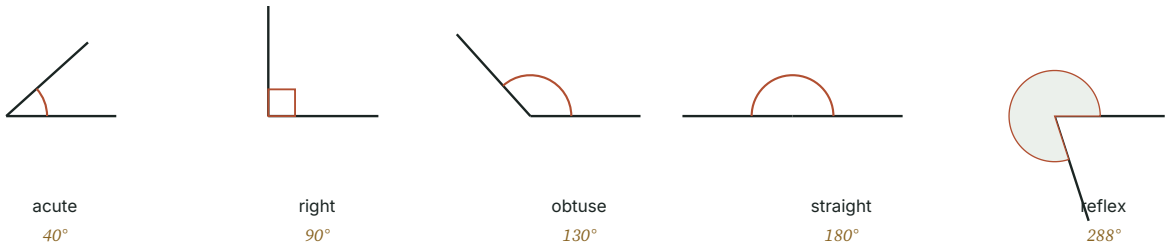


FIGURE 3 The angle families, drawn on the same baseline.

The facts you will use

Fact	Statement	Why it is true
Straight line	Angles on a straight line add to 180°.	A half turn is 180°.
Point	Angles at a point add to 360°.	A full turn is 360°.
Vertically opposite	Angles opposite each other at a crossing are equal.	Each is supplementary to the same angle.
Corresponding	Angles in the same position at each crossing of parallel lines are equal.	A translation maps one onto the other.
Alternate	Angles on opposite sides of the transversal, between the parallels, are equal.	Follows from corresponding angles and vertically opposite angles.
Co-interior	Angles between the parallels on the same side add to 180°.	They are supplementary.

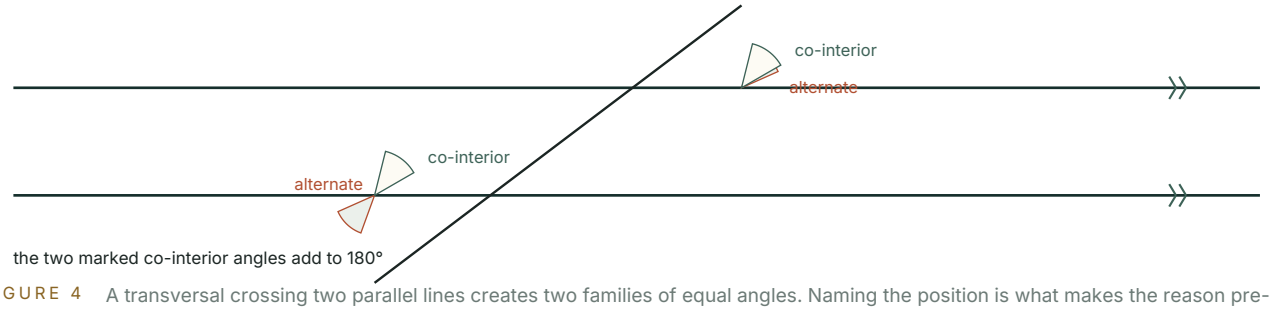


FIGURE 4 A transversal crossing two parallel lines creates two families of equal angles. Naming the position is what makes the reason precise.

WORKED EXAMPLE 7.1.1 — A STRAIGHT LINE AND A RIGHT ANGLE

PROBLEM

Two angles sit on a straight line. One is 118°. Find the other.

1. Angles on a straight line add to 180°.
2. $180° - 118° = 62°$.
3. Check: $118 + 62 = 180$.

ANSWER $62°$

The answer is acute, which matches the diagram ✓

WORKED EXAMPLE 7.1.2 — PARALLEL LINES

PROBLEM

Two parallel lines are crossed by a transversal. One angle is 73°. Find the alternate angle, the corresponding angle and the co-interior angle.

1. The alternate angle is equal: 73°.
2. The corresponding angle is equal: 73°.
3. The co-interior angle is supplementary: $180° - 73° = 107°$.
4. Each answer should be stated with the reason, not only the number.

ANSWER 73°, 73° and 107°.

All four angles at a crossing must add to 360°: 73 + 107 + 73 + 107 = 360 ✓

COMMON ERROR

Assuming lines are parallel because they look parallel

The parallel-line facts apply only when the lines are given as parallel, usually marked with arrows. Without the marking, an equal-looking angle proves nothing.

PRACTICE 7.1

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Classify each angle as acute, right, obtuse or reflex: 45°, 90°, 170°, 210°. F
2. Find the complement of 37° and the supplement of 37°. F
3. Two angles on a straight line are $3x°$ and 60°. Find x. F

PRACTICE 7.1 continued SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Angles at a point are 90°, 120° and y°. Find y. F
2. Explain why vertically opposite angles are equal. R
3. A gate brace makes a 65° angle with the horizontal. Find the angle it makes with the vertical post. P
4. Two parallel roads are crossed by a straight path at 112°. Find the co-interior angle and explain your reasoning. P
5. Can two angles on a straight line both be obtuse? Explain. E

7.2 Parallel lines and triangles

CRITERION A · ii · B · ii ◆ SYSTEM ◆ SPACE

The angle sum of a triangle is not an independent fact: it follows from the straight-line fact together with the parallel-line facts. Drawing a line parallel to one side through the opposite vertex makes the proof visible in a single diagram.

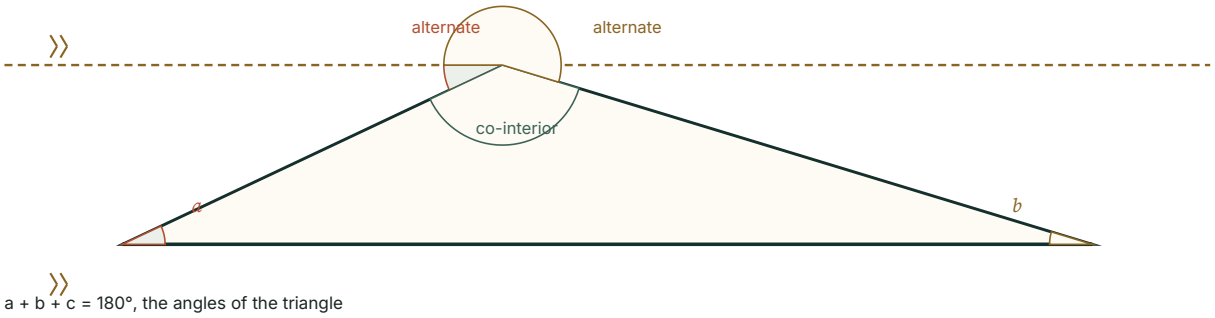


FIGURE 5 A line parallel to the base creates alternate angles equal to the base angles, so the three angles of the triangle fill a straight line.

KEY IDEA

The reasoning, in three steps

1 Draw a line through the top vertex parallel to the base. 2 The two angles between this line and the sides are alternate to the base angles, so they equal them. 3 The three angles at the top vertex now sit on a straight line, so they add to 180° — and those three angles are the three angles of the triangle.

WORKED EXAMPLE 7.2.1 — FINDING A MISSING ANGLE

PROBLEM

A triangle has angles of 47° and 68°. Find the third angle.

1. The three angles add to 180°.
2. 47 + 68 = 115.
3. 180 − 115 = 65°.

ANSWER 65°

47 + 68 + 65 = 180 ✓

WORKED EXAMPLE 7.2.2 — AN ISOSCELES TRIANGLE

PROBLEM

An isosceles triangle has two equal angles and a third angle of 40°. Find the equal angles.

1. The two equal angles share what is left: 180 − 40 = 140°.
2. Each is 140 ÷ 2 = 70°.
3. Check that all three are positive and add to 180.

ANSWER 70° each

70 + 70 + 40 = 180 ✓

PRACTICE 7.2 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the third angle: (a) 60° and 50° (b) 90° and 35°. F
2. A triangle has two equal angles of 55°. Find the third. F
3. Can a triangle have angles 100°, 50° and 40°? Explain. F
4. A right-angled triangle has one angle of 28°. Find the third. F
5. Explain why a triangle cannot have two right angles. R
6. A roof triangle has a base angle of 35° and is isosceles. Find all three angles. P
7. The angles of a triangle are x, $2x$ and $3x$. Find each angle. P
8. An exterior angle of a triangle equals the sum of the two opposite interior angles. Test this with two examples and explain why it works. E

7.3 Polygons and their angles

CRITERION A · iii · B · ii ◆ PATTERN ◆ SYSTEM

Any polygon can be cut into triangles by drawing diagonals from one vertex. A quadrilateral gives two triangles, a pentagon three, a

hexagon four — and the pattern is the rule.

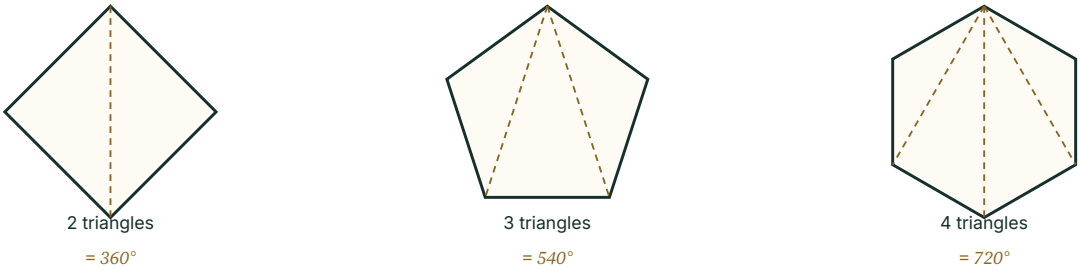


FIGURE 6 Cutting a polygon from one vertex into triangles gives the angle sum without measuring anything.

DEFINITION

Angle sum of a polygon

An n-sided polygon can be divided into $n - 2$ triangles from one vertex, so its interior angles add to $(n - 2) \times 180°$. For a quadrilateral this gives 360°; for a pentagon, 540°.

Polygon	Sides	Triangles	Angle sum
Triangle	3	1	180°
Quadrilateral	4	2	360°
Pentagon	5	3	540°
Hexagon	6	4	720°
Octagon	8	6	1 080°

WORKED EXAMPLE 7.3.1 — A REGULAR POLYGON

PROBLEM

Find the interior angle of a regular pentagon.

1. Angle sum of a pentagon: $(5 - 2) \times 180° = 540°$.
2. A regular pentagon has five equal angles.
3. $540° \div 5 = 108°$.

ANSWER 108°

$5 \times 108 = 540$ ✓

WORKED EXAMPLE 7.3.2 — WORKING BACKWARDS

PROBLEM

A regular polygon has an interior angle of 150°. How many sides has it?

1. Let the polygon have n sides: $(n - 2) \times 180°$ must equal $150n$.
2. Try values: a 12-sided polygon has angle sum 1 800°, and 1 800 ÷ 12 = 150°.
3. So the polygon has 12 sides.

ANSWER 12 sides (a regular dodecagon).

$(12 - 2) \times 180 = 1\,800$ and $1\,800 \div 12 = 150$ ✓

PRACTICE 7.3 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the angle sum of a quadrilateral and a hexagon. F
2. Find the interior angle of a regular hexagon. F
3. Three angles of a quadrilateral are 80°, 95° and 100°. Find the fourth. F
4. Find the interior angle of a regular octagon. F
5. Explain why the exterior angles of any polygon add to 360°. R
6. A regular polygon has an interior angle of 140°. How many sides has it? P
7. Explain why a regular pentagon cannot tile a floor, while a square can. E
8. Investigate the number of diagonals in an n-sided polygon and describe the pattern. E

7.4 Constructions with ruler and compass

CRITERION C · i, iii · D · ii ◆ SPACE ◆ SYSTEM

A construction uses only a straight edge and a pair of compasses, with no measurement. Its value is that every step is justified by a geometric fact, so the result is exact rather than close.

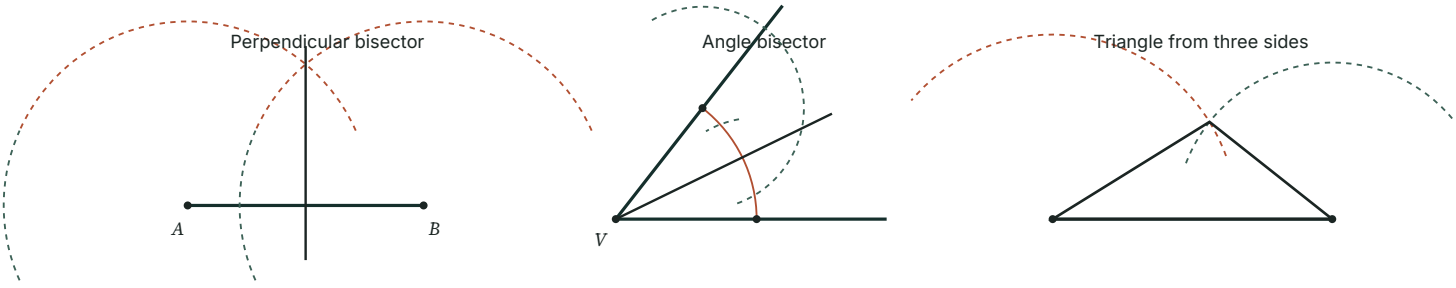


FIGURE 7 Three standard constructions: the perpendicular bisector, the angle bisector and a triangle from three sides.

Constructions you should be able to perform

Construction	Steps	Justification
Perpendicular bisector of AB	Arcs of equal radius from A and from B, crossing twice; join the crossings.	Every point on the bisector is equidistant from A and B.
Bisector of an angle	Arc across both arms; equal arcs from those two points; join the vertex to the crossing.	The two triangles formed are congruent.
Triangle from three sides	Draw one side; arcs of the other two lengths from its ends; join the crossing.	Two sides and the included construction fix the triangle.
Perpendicular at a point on a line	Equal marks either side of the point; equal arcs above; join.	The construction is symmetric about the point.

COMMON ERROR

Losing accuracy through a blunt pencil

In a construction the compass point must sit exactly on the intersection, and the pencil must be sharp. A 1 mm slip at the first arc can produce a visibly wrong bisector — and the geometry will be right even though the drawing is not.

WORKED EXAMPLE 7.4.1 — CONSTRUCTING A TRIANGLE

PROBLEM

Construct a triangle with sides 6 cm, 5 cm and 4 cm, and check whether a second triangle with the same sides could be a different shape.

1. Draw the 6 cm side as the base.
2. From one end, draw an arc of radius 5 cm; from the other, an arc of radius 4 cm.
3. Join each end of the base to the crossing point.
4. The three sides fix the triangle completely: any other arrangement gives a congruent triangle.

ANSWER One triangle shape only, up to reflection.

Measure the three angles; they are determined by the sides ✓

PRACTICE 7.4

SUPPORT 1–3 · CORE 2–6 · EXTENSION 5–7

1. Construct a triangle with sides 5 cm, 5 cm and 7 cm. Measure its angles. F
2. Construct the perpendicular bisector of a 7 cm segment and check that every point on it is equidistant from the ends. F
3. Bisect an angle of about 70° and check both halves with a protractor. F
4. Construct a triangle with sides 3 cm, 4 cm and 5 cm. What do you notice about one angle? R
5. Explain why the construction of a perpendicular bisector works, using congruent triangles. R
6. Plan a construction to draw an angle of exactly 60° without a protractor, and carry it out. P
7. Investigate whether a triangle can be constructed with sides 2 cm, 3 cm and 7 cm. Explain what happens. E
8. Construct an equilateral triangle of side 4 cm with compasses only. F
9. Construct the perpendicular bisector of a 6 cm segment and measure the distance from a point on it to each end. F
10. Explain in one sentence why a construction needs no measuring. R
11. Construct a 45° angle by bisecting a right angle, and check it with a protractor. P

04 · INVESTIGATE

INVESTIGATION 7

The triangle inequality

SPACE ◆ SYSTEM

Cut straws or strips into lengths of 2, 3, 4, 5, 6 and 8 units. Try to form a triangle from each triple. Record which triples work and which fail, then look for the rule.

Lengths	Triangle possible?	Reason
2, 3, 4	yes	…
2, 3, 5	…	…
2, 3, 7	no	…
3, 4, 5	yes	…
4, 5, 8	yes	…
5, 5, 10	…	…

Your investigation

1. Complete the table.
2. For each failing triple, add the two shorter lengths and compare with the longest.
3. Write the rule as a sentence about the three sides.
4. Test the rule on 4, 7, 11 and on 4, 7, 12.
5. Explain why the rule must hold: what happens to the two shorter sides as the angle between them opens?

KEY IDEA

The triangle inequality

A triangle can be made from three lengths exactly when the sum of any two is greater than the third. If two sides just reach the third, the triangle collapses into a straight line; if they fall short, they cannot meet.

ATL SKILL

ATL · Critical thinking

In order for a student to generalise from physical trials, students must record the boundary cases, because the rule is revealed where a figure just succeeds or just fails.

05 · APPLY

REAL-LIFE APPLICATION

The roof-truss workshop

CRITERION D · i–iv ◆ SPACE ◆ MODEL

A workshop is building a triangular roof truss with a horizontal span of 8 m. The two sloping rafters are equal in length and meet at 100° at the apex. The workshop needs the base angles and the rafter length.

Your task

1. Sketch the truss and mark the known angle.
2. Find the two base angles, stating the triangle fact you used.
3. Draw the truss to scale, 1 cm to 1 m, and measure the rafter length.
4. Explain why the truss must be a triangle rather than a quadrilateral for strength.
5. Calculate the total length of timber needed for the two rafters and the span, and add 10 % wastage.
6. State one assumption your model makes.

CRITERION

Criterion D · strands i–iv

- Identify the given and required quantities.
- Select angle facts and scale drawing as strategies.
- Apply them and convert units consistently.
- Justify the wastage allowance and evaluate the assumptions.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The geometry guide

CRITERION C · i–iv · A · i, ii

Context. A younger class is about to learn angle facts and keeps confusing alternate with corresponding angles. Produce a one-page guide that makes the distinction obvious.

1. Draw one diagram containing both pairs and shade them differently.
2. State each fact in words, using the shaded diagram as the reference.
3. Give one worked example per fact, with the reason named.
4. Add a warning about assuming lines are parallel.
5. Test the guide on a classmate: give them an unseen diagram and see whether they name the correct fact.
6. Evaluate: what did the diagram make easy, and what did it hide?

CRITERION

Criteria and strands developed

- Criterion A · Knowing and understanding — i, ii: correct statements and calculations.
- Criterion C · Communicating — i–iv: notation, diagram clarity and audience awareness.



FIGURE 8 How the guide might be laid out: one diagram of the pair being distinguished, then a short fact and a warning in words.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Angles on a straight line add to 180°; angles at a point add to 360°.
- Parallel lines create equal corresponding and alternate angles, and supplementary co-interior angles.
- The angles of every triangle add to 180°, and this follows from the earlier facts.
- The interior angles of an n-sided polygon add to $(n - 2) \times 180°$.
- Ruler-and-compass constructions are exact because each step is justified.

Vocabulary

Term	Meaning
acute / obtuse / reflex	Less than 90° / between 90° and 180° / between 180° and 360°.
complementary / supplementary	Adding to 90° / adding to 180°.
transversal	A line crossing two or more other lines.
alternate angles	Equal angles on opposite sides of a transversal.
corresponding angles	Equal angles in matching positions.
co-interior angles	Angles between parallels on the same side, adding to 180°.
bisector	A line dividing an angle or segment into two equal parts.

Misconceptions to watch

COMMON ERROR

Three that persist

- Lines that look parallel are parallel. Only the marking justifies the facts.
- All triangles have a 60° angle. Only equilateral triangles do.
- A bigger triangle has a bigger angle sum. The sum is 180° at every size.

Self-assessment

I can …	Not yet	Getting there	Confident
use straight-line and point facts	☐	☐	☐
identify parallel-line angle pairs	☐	☐	☐
find missing angles in a triangle	☐	☐	☐
find the angle sum of a polygon	☐	☐	☐
carry out the standard constructions	☐	☐	☐
give a reason for each step	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to give a complete answer, students must write the reason beside each calculated angle, even when the arithmetic was obvious.

Looking ahead · Chapter 8 · Sets and classification takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 7

SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. Find the supplement of 63° and the complement of 28°. F
2. Two angles on a straight line are 45° and a°. Find a. F
3. Find the third angle of a triangle with 72° and 39°. F
4. Find the interior angle of a regular hexagon. F
5. A quadrilateral has angles 90°, 90° and 120°. Find the fourth. F
6. Explain the difference between alternate and corresponding angles. R
7. Can a triangle have angles of 30°, 45° and 115°? Explain. R
8. A ladder makes a 70° angle with the ground. Find the angle it makes with a vertical wall. P
9. The angles of a pentagon are 100°, 110°, 120°, 130° and x°. Find x. P
10. Explain why the exterior angles of any polygon add to 360°, using a square-walking argument. R
11. Construct an angle of 90° at the end of a 6 cm segment, using only a ruler and compasses, and describe each step. P
12. A regular polygon has an exterior angle of 24°. How many sides has it? E
13. Investigate whether a triangle can be cut into two triangles each with the same angle sum, and explain the result. E
14. Design a truss with a span of 6 m and a height of 2 m. Find the rafter length to one decimal place and check it satisfies the triangle inequality. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–5	Revisit 7.1 to 7.3: name the fact before calculating.
6–7	Revisit 7.2: draw the parallel line through the vertex.
8–10	Revisit 7.4 and the investigation: boundary cases reveal the rule.
11–14	Revisit the key idea on constructions and the polygon pattern.



8

DESCRIBING SPACE AND BELONGING

Sets and the art of classification

A set is only a collection with a rule for belonging. Almost every mathematical idea begins with deciding what belongs together — and stating the rule precisely enough that nothing slips in unnoticed.

KEY CONCEPT · FORM

RELATED · REPRESENTATION

RELATED · SYSTEM

CONTEXT · IDENTITIES AND RELATIONSHIPS

INQUIRY

What makes a classification useful rather than merely tidy, and who decides where the boundary lies?

01 · MYP UNIT PLANNER

Statement of inquiry	Classifying objects into sets with precisely stated rules reveals structure that is invisible when the objects are considered one at a time.

KEY IDEA

Key and related concepts

Key concept · Form. A set is defined by the form of its rule, not by the objects it happens to collect.

Related · Representation. Venn diagrams, set brackets and number-system trees are three representations of the same relation.

Related · System. Number systems nest inside one another, and the nesting is itself a set relation.

NOTE

Global context · Identities and relationships

Exploration: how a community sorts its own members — by age, by trade, by household — and what a sorting rule includes and excludes.

Inquiry questions

- Factual: What does it mean for an element to belong to a set, and how is this written?
- Conceptual: Why does a precise rule matter more than a list?
- Debatable: Can any classification be neutral?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Use set notation accurately to describe a collection.
B · Investigating patterns	i, ii, iii	Find the pattern in the number of subsets of a set.
C · Communicating	i, ii, iii, iv	Draw a Venn diagram that another student can read correctly.
D · Applying mathematics	i, ii, iii, iv	Sort a real collection and justify the rule chosen.

ATL SKILL

ATL · Critical and creative thinking

In order for a student to classify without ambiguity, students must state the rule in a form that can be tested on a borderline object.

02 · ENGAGE

Everything in this room.

Sort everything in the room into two groups, then into three, then into five. Each time, write the rule you used. Now swap rules with another group and try to sort by their rule without asking what it means.



FIGURE 1 A sorting hall is a classification made physical: the shelves encode a rule that customers are expected to infer.
Original illustration created for this edition.

03 · EXPLORE

One rule, two results.

Take the numbers from 1 to 20. Sort them using the rule “even”, then using the rule “multiple of 3”, then using both rules at once.

1. Which numbers appear in both groups?
2. Which numbers appear in neither group?
3. Does every number belong to at least one group this time? Try a different pair of rules where it does not.
4. Write a rule that puts every number 1–20 into exactly one group.

1–20 by two rules

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

shaded: even · outlined: multiple of 3 · both: darker

FIGURE 2 The numbers 1 to 20 arranged by two rules at once. The overlap is the set of numbers satisfying both.

KEY IDEA

Classification needs a test

A classification is only as good as the test it implies. If you cannot say whether a borderline object belongs, the rule is not yet precise enough.

8.1 Sets, elements and notation

CRITERION A · i, ii · C · i ◆ FORM ◆ REPRESENTATION

DEFINITION

Set and element

A set is a well-defined collection of objects. Each object in it is an element (or member). We write $x \in A$ to mean that x belongs to set A, and $x \notin A$ to mean that it does not.

Two ways to name a set

Method	Example	When to use it
Listing (roster)	$A = \{2, 4, 6, 8\}$	The set is small or finite.
Rule (set-builder)	$B = \{x : x \text{ is an even number}, 1 \le x \le 8\}$	The pattern is clearer than the list.

Curly brackets always mean a set. Round brackets never do. The colon in a rule reads "such that", and the condition after it must be precise enough to decide any candidate object.

WORKED EXAMPLE 8.1.1 — READING SET NOTATION

PROBLEM

$A = \{2, 4, 6, 8, 10\}$. Decide whether each statement is true or false: $6 \in A$, $7 \in A$, $10 \notin A$.

1. 6 is listed in A, so $6 \in A$ is true.
2. 7 is not listed, so $7 \in A$ is false, which means $7 \notin A$ is true.
3. 10 is listed, so $10 \notin A$ is false.

ANSWER True, false, false.

Every element of A satisfies the rule "even, between 1 and 11" √

WORKED EXAMPLE 8.1.2 DESCRIBING A SET BY RULE

PROBLEM

Write the set of whole numbers between 5 and 15 that are divisible by 3, both by listing and by rule.

1. Candidates: 6, 9, 12, 15.
2. Listed: {6, 9, 12, 15}.
3. By rule: $\{x : x \text{ is a whole number}, 5 < x \leq 15, x \text{ is a multiple of } 3\}$.
4. The rule must state the range as well as the property, or it describes an infinite set.

ANSWER $\{6, 9, 12, 15\}$

Check the boundaries: 5 is excluded, 15 is included ✓

COMMON ERROR

Confusing $\in$ with $\subset$

$\in$ compares an object with a set; $\subset$ compares two sets. $3 \in \{1, 2, 3\}$ is true, but $\{3\} \subset \{1, 2, 3\}$ is a different statement about a set, not the number.

PRACTICE 8.1 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. List the set of vowels in the word MATHEMATICS. F
2. Decide whether $12 \in \{\text{multiples of } 4\}$. Explain. F
3. Write $\{3, 6, 9, 12\}$ by rule. F
4. Write the set of even numbers from 2 to 12 by listing. F
5. Explain the difference between $\in$ and $\subset$. R
6. Write a rule that describes $\{1, 4, 9, 16, 25\}$. P
7. A set $S = \{x : x \text{ is a letter in BANANA}\}$. List S and explain why it has fewer elements than the word has letters. P
8. Find a rule that is precise for whole numbers but ambiguous for decimals, and explain the ambiguity. E

8.2 Subsets, the universal set and the complement

CRITERION A · iii · C · ii ◆ SYSTEM ◆ REPRESENTATION

DEFINITION

Subset, universal set, complement

$A \subset B$ means every element of A is also in B. The universal set ξ is everything under discussion. The complement A' contains every element of ξ that is not in A.

The universal set is a decision, not a fact. If ξ is the whole numbers, the complement of the evens is the odds; if ξ is the digits 0–9, the complement is $\{1, 3, 5, 7, 9\}$. Always state ξ before using a complement.

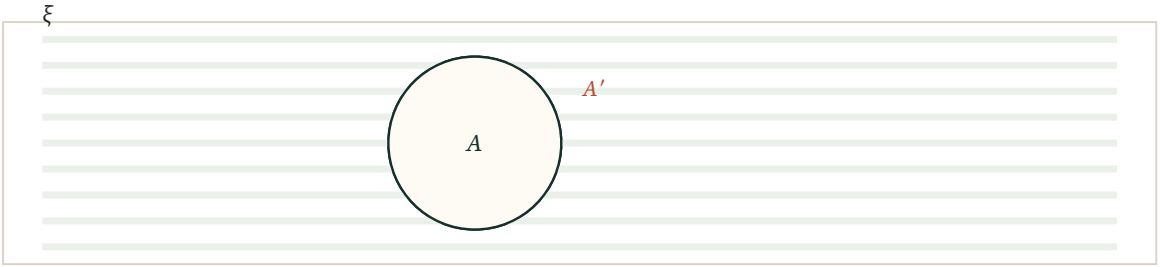


FIGURE 3 The shaded region is the complement of A inside the universal set ξ.

WORKED EXAMPLE 8.2.1 — LISTING SUBSETS

PROBLEM

List all subsets of $\{a, b\}$.

1. The empty set $\varnothing$ is a subset of every set.
2. Single-element subsets: $\{a\}$, $\{b\}$.
3. The whole set: $\{a, b\}$.
4. That is four subsets.

ANSWER $\varnothing$, $\{a\}$, $\{b\}$, $\{a, b\}$.

A set with 2 elements has $2^2 = 4$ subsets ✓

WORKED EXAMPLE 8.2.2 — USING THE COMPLEMENT

PROBLEM

$\xi = \{1, 2, 3, \ldots, 12\}$, $A = \{\text{multiples of 3}\}$. List A and A'.

1. $A = \{3, 6, 9, 12\}$.
2. A' contains everything in ξ not in A: $\{1, 2, 4, 5, 7, 8, 10, 11\}$.
3. Check that the two sets together give all twelve elements with no repeats.

ANSWER $A' = \{1, 2, 4, 5, 7, 8, 10, 11\}$

4 + 8 = 12 elements in total ✓

PRACTICE 8.2 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Is {2, 4} a subset of {1, 2, 3, 4}? Explain. F
2. List all subsets of $\{p, q, r\}$. F
3. $\xi = \{1, \ldots, 10\}$, $A = \{\text{primes}\}$. List A and A'. F
4. How many subsets has a set with 4 elements? F
5. Explain why the empty set is a subset of every set. R
6. Investigate the pattern in the number of subsets of a set with n elements and write the rule. P
7. A class of 30 students is the universal set; 12 play football, 9 play chess. Explain why you cannot yet find the complement of the football players without more information. E
8. Is every set a subset of itself? Justify your answer. E

8.3 Union, intersection and Venn diagrams

CRITERION A · ii · C · iii ◆ REPRESENTATION ◆ SYSTEM

DEFINITION

Union and intersection

The union $A \cup B$ contains everything in A or B or both. The intersection $A \cap B$ contains only what is in both. Two sets with no common element are disjoint.



FIGURE 4 Union and intersection shown on two identical circles. The shading, not the labels, carries the meaning.

WORKED EXAMPLE 8.3.1 — READING A VENN DIAGRAM

PROBLEM

In a class of 32, 18 study French, 15 study Spanish and 7 study both. Find the number who study at least one language and the number who study neither.

1. Both: 7.
2. French only: $18 - 7 = 11$.
3. Spanish only: $15 - 7 = 8$.
4. At least one: $11 + 7 + 8 = 26$.
5. Neither: $32 - 26 = 6$.

ANSWER 26 study at least one; 6 study neither.

$11 + 7 + 8 + 6 = 32$, the whole class ✓

WORKED EXAMPLE 8.3.2 — COUNTING WITHOUT A DIAGRAM

PROBLEM

$A = \{2, 4, 6, 8, 10\}$ and $B = \{4, 8, 12\}$. Find $A \cup B$ and $A \cap B$.

1. Intersection: elements in both — 4 and 8.
2. Union: everything in either, listed once — 2, 4, 6, 8, 10, 12.
3. Note that 4 and 8 appear once in the union, not twice.

ANSWER $A \cap B = \{4, 8\}$; $A \cup B = \{2, 4, 6, 8, 10, 12\}$

$|A| + |B| - |A \cap B| = 5 + 3 - 2 = 6 = |A \cup B|$ ✓

COMMON ERROR

Adding the overlap twice

Adding 18 and 15 in Example 8.3.1 gives 33, which is larger than the whole class. The seven students in both were counted twice; subtract them once.

PRACTICE 8.3

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$. Find $A \cup B$ and $A \cap B$. F
2. In a group of 20, 12 like tea, 9 like coffee, 5 like both. How many like neither? F
3. Draw a Venn diagram for two disjoint sets. F
4. $C = \{\text{vowels}\}$, $D = \{\text{letters in MATHEMATICS}\}$. Find $C \cap D$. F
5. Explain why $A \cap B$ is always a subset of A. R
6. In a survey of 40 students, 25 walk to school, 18 cycle, 10 do both. Draw the diagram and find how many do neither. P

PRACTICE 8.3 continued SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Two sets have 12 and 15 elements and their union has 20. Find the size of the intersection. P
2. Investigate whether $A \cup B$ can ever have fewer elements than A. Explain using a diagram. E

8.4 Classifying numbers and shapes

CRITERION A · i, iii · B · i ◆ FORM ◆ SYSTEM

Sets give classification a language. The number systems from Chapter 1 nest inside one another, and each nesting is a subset relation that can be drawn.

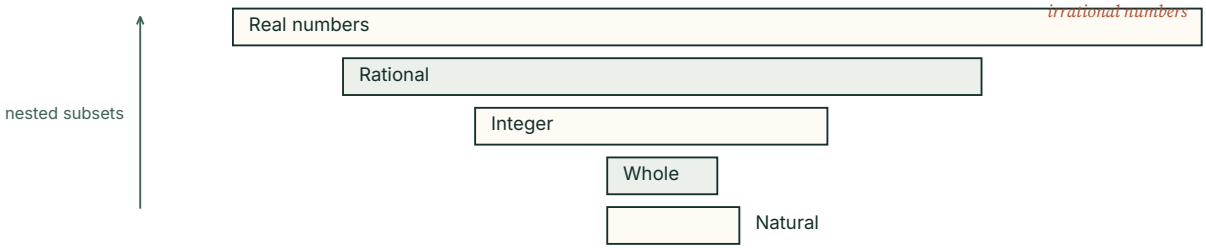


FIGURE 5 The number systems as nested sets. Every natural number is a whole number, every whole number an integer, and so on.

DEFINITION

Classifying by more than one property

A quadrilateral can be classified by two independent properties: the number of pairs of parallel sides, and whether the sides are equal. Drawing the two properties as a two-way table reveals the categories and their overlaps at the same time.

	One pair parallel	Two pairs parallel
All sides equal	—	rhombus (and square)
Opposite sides equal	trapezium	parallelogram / rectangle
No sides equal	trapezium	parallelogram

WORKED EXAMPLE 8.4.1 — CLASSIFYING A NUMBER

PROBLEM

Place −5 in as many number sets as it belongs to.

1. −5 is an integer, because it is a whole number with a sign.
2. It is rational, because it can be written −5/1.
3. It is real. It is not natural or whole, because those sets start at 0 or 1.

ANSWER Integer, rational, real.

Check each definition rather than the appearance of the number √

PRACTICE 8.4

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. List the number sets to which 0 belongs. F
2. Is every square a rectangle? Explain using sets. F
3. Sort these into rational and irrational: 0.25, √2, 7, π, −3/4. F
4. Name the quadrilateral with two pairs of parallel sides and four right angles. F
5. Explain why every natural number is an integer but not every integer is a natural number. R
6. Draw a two-way table classifying triangles by angle type and by side equality. P
7. Classify the numbers 1 to 20 by “prime” and “even” in a two-way table, and describe the empty cell. P
8. Investigate whether a classification can be complete and non-overlapping at the same time, using shapes as an example. E

04 · INVESTIGATE

INVESTIGATION 8

How many subsets?

FORM ◆ SYSTEM

A set with one element has two subsets. Build sets of two, three and four elements, list every subset and count them. Record the counts in a table and look for the pattern.

Elements	Number of subsets	Subsets
0	1	$\varnothing$
1	2	$\varnothing$, $\{a\}$
2	4	…
3	8	…
4	…	…

Your investigation

1. Complete the table by listing, not by guessing.
2. Describe the pattern in the middle column as a rule involving n.
3. Explain why doubling happens each time a new element is added: what can you do with every existing subset?
4. Predict the number of subsets of a five-element set and check by listing.
5. State the rule for the number of subsets of a set with n elements.

KEY IDEA

The result

A set with n elements has 2^n subsets. Each element is either in or out of a given subset, so there are two independent choices per element.

ATL SKILL

ATL · Transfer

In order for a student to generalise, students must move from counting cases to explaining why the count doubles.

05 · APPLY

REAL-LIFE APPLICATION

The clinic waiting list

CRITERION D · i–iv ◆ SYSTEM ◆ CHANGE

A clinic records which of two services each patient used last month: the general clinic (G) and the laboratory (L). In a sample of 60 patients, 38 used G, 27 used L and 15 used both. The clinic must justify its staffing to a funding committee.

Your task

1. Draw a Venn diagram and fill in all four regions.
2. Find the number who used exactly one service.
3. Find the number who used neither, and explain what that group could be.
4. Write two sentences the clinic director could use, each reporting a different true number in a way that supports a different staffing decision.
5. Explain why the same data can support two different conclusions.
6. State one way the survey could have produced misleading numbers.

CRITERION

Criterion D · strands i–iv

- Identify the given quantities and the universal set.
- Use the Venn diagram and subtraction as strategies.
- Apply the method accurately and check that the four regions sum to 60.
- Evaluate how presentation of the same data changes its effect.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The sorting rule

CRITERION C · i–iv · B · ii

Context. Your class library needs re-shelving. The current rule “fiction and non-fiction” no longer works for the collection.

1. Collect or imagine 20 items. State a sorting rule with two properties.
2. Draw the two-way table and place every item.
3. Identify any item that is hard to place and explain why.
4. Revise the rule so that the difficult item can be placed.
5. Test the revised rule on a classmate's item list.
6. Evaluate: which rule produced fewer empty cells, and did that make it a better classification?

CRITERION

Criteria and strands developed

- Criterion B · Investigating patterns — i–iii: the pattern in the number of subsets.
- Criterion C · Communicating — i–iv: table clarity, notation and justification of the rule.

One property runs across the top, the other runs down the side.

	Fiction	Not fiction
Hardback	*12 novels*	*4 atlases*
Paperback	*9 short-story collections*	*3 dictionaries*

Each cell is the intersection of the two properties. Every one of the 28 items lands in exactly one cell; an item that fits none tells you the rule needs revising.

FIGURE 6 A two-property classification drawn as a table: each cell is an intersection of the two rules.

KEY IDEA

An empty cell is evidence

A cell with no items is not a failure: it says no object in your collection satisfies both properties. Ask whether the property is badly chosen or the collection is genuinely limited, and say which in your write-up.

Before you hand this in

- Name both properties in the table heading.
- The four cell counts must add to the total.
- Show where your revised rule puts the awkward item.
- Say what the table revealed that a list would hide.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- A set is defined by a rule, and $\in$ tests membership of elements.
- Curly brackets name sets; the colon in a rule reads “such that”.
- $A \subset B$, the universal set ξ and the complement A' extend set vocabulary.
- $A \cup B$ and $A \cap B$ describe everything-in-either and everything-in-both.
- A set with n elements has 2^n subsets.

Vocabulary

Term	Meaning
element	An object belonging to a set.
subset	A set all of whose elements lie in another set.
universal set ξ	Everything under discussion in a particular problem.
complement	The elements of ξ that are not in the set.
union	Everything in either set, listed once.
intersection	Only the elements in both sets.
disjoint	Having no elements in common.
empty set $\varnothing$	The set with no elements; a subset of every set.

Misconceptions to watch

COMMON ERROR

Three that persist

- A list is a set. {1, 2, 2, 3} is the set {1, 2, 3}: repeats collapse.
- $\in$ and $\subset$ are interchangeable. They compare an object with a set and a set with a set.
- Sets are always numbers. A set can contain letters, shapes or people.

Self-assessment

I can …	Not yet	Getting there	Confident
read and write set notation	☐	☐	☐
list subsets of a set	☐	☐	☐
use a universal set and complement	☐	☐	☐
find unions and intersections	☐	☐	☐
complete a Venn diagram from a word problem	☐	☐	☐
justify a classification rule	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to state a rule, students must test it against one object that does and one that does not belong.

Looking ahead · Chapter 9 · Perimeter, area and surface area takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 8 — SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. List the set of factors of 12. F
2. Is 15 ∈ {multiples of 5}? Explain. F
3. Write {1, 8, 27, 64} by rule. F
4. How many subsets has a set with 5 elements? F
5. $\xi = \{1, \ldots, 9\}$, $A = \{1, 3, 5, 7, 9\}$. List A'. F
6. Explain what a universal set is and why it must be stated. R
7. $A = \{2, 3, 5, 7\}$, $B = \{1, 2, 3, 4\}$. Find $A \cup B$ and $A \cap B$. R
8. In a group of 50, 30 read the newspaper, 22 read magazines, 12 read both. Draw the diagram and find the number who read neither. P
9. A set has 64 subsets. How many elements has it? P
10. Explain why a Venn diagram with two circles has four regions, and draw a third circle to show how many regions appear. P
11. Sort the numbers 1 to 30 by the properties “prime” and “multiple of 3” in a two-way table. P
12. A survey of 40 people asked whether they owned a bicycle (B) and a helmet (H). 22 had B, 18 had H and 10 had neither. Find how many had both. E
13. Investigate whether the number of regions in a Venn diagram with n circles follows a simple rule, testing up to three circles. E
14. Design a classification of the quadrilaterals in this chapter using two properties, and state which cells are empty and why. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–5	Revisit 8.1 and 8.2: rule first, list second.
6–7	Revisit 8.3: name each region before adding.
8–11	Revisit 8.3 and the investigation: fill the overlap first.
12–14	Revisit 8.4: two properties give a table, not a list.



9

DESCRIBING SPACE AND BELONGING

Perimeter, area and the space inside

Perimeter is a length, area is a surface, volume is a space. Mixing them up is the most common error in measurement, and the units are what keep them apart: one dimension, two, three.

KEY CONCEPT · RELATIONSHIPS

RELATED · QUANTITY

RELATED · SPACE

CONTEXT · ORIENTATION IN SPACE AND TIME

INQUIRY

How do the units of a measurement show what kind of quantity is being measured?

01 · MYP UNIT PLANNER

Statement of inquiry	Measurement converts shapes into numbers, and the relationship between a shape and its dimensions determines which quantity is being described.

KEY IDEA

Key and related concepts

Key concept · Relationships. Each formula relates dimensions to a measured quantity; changing a dimension changes the quantity in a predictable way.

Related · Quantity. Length, area and volume are different kinds of quantity, not different sizes of the same one.

Related · Space. Area and volume describe how much space a figure occupies.

NOTE

Global context · Orientation in space and time

Exploration: how a builder estimates material — paint by area, edging by perimeter, concrete by volume — and what happens when the three are confused.

Inquiry questions

- Factual: What are the units of length, area and volume, and how do they relate?
- Conceptual: Why does doubling a length multiply the area by four?
- Debatable: Should a price be quoted per unit length, per unit area or per unit volume?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Select and apply the correct formula with correct units.
B · Investigating patterns	i, ii, iii	Investigate how area and volume change when dimensions change.
C · Communicating	i, ii, iii, iv	Show working with units carried through every step.
D · Applying mathematics	i, ii, iii, iv	Cost a real job and justify the estimate.

ATL SKILL

ATL · Organisation skills

In order for a student to measure accurately, students must record the unit at every stage, not only at the end of a calculation.

02 · ENGAGE

Three questions, one shape.

A rectangular courtyard is 8 m by 5 m. A tiler is asked how much edging is needed, how many square metres of paving are needed, and how much soil to fill a bed 30 cm deep over the whole area. The three answers use the same two numbers and are all different.

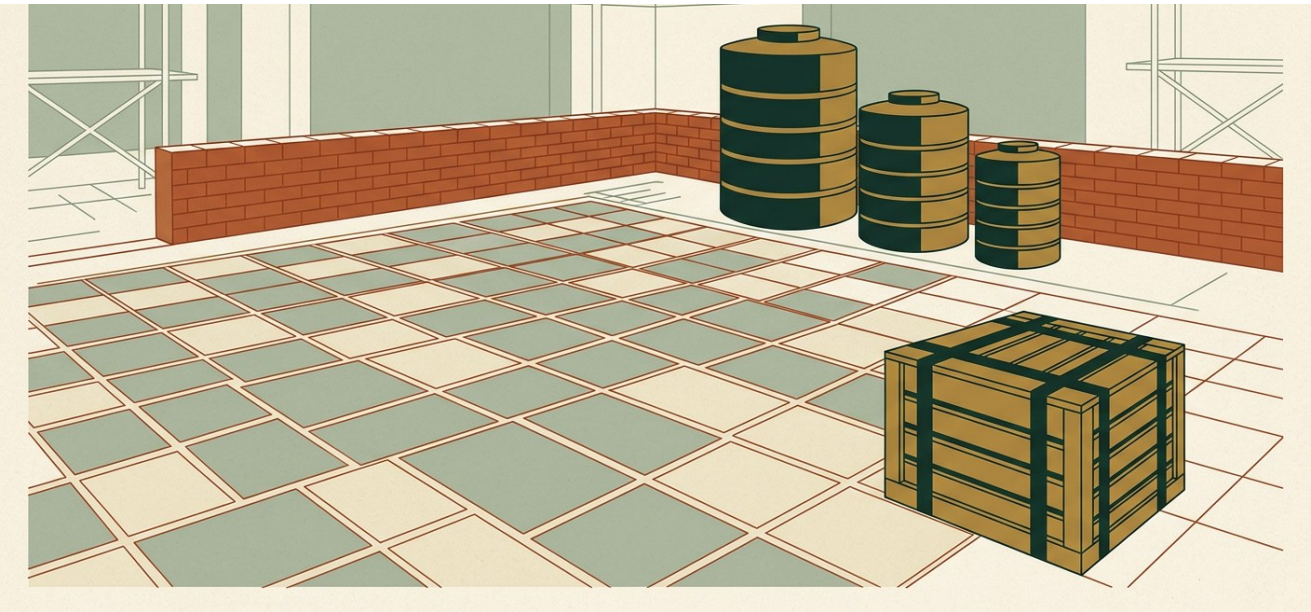


FIGURE 1 A tiled courtyard: the same rectangle, measured three different ways, gives a length, an area and a volume.

Original illustration created for this edition.

03 · EXPLORE

Count the squares.

Draw three rectangles on squared paper: 2 by 3, 4 by 3 and 4 by 6. Count the squares inside each and record the perimeter and area.

1. Which rectangle has twice the perimeter of the first?
2. Which has twice the area?
3. Double both dimensions of the 2 by 3 rectangle. What happens to the perimeter, and what happens to the area?
4. Write a sentence describing the difference.

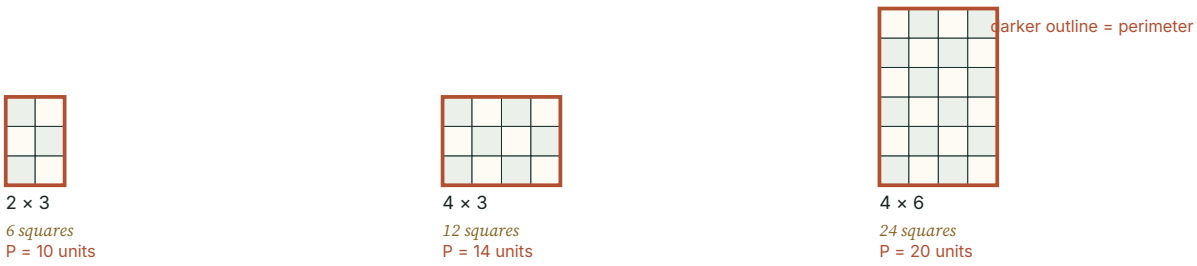


FIGURE 2 Perimeter counts the edge; area counts the squares inside. Doubling both dimensions doubles the perimeter but quadruples the area.

KEY IDEA

The distinction that matters

Perimeter is measured in units of length. Area is measured in square units. Volume is measured in cubic units. The unit is not decoration; it is how you know which question you answered.

9.1 Perimeter and circumference

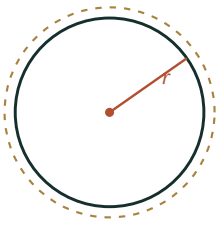
CRITERION A · i · C · ii ◆ QUANTITY ◆ SPACE

DEFINITION

Perimeter

The perimeter of a shape is the total length of its boundary, measured in units of length. For a circle the boundary is called the circumference, and $C = \pi d = 2\pi r$.

Shape	Perimeter
Rectangle	$P = 2(l + w)$
Square	$P = 4s$
Triangle	$P = a + b + c$
Circle	$C = \pi d = 2\pi r$



radius = half the diameter

$C = \pi\, d = 2\, \pi\, r$

every point on the circle is r from the centre

FIGURE 3 The radius, diameter and circumference of a circle. Every point on the circle is the same distance from the centre.

WORKED EXAMPLE 9.1.1 — PERIMETER OF A RECTANGLE

PROBLEM

Find the perimeter of a rectangle 8 m by 5 m.

1. $P = 2(l + w)$
2. $P = 2(8 + 5) = 2 \times 13 = 26$
3. The unit is metres, because perimeter is a length.

ANSWER 26 m

Adding all four sides gives 8 + 5 + 8 + 5 = 26 ✓

WORKED EXAMPLE 9.1.2 — CIRCUMFERENCE

PROBLEM

Find the circumference of a circle with radius 7 cm, using $\pi \approx 3.14$.

1. $C = 2\pi r$
2. $C = 2 \times 3.14 \times 7$
3. $C = 43.96$ cm.
4. Round sensibly: 44.0 cm to one decimal place.

ANSWER ≈ 44.0 cm

$3.14 \times 14 = 43.96$, so the estimate 44 is reasonable ✓

COMMON ERROR

Using diameter where radius belongs

$C = 2\pi r$ uses the radius. If the diameter is given, either halve it first or use $C = \pi d$. Using the diameter in the radius formula doubles the answer.

PRACTICE 9.1 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the perimeter of a rectangle 12 cm by 7 cm. F
2. Find the perimeter of a square of side 9 cm. F
3. Find the circumference of a circle of diameter 10 cm ($\pi \approx 3.14$). F
4. A triangle has sides 6 cm, 8 cm and 10 cm. Find its perimeter. F
5. Explain how to find the perimeter of a shape with a missing side length. R
6. A circular table has radius 60 cm. How much edging is needed to go round it? P
7. A rectangular garden is 14 m by 9 m. Fencing costs 6.50 per metre. Find the cost of fencing the whole boundary. P
8. Investigate: if you double every side of a rectangle, what happens to the perimeter? Explain why. E

9.2

Area of rectangles, triangles and parallelograms

CRITERION A · ii · B · i ◆ SPACE ◆ QUANTITY

DEFINITION

Area

Area is the amount of surface enclosed by a boundary, measured in square units. A square with side 1 cm has an area of 1 cm^2; 1 m^2 contains 10 000 cm^2.

Shape	Area	Why
Rectangle	$A = l \times w$	Counting unit squares in a grid.
Parallelogram	A = base × perpendicular height	A triangle cut from one end fits on the other, making a rectangle.
Triangle	$A = \frac{1}{2} \times$ base × perpendicular height	Two copies make a parallelogram.
Trapezium	$A = \frac{1}{2} \times (a + b) \times$ height	Two copies make a parallelogram with base $(a + b)$.

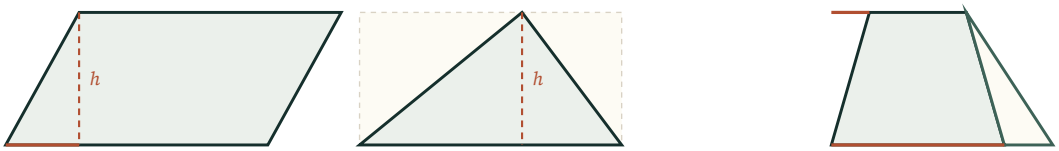


FIGURE 4 Each area formula is a rectangle in disguise: cut and rearrange, and the shape becomes $l \times w$.

WORKED EXAMPLE 9.2.1 — AREA OF A TRIANGLE

PROBLEM

Find the area of a triangle with base 9 cm and perpendicular height 4 cm.

1. $A = \frac{1}{2} \times$ base × height
2. $A = \frac{1}{2} \times 9 \times 4 = 18$
3. The unit is cm^2, because area is two-dimensional.

ANSWER 18 cm^2

The triangle fits twice into a 9 × 4 rectangle of area 36 √

WORKED EXAMPLE 9.2.2 — A COMPOSITE SHAPE

PROBLEM

A shape is made from a 6 m by 4 m rectangle with a right-angled triangle of base 6 m and height 3 m attached along the top. Find its area.

1. Rectangle: $6 \times 4 = 24$ m^2.
2. Triangle: $\frac{1}{2} \times 6 \times 3 = 9$ m^2.
3. Total: $24 + 9 = 33$ m^2.

ANSWER 33 m^2

Split the shape a different way and check the total is the same ✓

COMMON ERROR

Using a slant side as the height

The height in an area formula is always perpendicular to the base. In a parallelogram the slant side is longer than the height, so using it overestimates the area.

PRACTICE 9.2 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the area of a rectangle 15 cm by 8 cm. F
2. Find the area of a triangle with base 12 cm and height 5 cm. F
3. Find the area of a parallelogram with base 7 m and height 4 m. F
4. Convert 3 m^2 to cm^2. F
5. Explain why the area formula for a triangle contains a half. R
6. A trapezium has parallel sides 6 cm and 10 cm and height 5 cm. Find its area. P
7. A wall 4 m by 2.5 m has a window 1 m by 1 m. Find the area to be painted. P
8. Investigate: if the sides of a rectangle are both doubled, by what factor does the area increase? Test with three examples and explain. E

9.3 Circles and composite shapes

CRITERION A · iii · C · iii ◆ SPACE ◆ RELATIONSHIPS

DEFINITION

Area of a circle

$A = \pi r^2$. The area depends on the square of the radius, so doubling the radius multiplies the area by four — not by two.

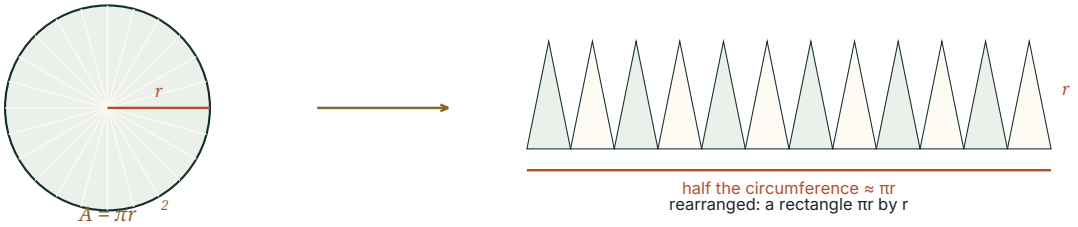


FIGURE 5 A circle cut into many thin sectors and rearranged approaches a rectangle of width πr and height r.

WORKED EXAMPLE 9.3.1 — AREA OF A CIRCLE

PROBLEM

Find the area of a circle of radius 5 cm, using $\pi \approx 3.14$.

1. $A = \pi r^2$
2. $A = 3.14 \times 5^2 = 3.14 \times 25$
3. $A = 78.5$ cm^2.

ANSWER 78.5 cm^2

The circle fits inside a 10 × 10 square of area 100 ✓

WORKED EXAMPLE 9.3.2 — A COMPOSITE SHAPE WITH A CIRCLE

PROBLEM

A circular pond of diameter 6 m sits in the centre of a square lawn 10 m by 10 m. Find the area of lawn visible.

1. Square: $10 \times 10 = 100$ m^2.
2. Radius of pond: 3 m, so area $= 3.14 \times 9 = 28.26$ m^2.
3. Lawn: $100 - 28.26 = 71.74$ m^2.

ANSWER ≈ 71.7 m^2

The pond is under a third of the square, so the answer must be over 66 m^2 ✓

PRACTICE 9.3 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the area of a circle of radius 4 cm ($\pi \approx 3.14$). F
2. Find the area of a circle of diameter 10 cm. F
3. Find the area of a semicircle of radius 6 cm. F
4. A circle has area 50.24 cm^2. Estimate its radius. F

PRACTICE 9.3 continued SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Explain why doubling the radius of a circle multiplies the area by four. R
2. A running track is a rectangle 60 m by 20 m with a semicircle at each end. Find the total area. P
3. A circular flower bed of diameter 4 m is surrounded by a path 1 m wide. Find the area of the path. P
4. Investigate: how does the area of a circle change if the radius is trebled? Give a general rule. E

9.4 Surface area and volume of prisms

CRITERION A · ii, iii · D · iii ◆ SPACE ◆ QUANTITY

DEFINITION

Surface area and volume

The surface area of a solid is the total area of all its faces, in square units. The volume of a prism is the area of its cross-section multiplied by its length, in cubic units: $V = A \times l$.

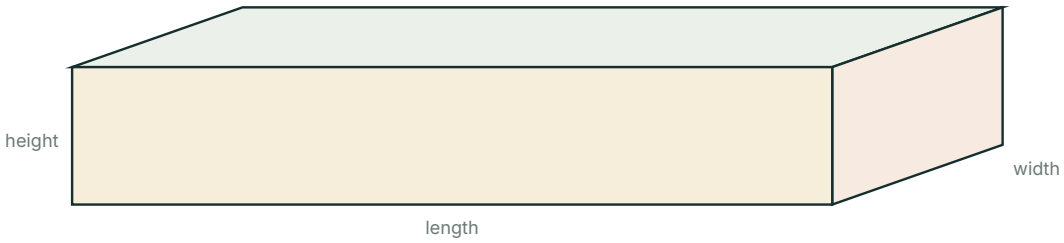


FIGURE 6 A prism has the same cross-section all along its length, which is why volume is cross-section area times length.

Solid	Volume	Surface area
Cuboid	$V = l \times w \times h$	$S = 2(lw + lh + wh)$
Cube	$V = s^3$	$S = 6s^2$
Triangular prism	$V = \frac{1}{2}bh \times l$	Two triangles plus three rectangles
Cylinder	$V = \pi r^2 h$	$S = 2\pi r^2 + 2\pi rh$

WORKED EXAMPLE 9.4.1 — VOLUME OF A CUBOID

PROBLEM

Find the volume of a box 30 cm by 20 cm by 15 cm.

1. $V = l \times w \times h$
2. $V = 30 \times 20 \times 15 = 9\,000$
3. The unit is cm^3, because volume is three-dimensional.

ANSWER $9\,000\ \text{cm}^3$

$9\,000\ \text{cm}^3 = 9$ litres, which is a plausible size for a storage box ✓

WORKED EXAMPLE 9.4.2 — SURFACE AREA OF A CUBE

PROBLEM

Find the total surface area of a cube of side 5 cm.

1. Each face is $5 \times 5 = 25\ \text{cm}^2$.
2. A cube has six congruent faces.
3. $S = 6 \times 25 = 150\ \text{cm}^2$.

ANSWER $150\ \text{cm}^2$

Six faces, all identical, so multiplying one face by six is exact ✓

COMMON ERROR

Mixing units across dimensions

$1\ \text{m}^2$ is $10\,000\ \text{cm}^2$, not $100\ \text{cm}^2$; $1\ \text{m}^3$ is $1\,000\,000\ \text{cm}^3$. The conversion factor is squared or cubed along with the unit.

PRACTICE 9.4

SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the volume of a cuboid 8 cm by 5 cm by 4 cm. F
2. Find the volume of a cube of side 6 cm. F
3. Find the surface area of a cube of side 3 cm. F
4. Convert $2\,000\ \text{cm}^3$ to litres. F
5. Explain the difference between surface area and volume using a box as an example. R
6. A cylindrical tank has radius 1 m and height 2 m. Find its volume in litres ($\pi \approx 3.14$). P
7. A fish tank 60 cm by 30 cm by 40 cm is filled to 80 % of its height. How many litres does it hold? P
8. Investigate: if every edge of a cube is doubled, by what factor do the surface area and the volume change? Explain. E

04 · INVESTIGATE

INVESTIGATION 9

Same perimeter, different area

SPACE ◆ PATTERN

Using whole-number sides, find every rectangle with a perimeter of 20 cm. Record each length, width and area, then look for the pattern.

Length (cm)	Width (cm)	Area (cm^2)
9	1	…
8	2	…
7	3	…
6	4	…
5	5	…

Your investigation

1. Complete the table.
2. Which rectangle has the largest area? The smallest?
3. Describe the pattern in the area column as the sides become more equal.
4. What happens with a perimeter of 24 cm? Test your description.
5. State the rule in words, and explain why a square gives the largest area.

KEY IDEA

The result

Among all rectangles with a given perimeter, the square has the largest area. This is why a fixed length of fencing encloses most ground when the enclosure is square — and why natural cells and bubbles tend towards round shapes, which do the same thing without corners.

ATL SKILL

ATL · Critical thinking

In order for a student to generalise, students must test the boundary of a pattern, not only its middle cases.

05 · APPLY

REAL-LIFE APPLICATION

The courtyard repair

CRITERION D · i–iv ◆ QUANTITY ◆ RELATIONSHIPS

A school courtyard measures 18 m by 12 m. It needs new edging along the whole boundary, new paving over the whole surface, and a raised planting bed 1.5 m wide along one 12 m side, filled to a depth of 40 cm.

Your task

1. Find the length of edging required.
2. Find the area to be paved, allowing for a planting bed of the size described.
3. Find the volume of soil needed for the bed, in cubic metres.
4. Convert the soil volume to litres and decide how many 50-litre bags to order.
5. Cost the job given: edging 9.20 per metre, paving 24 per m^2, soil 4.80 per bag.
6. State two assumptions and one reason the real cost may differ.

CRITERION

Criterion D · strands i–iv

- Identify which quantity each part of the job requires.
- Select perimeter, area or volume deliberately, with units.
- Apply correctly and convert units consistently.
- Justify the assumptions and evaluate the estimate.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The packaging problem

CRITERION C · i–iv · D · ii

Context. A small business packs soap in cuboids and wants to reduce cardboard without reducing the volume of the box.

1. Choose a cuboid with volume 1 000 cm^3 and find its surface area.
2. Find two other cuboids with the same volume and compare surface areas.

3. Which shape uses least cardboard? Relate this to the investigation.
4. Sketch the chosen net, labelling every dimension.
5. Calculate the cardboard saved compared with the first design, as a percentage.
6. Evaluate: what practical reasons might stop the business using the best mathematical shape?

CRITERION

Criteria and strands developed

- Criterion C · Communicating — i–iv: clear working, correct units, labelled diagram.
- Criterion D · Applying mathematics in real-life contexts — ii–iv: strategy choice, application and evaluation of the packaging decision.

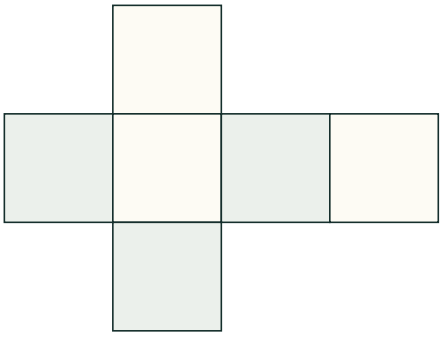


FIGURE 7 A net is the flat pattern a box folds from. The surface area of the box is the area of this net, and it is what the cardboard has to cover.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Perimeter is a length; circumference is the perimeter of a circle.
- Area is measured in square units, and each formula can be derived by rearranging a rectangle.
- The area of a circle is πr^2, so it grows with the square of the radius.
- Volume of a prism is cross-section area times length.
- Doubling every dimension multiplies area by 4 and volume by 8.

Vocabulary

Term	Meaning
perimeter	Total length of a boundary.
area	Surface enclosed, in square units.
volume	Space occupied, in cubic units.
cross-section	The shape you see when a solid is cut straight through.
prism	A solid with the same cross-section all along its length.
surface area	Total area of all the faces of a solid.
capacity	The volume a container can hold, often in litres.

Misconceptions to watch

COMMON ERROR

Three that persist

- Bigger perimeter means bigger area. A long thin rectangle can have a large perimeter and a small area.
- Area and perimeter share units. One is in cm, the other in cm^2.
- $1\ \text{m}^2 = 100\ \text{cm}^2$. It is $10\,000\ \text{cm}^2$, because both dimensions convert.

Self-assessment

I can …	Not yet	Getting there	Confident
find the perimeter of a shape	☐	☐	☐
use the circumference formula	☐	☐	☐
find the area of triangles and quadrilaterals	☐	☐	☐
find the area of a circle	☐	☐	☐
find the volume of a prism	☐	☐	☐
keep the units correct throughout	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to check a measurement answer, students must ask whether the size and the unit are both plausible for the object.

Looking ahead · Chapter 10 · Data, tables and graphs takes these ideas further. The self-assessment above tells you which to revisit first.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 9 — SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. Find the perimeter of a rectangle 11 cm by 6 cm. F
2. Find the area of a triangle with base 14 cm and height 6 cm. F
3. Find the area of a circle of radius 8 cm ($\pi \approx 3.14$). F
4. Find the volume of a cuboid 9 cm by 4 cm by 5 cm. F
5. Convert 5 m^2 to cm^2. F
6. Explain why the area of a circle quadruples when the radius doubles. R
7. A semicircle has diameter 12 cm. Find its perimeter and its area. R
8. A garden 15 m by 10 m has a path 1 m wide all round inside its boundary. Find the area of the path. P
9. A cylindrical drum of radius 40 cm and height 1 m is half full. How many litres does it contain? P
10. A cube has volume 64 cm^3. Find its surface area. P
11. A room is 4 m by 3 m and 2.5 m high. Find the area of the four walls, and the volume of the room. P
12. A tin of paint covers 12 m^2 per litre. How many litres are needed for two coats on four walls measuring 6 m by 2.5 m? E
13. Investigate: for a fixed volume, compare the surface area of a cube with that of a long thin cuboid. Which uses less material? E
14. Design a raised bed of volume 2 m^3 that fits in a 3 m by 2 m space. Justify the dimensions you chose. E

09 · CHECK YOUR PROGRESS

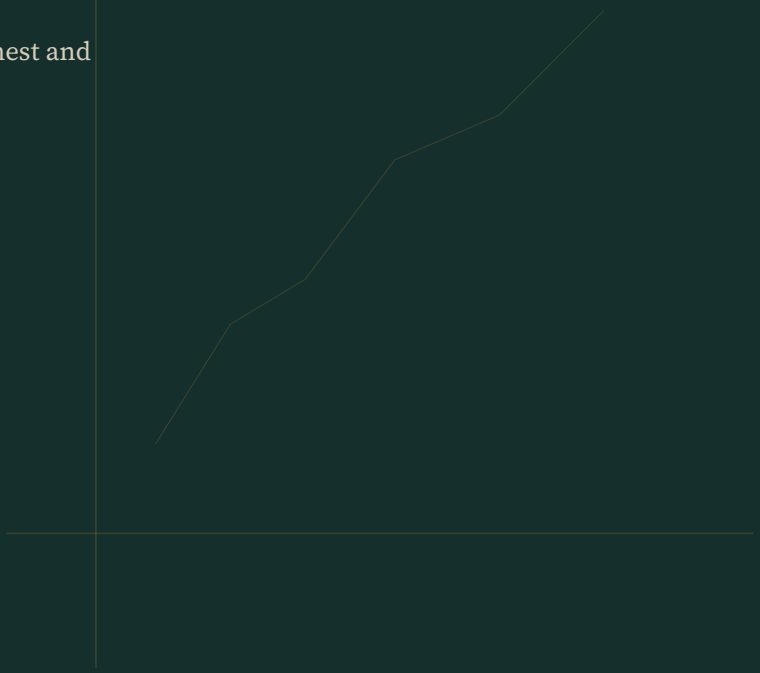
Question	If you cannot do this …
1–5	Revisit 9.1 and 9.2: decide the dimension before choosing a formula.
6–7	Revisit 9.3: the radius is squared, not doubled.
8–11	Revisit 9.4 and the composite-shape examples.
12–14	Revisit the investigation and the packaging task.

PART V

Reading the story in data

Every table and graph is a set of choices, and every choice decides what a reader notices.

Data becomes evidence only when the display is honest and the summary is appropriate.



10

READING THE STORY IN DATA

Data, tables and the shape of a graph

A table is a set of decisions about what to record. A graph is a set of decisions about what to show. Neither is neutral, and both can be read more carefully than they are usually read.

KEY CONCEPT · RELATIONSHIPS

RELATED · REPRESENTATION

RELATED · CHANGE

CONTEXT · FAIRNESS AND DEVELOPMENT

INQUIRY

How can the same data be displayed so that it tells two different stories, and how does a reader tell which one is honest?

01 · MYP UNIT PLANNER

Statement of inquiry	Data becomes evidence only when its collection, summary and display are chosen deliberately enough to be examined by someone else.

KEY IDEA

Key and related concepts

Key concept · Relationships. A graph asserts a relationship between quantities; the shape of the graph is the claim.

Related · Representation. Table, bar chart, dot plot and pie chart each hide something and reveal something else.

Related · Change. A time-series graph is about change, and its scale decides whether change looks dramatic.

NOTE

Global context · Fairness and development

Exploration: how statistics about a community are used to argue for resources, and what a well-chosen average can conceal about the people behind it.

Inquiry questions

- Factual: What are the mean, median, mode and range, and how is each calculated?
- Conceptual: Why can the mean be a misleading summary of a data set?
- Debatable: When does a graph become a form of persuasion rather than a report?

Criterion	Strands	Evidence in this chapter
A · Knowing and understanding	i, ii, iii	Calculate and interpret the three averages and the range.
B · Investigating patterns	i, ii, iii	Investigate how one extreme value changes each average.
C · Communicating	i, ii, iii, iv	Choose and justify a display for a given data set.
D · Applying mathematics	i, ii, iii, iv	Report a class survey honestly, including its limitations.

ATL SKILL

ATL · Media literacy skills

In order for a student to read a graph critically, students must check the axes before reading the shape.

02 · ENGAGE

The best class in the school.

Two classes compare their test scores. Class A has a mean of 62 % and Class B a mean of 58 %. Class B claims to be better because it has more students above 70 % and one student scored 100 %. Both claims use the same numbers.

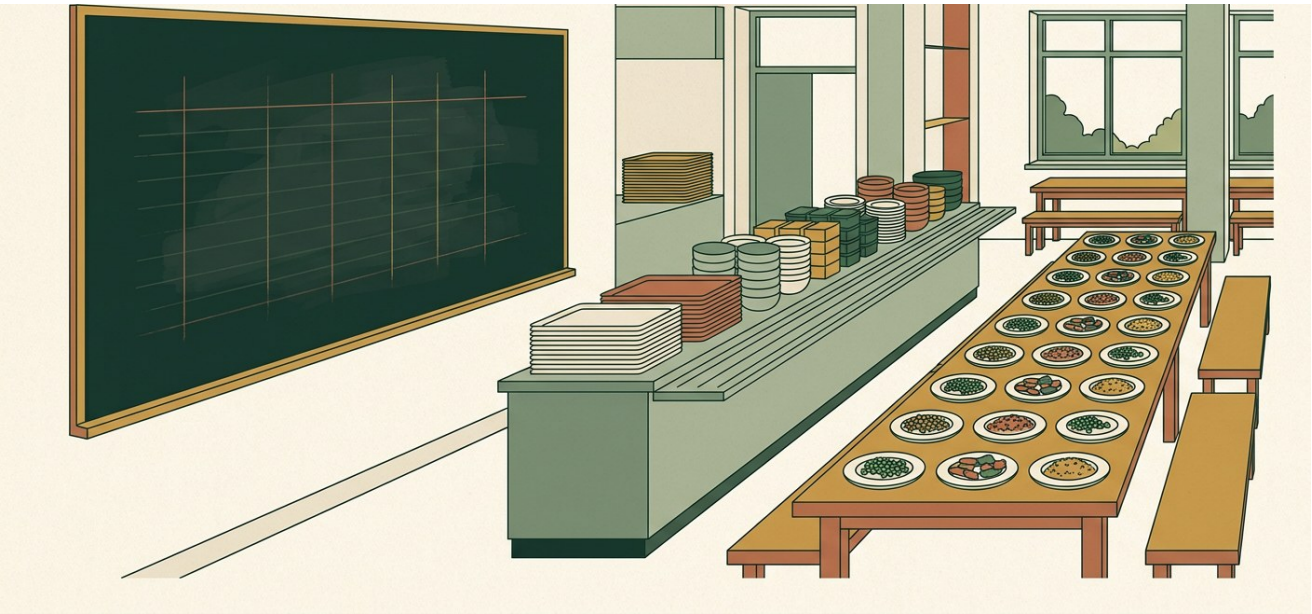


FIGURE 1 A canteen ledger: every record is a decision about what to count and how to count it.

Original illustration created for this edition.

03 · EXPLORE

Twenty numbers.

Collect twenty numbers — shoe sizes, minutes spent on homework, or scores from a quick quiz. Record them in a table with a tally column.

1. Which value occurs most often?
2. Arrange the values in order. Which value sits in the middle?
3. Add them all and divide by twenty. What do you get?
4. Which of the three numbers best describes the group, and why?

Value	Tally	Frequency
3	𝍸	3
4	𝍸 I	5
5	𝍸	4
6	II	2
7	I	1

total = 15 observations

the list becomes a shape you can see at once

FIGURE 2 A frequency table turns a list into a shape: the tally column is the data, the frequency column is the summary.

KEY IDEA

Three answers to one question

"What is typical?" has at least three mathematical answers: the mode (most common), the median (middle in order) and the mean (share out equally). They answer different questions and can disagree.

10.1 Collecting and organising data

CRITERION A · i · C · i ◆ REPRESENTATION ◆ RELATIONSHIPS

DEFINITION

Data vocabulary

Data are observations. A variable is what is being recorded; it is categorical if it names a group and numerical if it is a number. Discrete data can be counted exactly; continuous data are measured and depend on the accuracy of the instrument.

Question	Variable	Type
What is your favourite subject?	subject	categorical
How many siblings have you?	number of siblings	numerical, discrete
How tall are you?	height	numerical, continuous
How do you travel to school?	mode of travel	categorical

A good survey question has one meaning. "How much screen time do you have?" can be answered in hours, minutes, or by device, and the answers will not be comparable. Before collecting data, decide the unit and the categories.

WORKED EXAMPLE 10.1.1 — BUILDING A FREQUENCY TABLE

PROBLEM

The number of goals scored in twelve matches: 2, 1, 3, 0, 2, 4, 1, 2, 0, 3, 2, 1. Build a frequency table.

1. List the possible values: 0, 1, 2, 3, 4.
2. Count each: 0 appears twice, 1 three times, 2 four times, 3 twice, 4 once.
3. Check the frequencies add to 12, the number of matches.

ANSWER $0 \to 2,\ 1 \to 3,\ 2 \to 4,\ 3 \to 2,\ 4 \to 1$

$2 + 3 + 4 + 2 + 1 = 12$ ✓

COMMON ERROR

Losing an observation

If the frequencies do not add to the number of observations, one entry has been counted twice or missed. Always check the total before calculating anything from the table.

PRACTICE 10.1 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Classify each variable: eye colour, number of pets, mass, month of birth. F
2. Build a frequency table for: 4, 5, 4, 3, 5, 4, 6, 5, 4. F
3. Which of these is discrete: shoe size, height, temperature? F
4. Write a better version of the question “Do you like sport?” F
5. Explain why the unit must be fixed before a survey begins. R
6. Design a short survey to find how students travel to school, with at most five categories. P
7. A survey records “about 2 hours” of homework for several students. Explain why this weakens the data. P
8. Investigate two ways the same question could be worded to produce different answers, and explain the effect. E

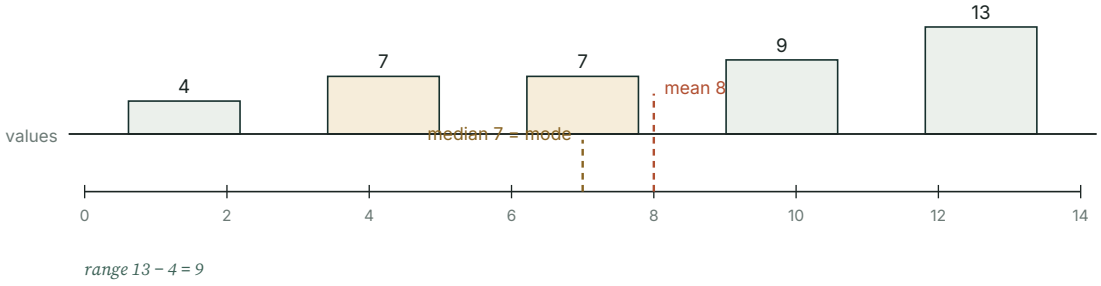
10.2 Averages and range

CRITERION A · ii · C · ii ◆ QUANTITY ◆ CHANGE

DEFINITION

Mean, median, mode and range

Mean = total ÷ number of values. Median = the middle value of the ordered list. Mode = the most frequent value. Range = largest − smallest, a measure of spread, not of average.



range 13 − 4 = 9

FIGURE 3 Five values, summarised three ways. Each marker sits at a different number, and each number answers a different question.

WORKED EXAMPLE 10.2.1 — ALL THREE AVERAGES

PROBLEM

Find the mean, median and mode of 4, 7, 7, 9, 13.

1. Mean: (4 + 7 + 7 + 9 + 13) ÷ 5 = 40 ÷ 5 = 8.
2. Median: the ordered list is unchanged; the middle value is 7.
3. Mode: 7 appears twice, more than any other value.
4. Range: 13 − 4 = 9.

ANSWER Mean 8, median 7, mode 7, range 9.

The mean is larger than the median because 13 pulls it upward √

WORKED EXAMPLE 10.2.2 — AN EVEN NUMBER OF VALUES

PROBLEM

Find the median of 3, 5, 8, 10, 12, 15.

1. There is no single middle value with six numbers.
2. The two middle values are 8 and 10.
3. Median = (8 + 10) ÷ 2 = 9.

ANSWER 9

Three values lie below 9 and three above √

COMMON ERROR

Taking the median of an unordered list

The median must be found after ordering. In 10, 2, 6, 4 the middle of the written list is 6, but the ordered list is 2, 4, 6, 10 and the median is (4 + 6) ÷ 2 = 5.

PRACTICE 10.2 — SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Find the mean of 6, 8, 10, 12. F
2. Find the median of 3, 9, 4, 7, 5. F
3. Find the mode of 2, 3, 3, 5, 7, 3. F
4. Find the range of 14, 6, 9, 21, 11. F
5. Explain why the median is often used for house prices. R
6. A data set has mean 12 and six values. If five are 8, 10, 11, 13 and 15, find the sixth. P
7. Scores: 45, 52, 58, 61, 63, 88. Find all three averages and say which best represents the group. P
8. Investigate: add one very large value to a small data set and describe what happens to the mean and to the median. E

10.3

Bar charts, dot plots and pie charts

CRITERION C · iii · B · i ◆ REPRESENTATION ◆ RELATIONSHIPS

A display is a decision. A bar chart compares categories, a dot plot shows every observation and how they cluster, and a pie chart shows each category as a share of a whole. Choosing the wrong one is not a small mistake: it can make a comparison invisible.

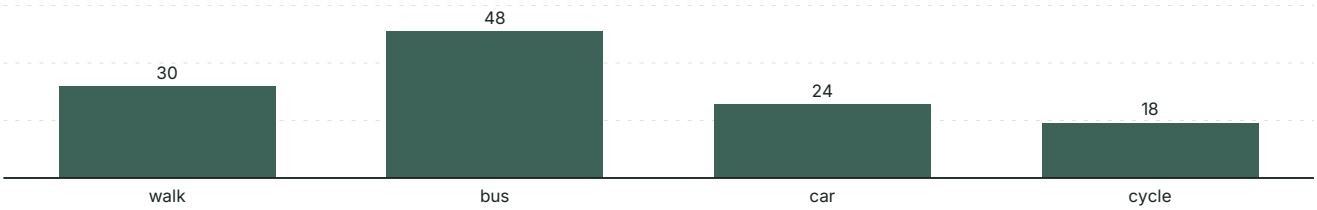
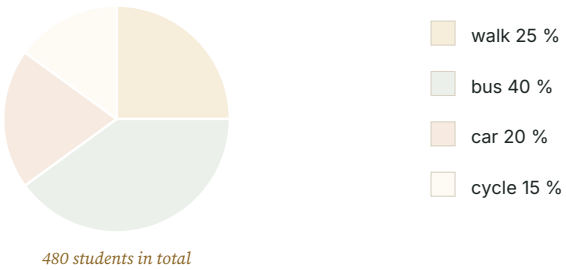


FIGURE 4 A bar chart of categorical data. The bars are separate because the categories are separate.



FIGURE 5 A dot plot of the same values shows the shape of the data: where it clusters and where it spreads.



480 students in total

FIGURE 6 A pie chart shows shares of one whole. It is the wrong choice when the parts do not add to a meaningful total.

WORKED EXAMPLE 10.3.1 READING A PIE CHART

PROBLEM

A pie chart shows 25 % walking, 40 % by bus, 20 % by car and the rest by bicycle. How many students cycle in a school of 480?

1. The four shares must total 100 %: 25 + 40 + 20 = 85, so cycling is 15 %.
2. 15 % of 480 = 0.15 × 480 = 72.
3. Check: 120 + 192 + 96 + 72 = 480.

ANSWER 72 students

The four numbers reproduce the whole school ✓

COMMON ERROR

A pie chart of a total that means nothing

Pie charts require parts of one whole. A pie chart of "number of goals per player" is meaningless once you realise the goals are not a fixed total being shared out.

PRACTICE 10.3 SUPPORT 1–4 · CORE 3–7 · EXTENSION 6–8

1. Which display best compares five categories? Give a reason. F
2. A bar chart shows 12, 9, 15 and 6. Find the total. F
3. In a pie chart, one sector is 90°. What percentage is it? F
4. Draw a dot plot for 2, 3, 3, 4, 4, 4, 5. F
5. Explain why pie charts are unsuitable for data that are not parts of a whole. R
6. A survey of 200 students shows 30 % prefer cricket, 45 % football and the rest athletics. Draw a pie chart. P
7. The same data are shown once as a bar chart starting at 0 and once starting at 50. Describe the difference in impression. E
8. Investigate which display makes a small difference between two groups easiest to see, using your own data. E

10.4 Interpreting graphs critically

CRITERION C · iv · D · iv ◆ CHANGE ◆ REPRESENTATION

Every graph contains choices: which axis, which scale, where the vertical axis starts, which summary is quoted. A reader who checks those choices can tell the difference between a real change and a drawn one.



FIGURE 7 The same data drawn twice. Left: the vertical axis starts at zero. Right: it starts at 90, and the increase looks enormous.

Four questions to ask of any graph

Check	What to look for
The axes	What is on each axis, and in what unit?
The scale	Does the vertical axis start at zero? Are the steps equal?
The sample	How many observations, and who was asked?
The summary	Which average is quoted, and why that one?

WORKED EXAMPLE 10.4.1 — A MISLEADING CLAIM

PROBLEM

A shop advertises “average customer spending of 45” after recording spending of 8, 9, 10, 12, 160. Is the claim fair?

1. Total = 8 + 9 + 10 + 12 + 160 = 199, so mean = 39.8 — not 45.
2. The 160 is an outlier and pulls the mean upward.
3. The median is 10, which represents four of the five customers far better.
4. The claim is arithmetically close but statistically misleading.

ANSWER Not fair: the median is 10; the mean is distorted by one outlier.

Four of five customers spend under 13 √

PRACTICE 10.4

SUPPORT 1–3 · CORE 2–6 · EXTENSION 5–7

1. A graph's vertical axis starts at 100. What effect does this have? F
2. Which is affected more by an outlier: the mean or the median? F
3. Name one question you should ask about a survey's sample. F
4. Explain why “average” alone is an incomplete statement. R
5. Find one misleading graph in a news source and describe the choice that misleads. P
6. Redraw a misleading graph with a zero axis and describe how the message changes. P
7. Investigate: how many ways can the same five numbers be summarised to support two opposite conclusions? E

04 · INVESTIGATE

INVESTIGATION 10

What one outlier does

CHANGE ◆ QUANTITY

Take the data set 5, 6, 6, 7, 8, 9, 10. Calculate the mean and median. Now add one extra value of 100 and recalculate. Record everything in a table.

Data set	Mean	Median	Range
5, 6, 6, 7, 8, 9, 10	…	…	…
with 100 added	…	…	…
with 1000 added	…	…	…

Your investigation

1. Complete the table, showing your working for the means.
2. Which measure changes most when one value is added?
3. Which measure changes least?
4. Write a sentence describing when the median is the better summary.
5. Give a real example where an outlier is the most important value in the set.

KEY IDEA

The result

The mean uses every value, so a single extreme value can move it a long way. The median depends only on position, so it barely moves. Neither is “correct”: the choice depends on what the summary is for.

ATL SKILL

ATL · Critical thinking

In order for a student to choose a summary, students must ask what decision the summary will support.

05 · APPLY

REAL-LIFE APPLICATION

The school travel survey

CRITERION D · i–iv ◆ REPRESENTATION ◆ FAIRNESS

Your year group is asked how students travel to school, so that the school can argue for a covered bicycle shelter. You have one week and 120 students to survey.

Your task

1. Define the categories and the unit before collecting anything.
2. Collect and tabulate the responses, including a "not recorded" row.
3. Choose a display and justify the choice against one alternative.
4. Calculate one average appropriate to the data and explain why the others are not.
5. Write two sentences of findings for the school council, and one limitation of your evidence.
6. Explain how a different choice of categories could have changed the conclusion.

> CRITERION
>
> **Criterion D · strands i–iv**
>
> - Identify the variable, categories and sample.
> - Select a display and a summary deliberately.
> - Apply them accurately and total the frequencies.
> - Evaluate the evidence and state its limits.

06 · CRITERION-MAPPED MINI-TASK

MINI-TASK

The honest infographic

CRITERION C · i–iv · D · iv

Context. A community group has one page to show how its after school programme has changed attendance over five years, and it must not mislead.

1. Invent five years of attendance figures with a real but modest rise.
2. Draw the graph twice: once with a zero axis, once with a truncated axis.
3. Write the caption each version would carry in a leaflet.
4. Explain which version you would publish and why.
5. Add one sentence about the size of the sample behind the figures.
6. Evaluate: is a true graph with a truncated axis still honest?

CRITERION

Criteria and strands developed

- Criterion C · Communicating — i–iv: axes, labels, scale and caption clarity.
- Criterion D · Applying mathematics in real-life contexts — ii–iv: choice of display and evaluation of its honesty.



FIGURE 8 The two versions of one data set, side by side, with the claim each one makes.

07 · REFLECT AND CONSOLIDATE

REVIEW

Chapter review

What this chapter established

- Data are classified by type before they are displayed.
- The mean, median and mode answer different versions of “typical”.
- The range describes spread, not average.
- Bar charts compare categories, dot plots show shape, pie charts show shares.
- A graph's scale and a summary's choice can change the claim it makes.

Vocabulary

Term	Meaning
data	Observations collected for a purpose.
variable	The quantity or category being recorded.
frequency	How many times a value occurs.
mean	Total divided by the number of values.
median	The middle value of an ordered list.
mode	The most frequent value.
range	Largest value minus smallest value.
outlier	A value far from the rest of the data.

Misconceptions to watch

COMMON ERROR

Three that persist

- The mean is the typical value. With an outlier it may describe nobody in the set.
- The range is an average. It measures spread, not centre.
- A pie chart suits any data. It needs parts of one whole.

Self-assessment

I can …	Not yet	Getting there	Confident
build a frequency table	☐	☐	☐
calculate mean, median and mode	☐	☐	☐
choose an appropriate display	☐	☐	☐
read a graph and check its axes	☐	☐	☐
explain what a summary conceals	☐	☐	☐
report data honestly to an audience	☐	☐	☐

CRITERION

Reflect on the criteria

- Criterion A: which calculation in this chapter would you now do without checking?
- Criterion C: where did notation or a diagram carry an argument for you?
- Criterion D: which real context changed how you understood the mathematics?

ATL SKILL

ATL · Reflection

In order for a student to report data responsibly, students must state what the data cannot show as clearly as what it can.

A note to your teacher

Two sentences: what you can now do that you could not do at the start of the chapter, and which single idea you would still like explained.

08 · MIXED PRACTICE

PRACTICE Mixed practice 10 — SUPPORT 1–5 · CORE 4–11 · EXTENSION 9–14

1. Find the mean of 7, 9, 11, 13. F
2. Find the median of 4, 8, 2, 9, 6. F
3. Find the mode of 5, 5, 6, 7, 5, 8. F
4. Find the range of 3, 17, 9, 12. F
5. Classify the variable “number of books read” as discrete or continuous. F
6. Explain why an outlier affects the mean more than the median. R
7. A pie chart shows 30 % of 240 students. How many students is that? R
8. Scores: 12, 15, 15, 18, 20, 45. Find mean and median and say which better represents a typical score. P
9. A bar chart axis starts at 200 and the bars are 210 and 220. Describe the impression and how to correct it. P
10. A data set of eight values has mean 15. Find the total. P
11. Design a frequency table for the letters in MATHEMATICS. P
12. A shop claims “average spend 60” from data 15, 20, 25, 30, 210. Evaluate the claim. E
13. Investigate: construct two data sets with the same mean but different spreads. E
14. Write a short paragraph advising a student how to read a graph in a news report critically. E

09 · CHECK YOUR PROGRESS

Question	If you cannot do this …
1–5	Revisit 10.1 and 10.2: type first, then calculate.
6–7	Revisit 10.3: which display answers which question.
8–11	Revisit 10.4 and the investigation.
12–14	Revisit the honest-infographic task and the four checks.

KEY IDEA

Closing the course

Nine chapters ago a number was a written symbol. It is now a position, a power, a factor, a fraction, a percentage, a variable, a length, an area, a volume and finally a data point — and each of those is a choice about what to notice. Mathematics is the discipline of making those choices explicit enough that someone else can check them.

REFERENCE

Answers and brief solutions

An answer ends a calculation; a solution records a decision. Where a question asks for reasoning, the entry below names the idea that earns the mark rather than reproducing a full response.

Chapter 1 · Number systems and place value

Ref	Answer
Prereq 1	4 060 — the zeros hold the hundreds and ones columns open.
Prereq 2	3.5 > 3.45: compare tenths first, 5 > 4.
Prereq 3	56 and 8.
Prereq 4	3 050.
Prereq 5	0.75.
Prereq 6	0.06 < 0.6 < 0.66.
1.1	1 (a) −2 (b) −7. 2 −7 < −3. 3 −3, −2, −1, 0, 1. 5 −2.05 is greater (−2.05 lies to the right of −2.5). 6 First submarine: −25 m; second: −32 m, so the second is 7 m deeper. 8 −6 and 1 (or any pair summing to −5 with difference 7).
1.2	1 (a) 2 007 (b) 940 (c) 6 210. 2 (a) 700 (b) 7 (c) 7 (d) 0.07. 3 6 000 + 0 + 40 + 5 = 6 × 1000 + 4 × 10 + 5 × 1. 4 3 or 4 (5 302 to 5 492). 5 0.7 = 0.70 > 0.65. 6 Estimate 250; exact 225. 7 2 400 kg. 9 0.003 means three thousandths; 0.300 means three tenths.
1.3	1 (a) 8 000 + 500 + 2 (b) 7 + 0.04 (c) 60 + 0.5. 2 (a) 9 307 (b) 20.5. 3 Thirty thousand, four hundred and seven; 3 × 10 000 + 0 × 1000 + 4 × 100 + 0 × 10 + 7 × 1. 4 604 is larger (600.4). 5 3 × 10 must be 3 × 100: 2 300, not 2 030. 6 1 205.09; the zero holds the tenths column open.
1.4	1 (a) > (b) < (c) <. 2 (a) 7 (b) 13 (c) 1. 3 45 680; 45 700; 46 000. 4 3.04 < 3.4 < 3.404 < 3.44. 5 10.00 — the hundredths digit 6 rounds 9.99 up. 7 5.64 rounds to 5.6 but to 5.64 at two places. 9 No: a number cannot be both about 100 and about 0.1.
1.5	1 (a) 1 200 (b) 30 (c) 700. 2 100; 98.98. 3 About 21 packets (500 ÷ 24 = 20.8). 4 4 000 × 0.5 = 2 000. 5 Estimate 5 × 3 = 15, so 161.2 is impossible; 5.2 × 3.1 = 16.12. 7 1 250 ÷ 18 ≈ 69 minutes.
Investigation 1	Each row multiplies the previous row by 10. With a leading zero allowed the count is 10^{n}; without it, $9 \times 10^{n-1}$. The difference is the 10^{n-1} strings that begin with zero.
Mixed 1	1 Eight million, seven thousand and forty. 2 0.06. 3 6.5 and 6. 4 (a) < (b) <. 5 7 092.005. 6 The zero holds the tenths column; the number is 2.05. 7 −0.5 < −0.05 < 0 < 0.05 < 0.5. 8 Estimate 2 000; exact 1 938. 9 Estimate 94.50; exact 94.50 — 6.75 × 14 = 94.50. 10 11.5 km to 12.5 km. 12 Largest 7 530.0, smallest 0.035 7 (with one decimal point only). 13 About 22 minutes.

Chapter 2 · Powers and order of operations

Ref	Answer
2.1	1 (a) 4^3 (b) 9^2 (c) 6^1. 2 (a) 8 (b) 25 (c) 10 000 (d) 1. 3 (a) 49 (b) 128 (c) 64. 4 $3^4 = 81 = 9^2$. 5 $4^3 = 64$, not 12. 7 $2^5 = 32$ people. 8 $2^{10} = 1\,024 > 100$. 9 32 and 64.
2.2	1 (a) $3 \times 10^3 + 4 \times 10^2$ (b) $7 \times 10^4 + 2 \times 10^2 + 5 \times 10^0$ (c) 8×10^6. 2 (a) 5 200 (b) 900 000. 3 4.5×10^4. 4 5.8×10^5 is larger: a higher power of ten dominates. 5 3 000 = 3×10^3: three zeros means exponent 3. 6 150 000 000 km; eight zeros. 7 $2 \times 10^3 + 0 \times 10^2 + 2 \times 10^1 + 6 \times 10^0$.
2.3	1 (a) 81 (b) 121 (c) 8 (d) 125. 2 (a) 8 (b) 11 (c) 5. 3 64, 81, 100. 4 49 and 64. 5 30 lies between 25 and 36. 6 14 cm; 56 cm. 7 80 cm. 9 Add 14 (64) and subtract 1 (49).
2.4	1 18. 2 8. 3 (a) 22 (b) 100 (c) 17. 4 (a) 12 (b) 3 (c) 9. 5 31. 6 (a) $(3 + 5) \times 2$ (b) $(12 - 4) \times 2$ (c) $18 \div (3 + 6)$. 7 $6 \div 3 = 2$ first, so $12 + 2 = 14$.
2.5	1 (a) 9 (b) 36 (c) 100. 2 (a) 18 (b) 72 (c) 200 — each is double question 1. 3 29 and 49; not equal. 4 81 cm^2. 5 (a) 27 cm^3 (b) 125 cm^3. 6 (a) 20 m (b) 80 m; doubling t multiplies d by 4 because t is squared. 7 (a) $(2 + 3)^2$ (b) $(4 \times 3)^2$. 8 11 cm; new area 1 089 $cm^2 = 9 \times 121$.
Investigation 2	10 folds: 1 024 layers, 102.4 mm. 20 folds: $2^{20} \times 0.1$ mm ≈ 104.9 m. Thickness after n folds = 0.1×2^n mm. About 15 folds exceeds 3 m.
Mixed 2	1 (a) 64 (b) 1 000 (c) 10 (d) 2. 2 $5^4 = 625$. 3 19. 4 $9 \times 10^4 + 0 \times 10^3 + 3 \times 10^2$. 5 (a) 32 (b) 72 (c) 84. 6 $(2 + 4) \times 3$. 8 60 m. 9 56. 10 81.92 mm ≈ 0.082 m. 12 $64 = 2^6 = 4^3 = 8^2$. 14 Cycle 2, 4, 8, 6; 2^{20} ends in 6.

Chapter 3 · Factors, multiples and primes

Ref	Answer
3.1	1 (a) 1×18, 2×9, 3×6 (b) 1×28, 2×14, 4×7 (c) 1×45, 3×15, 5×9. 2 (a) 7, 14, 21, 28, 35 (b) 11, 22, 33, 44, 55. 3 Nine factors: 1, 2, 3, 4, 6, 9, 12, 18, 36. 4 21, 22, 26, 27, 33, 34, 35, 38, 39. 6 12 seconds. 8 12 (factors 1, 2, 3, 4, 6, 12).
3.2	1 114, 321, 891. 2 234, 567. 3 Divisible by 2, 3, 4, 5, 6 and 10. 4 Yes: even, and digit sum 18. 6 The digit sum is a multiple of 3 but not of 9. 7 120.
3.3	1 23, 29, 31, 37, 41, 43, 47. 2 53, 59, 61. 4 $91 = 7 \times 13$, not prime. 5 7 + 23, 11 + 19, 13 + 17: three pairs. 7 101.
3.4	1 (a) $2^2 \times 3^2$ (b) $3^2 \times 5$ (c) $2^2 \times 5^2$. 2 $2^3 \times 3^2$. 3 2^7. 5 $210 = 2 \times 3 \times 5 \times 7$ and $14 = 2 \times 7$, so yes. 7 30. 8 $2^2 \times 3 = 12$ or $2 \times 3^2 = 18$.
3.5	1 (a) 4 (b) 9. 2 (a) 12 (b) 24. 3 HCF 12, LCM 72. 5 10:00. 6 12 packs. 8 4 and 60, or 12 and 20.
Investigation 3	Column 1 is fully shaded because 1 divides everything. Prime rows have exactly two shaded cells (1 and the number). Row n is shaded every n columns.
Mixed 3	1 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60. 2 15. 3 24. 4 $2^2 \times 3^2 \times 5$. 5 97 and 131. 7 A square of a prime: 4, 9, 25. 8 180 seconds. 9 36 pencils and 24 pens. 10 Yes: divisible by 3 (digit sum 18) and by 4 (last two digits 56). 11 97. 13 16 and 12, or 48 and 4. 14 12-tile squares; $7 \times 5 = 35$ tiles.

Chapter 4 · Fractions and decimals

Ref	Answer
4.1	1 e.g. (a) 2/4, 3/6 (b) 6/10, 9/15 (c) 14/20, 21/30. 2 (a) 2/3 (b) 3/5 (c) 3/4. 3 24/42. 4 8/12 and 6/9. 6 9/24: the bag is cut into 24 parts and 9 are used. 7 No: 6/8 is useful when the whole is cut into eight parts. 8 14/21. 9 1/3, 2/6, 3/9, 4/12, 5/15, 6/18, 7/21, 8/24, 9/27.
4.2	1 (a) < (b) <. 2 3/8 < 2/5 < 7/16 < 1/2. 3 5/9 (36/63 > 35/63). 4 7/12. 6 Comparing numerators and denominators separately ignores that the denominator is the unit. 7 2/5 km (0.4 km). 8 5/12.
4.3	1 (a) 3/5 (b) 1/2 (c) 2/5. 2 (a) 5/6 (b) 11/12 (c) 1/4. 3 29/24 = 1 5/24. 4 (a) 1 2/5 (b) 6 3/4. 6 7/8 full; 1/8 remains. 7 2/3 and 1/3. 8 7/8. 9 5/6 − 2/6 = 3/6 = 1/2.
4.4	1 (a) 1/3 (b) 2/3 (c) 1/4. 2 (a) 4/3 (b) 1/5 (c) 7. 3 (a) 2 (b) 2 (c) 6. 4 30. 6 1 1/8 cups (9/8). 7 10 pieces. 9 3/4 × 5/6 < 3/4 because 5/6 < 1.
4.5	1 (a) 5.05 (b) 5.3 (c) 0.46. 2 (a) 0.3 (b) 0.36 (c) 1. 3 (a) 8 (b) 30. 4 0.04 < 0.4 < 0.404 < 0.44. 6 9.00. 7 5 cups.
4.6	1 (a) 0.75 (b) 0.7 (c) 0.625. 2 (a) 3/5 (b) 1/4 (c) 1/20. 3 80 %. 4 1/4 and 3/8. 6 No: 0.33 is 33/100, which is less than 1/3. 7 2/5 < 0.41 < 3/7. 8 e.g. 1/11 = 0.0909…
Investigation 4	Terminating: 1/2, 1/4, 1/5, 1/8, 1/10. Recurring: 1/3, 1/6, 1/7, 1/9, 1/11, 1/12. Denominators of terminating decimals contain only 2s and 5s.
Mixed 4	1 (a) 2/3 (b) 4/7 (c) 3/4. 2 5/6. 3 1/6. 4 4/3 = 1 1/3. 5 7/20; 0.35. 6 11/20 < 0.58 < 3/5. 8 5/3 cups = 1 2/3. 9 7/32 m. 10 1.00. 11 9/20. 14 600/7 ≈ 85.7 litres.

Chapter 5 · Percentages

Ref	Answer
5.1	1 (a) 60 % (b) 8 % (c) 125 %. 2 (a) 0.45 (b) 0.07 (c) 1.2. 3 (a) 1/4 (b) 3/5 (c) 1/8. 4 35 %. 6 30 %. 7 35 % of 80 = 28; 40 % of 70 = 28: equal. 8 66.7 % (not exact).
5.2	1 (a) 9 (b) 21 (c) 3.5. 2 (a) 36 (b) 91 (c) 27. 3 12. 4 65 %. 6 75 %. 7 345 boys. 8 56. 9 The products are equal but the statements describe different bases.
5.3	1 88. 2 200. 3 (a) 1.35 (b) 0.88 (c) 2.5. 4 1 872. 6 244.80. 7 60. 8 13.4 %. 9 0.90, a 10 % decrease.
5.4	1 51. 2 49.50. 3 540. 4 100.80. 6 621. 7 2 000. 8 Shop A: 140; shop B: 144; shop A is cheaper by 4.
Investigation 5	Every starting price gives 36 %: 0.8 × 0.8 = 0.64. For p % twice, total discount = $100(1 - (1 - p/100)^2)$.
Mixed 5	1 85 %; 0.06. 2 48. 3 201.60. 4 63. 5 75 %. 6 48: 48 × 1.25 × 0.8 = 48, unchanged. 8 15 %. 9 264. 10 36.80; 4.80. 11 25 % of cost; 20 % of selling price. 12 80. 14 2 704, which is more than 2 700.

Chapter 6 · Patterns and equations

Ref	Answer
6.1	1 (a) 15, 19, 23 (b) 11, 8, 5. 2 (a) $3n - 1$ (b) $4n + 6$. 3 $5n - 4$. 4 53. 6 $5n + 7$. 7 The 12th term.
6.2	1 15. 2 7. 3 16. 4 (a) $n + 5$ (b) $3m$ (c) $2p - 8$. 6 $12 + 0.05m$. 7 $4x + 6$. 8 $n = 4$.
6.3	1 (a) $7x$ (b) $7y$ (c) $7a$. 2 (a) $8x + 2$ (b) $6m - 3$. 3 (a) $3x + 6$ (b) $10y - 5$. 4 $5x + 11$. 6 $2x + 10$. 8 e.g. $2(2x + 3)$ and $4x + 2 + 4$.
6.4	1 (a) 7 (b) 12 (c) 7. 2 (a) 5 (b) 5. 3 −3. 4 4. 6 10 km. 7 $4s = 44$, $s = 11$ cm. 8 $x = 4$. 9 e.g. $2(x + 1) + 4 = 16$.
Investigation 6	1, 2, 3, 4 squares use 4, 7, 10, 13 matchsticks. Rule $3n + 1$. 20 squares: 61. 100 matchsticks: $3n + 1 = 100$ gives $n = 33$.
Mixed 6	1 21, 26. 2 $3n + 1$. 3 11. 4 $4a + 5b$. 5 6. 6 $8x - 12$. 7 9. 9 $25 + 18h$; 79. 10 The 23rd term. 11 $x = 5$. 12 The numbers are 23, 24, 25. 13 $4x + 6$; 30.

Chapter 7 · Lines, angles and constructions

Ref	Answer
7.1	1 Acute, right, obtuse, reflex. 2 53°; 143°. 3 $x = 40$. 4 $y = 150$. 6 25°. 7 68° (co-interior).
7.2	1 (a) 70° (b) 55°. 2 70°. 3 No: the sum is 190°. 4 62°. 6 35°, 72.5°, 72.5°. 7 30°, 60°, 90°.
7.3	1 360°; 720°. 2 120°. 3 85°. 4 135°. 6 9 sides. 8 $n(n - 3)/2$.
7.4	1 Base angles 45.6° each, apex 88.8° (to 1 d.p.). 2 Every point on the bisector is the same distance from both ends of the segment. 3 About 35° each. 4 A right angle: $5^2 = 3^2 + 4^2$. 6 Construct an equilateral triangle on any segment; each angle is 60°. 7 Impossible: $2 + 3 < 7$, so the two short sides cannot meet.
Investigation 7	2, 3, 5 fail (they form a straight line); 2, 3, 7 fail; 5, 5, 10 fail; 4, 7, 12 fail. A triangle needs the sum of any two sides to exceed the third.
Mixed 7	1 117°; 62°. 2 135. 3 69°. 4 120°. 5 60°. 7 No: the sum is 190°. 8 20°. 9 80°. 12 15 sides. 14 Rafter = $\sqrt{(3^2 + 2^2)} \approx 3.6$ m; $3.6 + 3.6 > 6$.

Chapter 8 · Sets and classification

Ref	Answer
8.1	1 {A, E, I}. 2 No: 12 is a multiple of 4, so the statement is true only if $12 \in$ {4, 8, 12, ...}. 3 $\{3x : 1 \le x \le 4\}$. 4 {2, 4, 6, 8, 10, 12}. 6 Square numbers. 7 {B, A, N}: 3 elements.
8.2	1 Yes: both elements are in the larger set. 2 $\varnothing$, {p}, {q}, {r}, {p,q}, {p,r}, {q,r}, {p,q,r}. 3 {2, 3, 5, 7}; {1, 4, 6, 8, 9, 10}. 4 16.
8.3	1 $\cup$ = {1, 2, 3, 4, 5}; $\cap$ = {3, 4}. 2 4. 4 {A, E, I}. 6 7 do neither. 7 7.
8.4	1 Natural, whole, integer, rational, real. 2 Yes: a square satisfies the definition of a rectangle, so squares $\subset$ rectangles. 3 Rational: 0.25, 7, $-3/4$; irrational: $\sqrt{2}$, π. 4 Rectangle.
Investigation 8	Subsets: 1, 2, 4, 8, 16, 32 for $n = 0, 1, 2, 3, 4, 5$. Each new element doubles the count, because every subset either contains it or does not: 2^n.
Mixed 8	1 {1, 2, 3, 4, 6, 12}. 2 Yes. 3 $\{n^3 : 1 \le n \le 4\}$. 4 32. 5 {2, 4, 6, 8}. 7 $\cup$ = {1, 2, 3, 4, 5, 7}; $\cap$ = {2, 3}. 8 10 read neither. 9 6 elements. 12 10 had both. 13 Regions 2, 4, 8 for 1, 2, 3 circles.

Chapter 9 · Perimeter, area and volume

Ref	Answer
9.1	1 38 cm. 2 36 cm. 3 31.4 cm. 4 24 cm. 6 377 cm (≈ 3.77 m). 7 46 m, so 299.00.
9.2	1 120 cm^2. 2 30 cm^2. 3 28 m^2. 4 30 000 cm^2. 6 40 cm^2. 7 9 m^2.
9.3	1 50.24 cm^2. 2 78.5 cm^2. 3 56.52 cm^2. 4 4 cm. 6 1 200 + 314 = 1 514 m^2. 7 15.7 m^2 (28.26 − 12.56).
9.4	1 160 cm^3. 2 216 cm^3. 3 54 cm^2. 4 2 litres. 6 6 280 litres. 7 57.6 litres.
Investigation 9	Areas with perimeter 20 cm: 9, 16, 21, 24, 25 cm^2. The square gives the largest area; the flatter the rectangle, the smaller the area.
Mixed 9	1 34 cm. 2 42 cm^2. 3 200.96 cm^2. 4 180 cm^3. 5 50 000 cm^2. 7 30.84 cm; 56.52 cm^2. 8 46 m^2. 9 251.2 litres. 10 96 cm^2. 11 35 m^2; 30 m^3. 12 2.5 litres. 13 The cube.

Chapter 10 · Data, tables and graphs

Ref	Answer
10.1	1 Categorical, discrete, continuous, categorical. 2 3 → 1, 4 → 4, 5 → 3, 6 → 1. 3 Shoe size.
10.2	1 9. 2 5. 3 3. 4 15. 6 15. 7 Mean 54.5, median 59.5, mode none; the median represents the group better because 88 is an outlier.
10.3	1 A bar chart. 2 42. 3 25 %. 5 Because the parts must share one whole. 6 Cricket 108°, football 162°, athletics 90°.
10.4	1 It exaggerates small differences. 2 The mean. 3 Who was asked, and how many.
Investigation 10	Mean and median of 5, 6, 6, 7, 8, 9, 10 are 7.29 and 7. With 100: mean 18.9, median 7.5. With 1000: mean 130.6, median 7.5. The mean changes most; the median barely moves.
Mixed 10	1 10. 2 6. 3 5. 4 14. 6 The median depends only on position. 7 72 students. 8 Mean 20.8, median 16.5; the median is more representative. 10 120. 11 A 2, C 1, E 1, H 1, I 1, M 2, S 1, T 2. 12 Mean 60, median 25: the 210 is an outlier, so the claim overstates a typical spend.

REFERENCE

Formula bank

Every formula used in the course, grouped by part. A formula is not a fact to memorise: each one can be rebuilt from a diagram or a definition.

Number and proportion

Rule	Statement
A value in expanded form	digit × place value, summed over all columns
Percentage of a quantity	percentage ÷ 100 × quantity
Percentage change	(change ÷ original) × 100 %
Adding fractions	a/b + c/d = (ad + bc) ÷ bd
Multiplying fractions	a/b × c/d = ac ÷ bd
Dividing fractions	a/b ÷ c/d = a/b × d/c
Simplifying a ratio	divide every part by the same common factor
Unitary method	find the value of one unit, then multiply
Order of operations	brackets, indices, multiplication and division, then addition and subtraction
Dividing by a decimal	multiply both numbers by the same power of ten first

Algebra

Rule	Statement
nth term of a linear sequence	term = first term + (n − 1) × common difference
Expanding	a(b + c) = ab + ac
Solving two-step equations	undo addition or subtraction first, then multiplication or division
Collecting like terms	add or subtract the coefficients of matching terms
Substituting	replace each letter with its value, then evaluate
Linear sequence	constant difference between consecutive terms

Shape and space

Rule	Statement
Angles on a straight line	sum to 180°
Angles at a point	sum to 360°
Angle sum of a triangle	180°
Angle sum of an n-sided polygon	(n − 2) × 180°

Rule	Statement
Area of a rectangle	A = length × width
Area of a triangle	A = ½ × base × perpendicular height
Area of a parallelo-gram	A = base × perpendicular height
Area of a trapezium	A = ½ × (a + b) × h
Circumference of a circle	C = πd = 2πr
Area of a circle	A = πr^2
Volume of a prism	V = area of cross-section × length
Volume of a cuboid	V = length × width × height
Surface area of a cuboid	S = 2(lw + lh + wh)
Capacity	1 litre = 1 000 cm^3
Angles in parallel lines	corresponding and alternate angles are equal
Perimeter	the sum of all side lengths
Units of area	1 m^2 = 10 000 cm^2

Data

Rule	Statement
Mean	sum of values ÷ number of values
Range	largest value − smallest value
Median	middle value of an ordered set
Mode	the value that occurs most often
Frequency	the number of times a value occurs
Angle of a pie-chart sector	frequency ÷ total × 360°
Comparing averages	use the mean with the range, and the median when out-liers are present

REFERENCE

Glossary

Definitions in the form the course uses them. Where a word has a different meaning outside mathematics, the mathematical meaning is the one given.

A	
Acute angle	An angle smaller than 90°.
Algebraic expression	A combination of numbers, letters and operations with no equals sign.
Alternate angles	Equal angles on opposite sides of a transversal crossing two parallel lines.
Angle	The amount of turn between two lines that meet at a point.
Approximation	A value used in place of an exact value, close enough for the purpose.
Area	The amount of surface enclosed by a boundary, measured in square units.
Average	A single value used to summarise a set; mean, median and mode are the averages used in this course.

B	
Bar chart	A display in which the height of each bar represents the frequency of a category.
Base	In a power, the number being multiplied repeatedly; in a place-value system, the number of digits used.

C	
Capacity	The volume a container can hold, often measured in litres.
Composite shape	A shape made by joining simpler shapes.
Congruent	Exactly the same shape and size.
Constant	A quantity whose value does not change in a given expression.
Coordinate	A pair of numbers locating a point relative to two axes.
Cross-section	The face revealed by cutting a solid with a plane.
Cube	A solid with six congruent square faces; also a number multiplied by itself twice.
Cumulative frequency	A running total of frequencies.

D	
Decimal	A number written using place value to the right of a decimal point.
Denominator	The number below the line in a fraction: the number of equal parts in the whole.
Diameter	A straight line through the centre of a circle, joining two points on the circumference.
Difference	The result of subtracting one number from another.
Digit	One of the ten symbols 0–9 used to write numbers in base ten.
Divisible	A whole number a is divisible by b if a ÷ b has no remainder.

E	
Equation	A statement that two expressions are equal.
Equivalent	Having the same value, however written.
Estimate	A deliberately simplified calculation used to judge size.
Expanded form	A number written as the sum of its place values.
Exponent	The number that shows how many copies of the base are multiplied.
Expression	A mathematical phrase that can be evaluated but is not a statement of equality.

F	
Factor	A whole number that divides another exactly.
Formula	A rule written as an equation relating quantities.
Fraction	A number written as one integer divided by another.
Frequency	The number of times a value occurs.

G	
Generalise	To state a rule that covers every case, not just the ones examined.

H	
Highest common factor	The largest number that divides two or more numbers exactly.
Histogram	A display of continuous data in which bars touch, because the classes are continuous.

I	
Improper fraction	A fraction whose numerator is greater than or equal to its denominator.
Index	Another name for exponent.
Integer	A whole number or the negative of a whole number.
Intersection	The set of elements belonging to both of two sets.
Investigation	A structured exploration of cases leading to a conjecture and a general rule.

L	
Lowest common multiple	The smallest number that is a multiple of two or more given numbers.

M	
Mean	The sum of the values divided by the number of values.
Median	The middle value when the data are in order.
Mode	The value that occurs most often.
Multiple	The result of multiplying a number by a whole number.

N	
Natural number	A counting number: 1, 2, 3, …
Net	A flat arrangement of faces that folds to make a solid.
Numerator	The number above the line in a fraction: how many equal parts are taken.

O	
Order of operations	The agreed sequence for evaluating expressions: brackets, indices, multiplication and division, addition and subtraction.

P	
Parallel	Lines in the same plane that never meet.
Percentage	A fraction or ratio expressed as a number of hundredths.
Perimeter	The distance around a closed shape.
Perpendicular	Meeting at a right angle.
Pi (π)	The ratio of a circle's circumference to its diameter, approximately 3.142.
Place value	The value a digit carries because of its position.
Power	A product of equal factors, written with a base and an exponent.
Prime number	A whole number greater than 1 with exactly two factors: 1 and itself.
Prime factorisation	A number written as a product of prime factors.
Prism	A solid with a constant cross-section and flat faces.
Proportion	A statement that two ratios are equal.

R	
Radius	The distance from the centre of a circle to its circumference.
Range	The difference between the largest and smallest values in a set.
Ratio	A comparison of two quantities by division.
Reciprocal	The number you multiply by to get 1; the reciprocal of a/b is b/a.
Recurring decimal	A decimal with a repeating pattern of digits.
Rounding	Replacing a number with a nearby value at a stated precision.

S	
Set	A collection of objects, called elements.

S	
Simplify	To write in an equivalent but shorter form.
Solution	A value that makes an equation true.
Square number	The product of a number with itself.
Square root	A number which, multiplied by itself, gives the given number.
Standard form	A number written with a single digit before the decimal point and a power of ten.
Statement of inquiry	A sentence expressing the conceptual understanding a unit develops.
Subset	A set every element of which belongs to another set.
Substitution	Replacing a letter in an expression with a number and evaluating.
Surface area	The total area of all the faces of a solid.

T	
Term	A single number, letter or product in an expression.
Transversal	A line that crosses two or more other lines.

U	
Union	The set of elements belonging to either or both of two sets.

V	
Variable	A letter standing for a number that can change or is not yet known.
Venn diagram	A diagram using overlapping regions to represent sets.
Volume	The amount of space a solid occupies.

W	
Whole number	A natural number or zero.

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REFERENCE

References and acknowledgements

Sources consulted for framework alignment, and the provenance of every asset in this edition.

Curriculum framework

- International Baccalaureate Organization. *MYP: From principles into practice.* Consulted for the structure of key and related concepts, global contexts, statements of inquiry and approaches to learning. This book is not published or endorsed by the IB.
- International Baccalaureate Organization. *Mathematics guide: Middle Years Programme.* Consulted for the four assessment criteria and their strands and for the Year 1 scope of number, algebra, geometry and statistics.
- Oxford University Press, *MYP Mathematics* series. Consulted as pedagogical inspiration only: the sequence of inquiry, the balance of representation and the emphasis on conceptual framing. No text, exercise, diagram or layout has been copied.

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NOTE

A note on the illustrations

The illustrations in this edition are drawn rather than photographed. They were made to a single palette so that they sit inside the page design instead of interrupting it, and every one of them depicts a situation in which the chapter's mathematics is actually used.

Mathematics is a way of seeing, not a list of rules.

Ten chapters build Year 1 mathematics from the ground up: how our number system encodes quantity; why fractions, decimals and percentages describe the same wholes in different ways; how algebra compresses a pattern into a rule; how shape and measurement answer practical questions; and how data becomes evidence — or misleading noise — depending on how it is displayed.

Every chapter moves through one rhythm: engage, explore, explain, work through examples, practise, investigate, and reflect. Each section names the MYP criterion it develops, and each chapter closes with a review, mixed practice and a criterion-mapped mini-task.

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